

The economy has memory as well

The impacts of Argentina's last military dictatorship on economic growth

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Abstract

This research analyses the impacts of Argentina's last military dictatorship on economic growth. It is framed within what some authors call the "Argentine failure": the divergence of Argentina's per capita output from that of some countries that are today developed. The second chapter of this thesis systematises this discussion according to the characteristics of the economic theory underlying the different hypotheses on the determinants of the divergence. The third chapter analyses, from a classical-Keynesian theoretical framework, how the economic policies and institutional reforms carried out by the dictatorship combined with the international context to produce a break in the economy's growth trend. The magnitude of the growth divergence produced by the dictatorship's policies is estimated in the fourth chapter, using an econometric method known as "synthetic control". Finally, the implications of these results for the discussion on the "failure" are assessed.

The explanations for the divergence surveyed here assume that economic growth is determined by the supply of productive factors, on the assumption that market mechanisms exist which guarantee a tendency towards the full employment of these factors. In contrast, the analysis proposed here sheds light on the channels through which the military dictatorship constrained the growth of demand, mainly through its persistent impacts on the current account, public spending and wages. Without the profound changes brought about by the military dictatorship, Argentina could have reached a per capita output between 43% and 59% higher in 1991, growing at an average annual rate of between 1.4% and 1.9%, in line with countries such as Brazil, Australia, Mexico or Uruguay. These results are robust to a wide range of sensitivity analyses. The estimates suggest that the country did not undergo a break in its growth trend before 1976, so that these impacts of the policies of the last military dictatorship are the main factor behind the divergence in Argentina's economic growth during the twentieth century.

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To those who shaped me as a person. My family and the friends who have been with me since childhood (and, I hope, into old age).

To our people, who irreverently resist the neoliberal programmes of structural reform, and above all to the detained-disappeared comrades, present, now and always!

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1. Introduction

“Frustration at the failure of a destiny of greatness may be a consequence not so much of that failure as of the initial decision that such a destiny of greatness did in fact exist. [...] The end of all that caused a very understandable bitterness, but it does not pose an overly complicated historical problem.”

Tulio Halperín Donghi, in Trímboli (2023)

The discipline of history warns us about the risks of venturing counterfactuals when searching for causes and effects from a long-run perspective. In a deliberate methodological effort, historians tend to avoid asking questions such as: what would have become of history had events unfolded differently? Indeed, in the epigraph above Tulio Halperín Donghi maintains that the discussions on what in chapter 2 we will call “the Argentine failure” are of no particular interest to historians, because of their focus on a destiny that never existed. The absence of counterfactual scenarios, however, prevents the quantification of the implications of particular actions or policies, which explains a certain irreverence on the part of economists, statisticians and other scientists in resorting to them despite the warnings. This thesis is an effort to walk the fine line between understanding the long-run effects of a given set of policies and rejecting reckless counterfactuals. Its aim is to assess the impacts of the economic policies of Argentina’s last military dictatorship on economic growth. Specifically, it seeks to estimate the magnitude of the divergence caused by the dictatorship and to shed light on the channels through which its policies structurally conditioned growth.

The object of research is framed within multidisciplinary discussions on the trajectory of the Argentine economy during the twentieth century, which are reviewed in chapter two. While some authors have rejected the search for turning points in growth, others have identified critical junctures at which the actions of particular groups —especially the state, through economic policy— had persistent implications that outlasted a single government. One of these critical junctures opened with the 1976 coup d’état by the Armed Forces and the establishment of a military regime that suppressed political freedom and, with it, the resistance that the rest of society might have offered to its economic policies. The remaining chapters concentrate on this period and its consequences.

The dictatorship explicitly set out to inflict major transformations on the economic, social and political structure, in an attempt to refound the nation (Águila, 2023). In fact, despite the contradictions and overlapping projects that characterised its administration, the reforming drive subordinated other policy objectives, in particular the attempts at stabilisation (Zack & Pryluka, 2022). The economic policies and institutional reforms formulated with that objective interacted with changes that were taking place simultaneously in the international division of labour and the international financial architecture. The main impacts of this neoliberal economic programme arose from trade liberalisation, the reform of financial institutions and

the structural change in the political power of workers, as discussed in chapter three in the light of a model of classical-Keynesian inspiration.

These persistent effects that the military government sought to bring about in society, the state and the economy had severe consequences for economic growth. The fourth chapter quantifies the divergence in long-run economic growth produced by the policies reviewed, drawing once again on the classical-Keynesian theoretical framework as the theoretical foundation for the empirical estimation. This estimation is carried out by applying a synthetic control, a method that makes it possible to approximate the dynamics the Argentine economy would have followed in the absence of these reforms, even after taking global transformations into account. The exercise thus seeks to gauge the extent to which Argentina's economic performance in the last quarter of the century can be attributed to the military dictatorship and to its specific behaviour relative to other governments in the region (democratic or not) during the same period.

The proposed itinerary aims to offer a new perspective on an aspect already assessed by historians using other methodologies: the implications of the last military dictatorship for the subsequent period in economic terms, and the channels through which these materialised. It is hoped to make a contribution "from economics" as a discipline, based on the formulation of a counterfactual which, as is argued, is not "neutral" but rests on a theoretical framework, made explicit here. Quantifying the break in the long-run growth trend also makes it possible to place the results in the broader context of the discussion on the "Argentine failure". These discussions are addressed in the fifth and final chapter of this thesis.

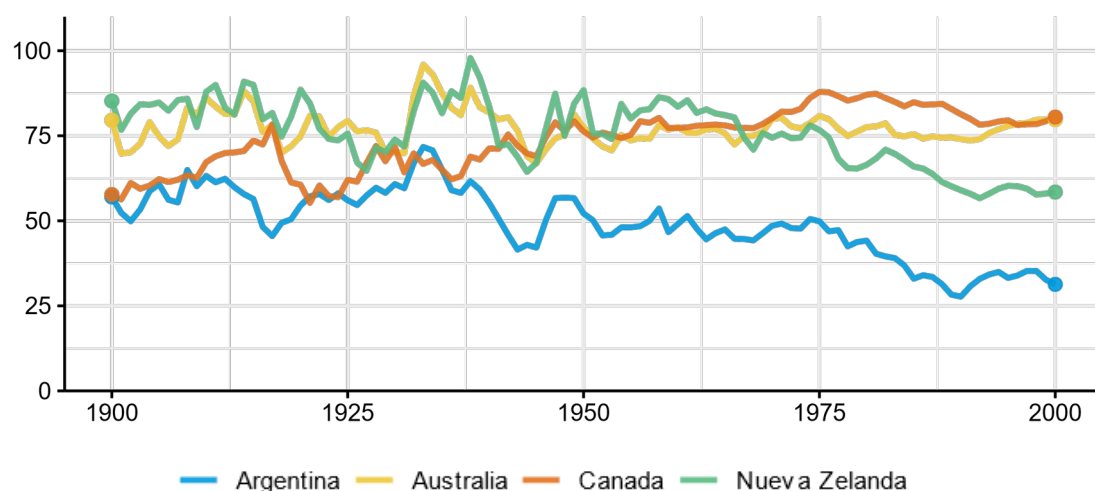
Only one reflection remains to conclude this introduction. These pages could never suffice for an economist such as the present author to convey the gravity of the atrocities committed by the military regime and the depth of their social implications. In this space we confine ourselves to estimating their consequences in one particular dimension —economic growth as measured by per capita output—. Although, as will be seen, the economic results entailed a considerable deterioration in living conditions and persistently restricted the possibilities of progress under democracy, no positive effect could ever have justified the systematic repression and the suppression of political activity witnessed between 1976 and 1983. Memory, truth and justice cannot be renounced.

2. Breaks in Argentine economic growth during the twentieth century

2.1. A framework for interpreting the discussion

As anticipated in the introduction, one of the recurring topics in Argentine economic historiography is the discussion around a stylised fact known as “the Argentine failure” (Míguez, 2005): the divergence of its growth from that of the core countries. While, according to data from the Maddison Project Database (2020), at the beginning of the twentieth century the country was among those with the highest per capita income in the world, with levels similar to those of Austria and higher than those of countries such as Finland or Japan, by the end of the century those countries had twice Argentina’s per capita output. Moreover, had the country sustained a growth rate equal to that of supposedly comparable countries, such as Australia, it would have ranked among the highest incomes in the world by the end of the period.

Figure 1. The “Argentine failure”. Per capita GDP of selected countries as a percentage of that of the United States



Source: Author’s elaboration based on Maddison.

The “stylised fact” itself, or at least its formulation in terms of failure, has not been free from controversy. On the one hand, some authors argue that at the beginning of the century Argentina was going through a boom favoured by unrepeatable factors (the recent development of the railways and the availability of land with which to expand its agricultural frontier), so that its income level did not adequately reflect its potential to keep growing (Llach, 2013). New data show that the improvement in the terms of trade experienced up to that point was even greater than previously thought, which reinforces the argument that Argentina was never a developed country like the European ones, but rather that its high level of per capita output was explained by conjunctural factors (Francis, 2013). As Francis shows, although the country’s per capita wealth stood at high levels, other indicators of the level of development in

dimensions such as education, health, political institutions and wage levels lagged behind those of the core countries. This author has also critically analysed the available estimates of Argentine per capita GDP (including those of the Maddison project), concluding that they may be significantly overestimating the country's income level (for further detail see section 4.5.7).

On the other hand, it could be the case that Argentina was part of a group of countries which, similarly affected by changes in the global economy, suffered a decline in their growth regardless of their individual idiosyncrasies. In that case, the divergence from other countries would be explained by these global changes, and one could not speak of a “failure”. Assessing the country's growth therefore requires establishing some point of comparison—or counterfactual—which raises the question of which country could be considered comparable to Argentina. Part of the discussion centres on this question, opening up a range of comparative studies with Australia, other countries in the region, and so on.

The greatest controversies, however, have centred on the causes of the “Argentine failure”, and consequently on the date at which the trend break occurs. While some authors conceive Argentina's relative stagnation as the outcome of an unfolding of multiple factors and decisions (Gerchunoff & Hora, 2021; Gerchunoff & Llach, 2019), others have pointed to different specific moments at which breaks in the growth trend are said to have occurred.

As detailed in the following section, one of the candidates singled out by various authors is 1914, when the outbreak of the First World War interrupted international economic flows, against the backdrop of the fall of Great Britain as the leading world power—in favour of the United States, a country which, instead of demanding Argentina's export products, competed with them—. This change at the global level coincided with the passing of the Sáenz Peña Law of 1912, which made the secret ballot universal for men and which, as we shall see, some authors identify as the beginning of the divergence.

The combination of these external and internal factors brought to an end a period of dizzying growth under a single pattern of development—the Agro-Export Model (MAE, by its Spanish acronym)—, politically marked by the hegemony of a liberal elite organised in the Partido Autonomista Nacional (PAN). Between 1880 and 1914 the national state was consolidated on the basis of a liberal constitution and a pattern of specialisation centred on the export of meat and cereals from the humid pampas (Hora, 2010). Some authors mark the abandonment of this model as the beginning of the divergence. But the date of the MAE's demise is also part of the historiographical controversies.

The Sáenz Peña Law put an end to the oligarchic order, while the world war ended the favourable terms of trade that had financed the expansion of the export model. But it was only in 1930 that the “forced industrialisation” brought about by the collapse of world trade during the Great Depression began. That year coincided with the coup d'état against the Yrigoyen government and the beginning of the “Infamous Decade”, a period in which conservative military governments succeeded one another through elections marked by fraud. The succession of coups d'état and military governments would continue until 1983, an aspect

emphasised by the hypothesis that the institutional instability of the period opened in 1930 was the cause of the divergence. Some historians interpreted 1930 as the beginning of an attempt at conservative restoration by the traditional privileged classes associated with the earlier export model, while others emphasised the state and societal innovations that would be the prelude to a new order (Hora, 2018). In economic matters, a public and later historical discussion arose over the characterisation of this interregnum between the MAE and that new order associated with industrialisation (Llach, 1985), and its implications for the rest of the twentieth century. Alejandro Bunge, a contemporary of the interwar period, put forward the hypothesis that the exhaustion of the MAE from 1914 onwards required a change in the role of the state—more favourable to deliberate industrialisation—, which did not materialise owing to political and institutional factors, in particular the opposition of the main economic groups (which were importers and landowners, rather than agrarian and industrial capitalists). Later, Di Tella and Zymelman (1967) christened this hypothesis the “theory of the delay”: the delay in starting industrialisation between 1914 and 1930 would explain the Argentine divergence.

Following the chronological order, the third major candidate is reached around 1946. During the post-war period there was, at the global level, a retreat of economic liberalism, the creation of the European welfare states and the “thirty golden years” of capitalism. In Argentina, the salient development was the rise of Peronism, a social and political movement that would condition politics up to the present day. Peronism would inaugurate “import-substitution industrialisation” as state policy. For some authors, as we shall see, it is the characteristics of Peronism itself and its consequences for the politics of the rest of the century that explain the divergence. For others, the crux lies in industrialisation, its *stop & go* economic cycles and its intention of moving away from Argentina’s “comparative advantages”. With heterogeneous strategies, state-led industrialisation would not be abandoned until 1976.

Finally, then, the years 1975/1976 are singled out as a new critical juncture. In a reading more centred on macroeconomics, the “Rodrigazo”¹ is identified as the breaking point, for having given way to a regime of high inflation and high economic instability that would last some 15 years. Other authors, in line with this thesis, focus on the beginning of the last military dictatorship and the abandonment of state-led industrialisation. The events of this period are presented in greater detail in chapter 3.

Some studies have systematised the available works in different ways, with a view to identifying alternative explanations. Both Míguez (2005) and Sánchez (2018) draw a first distinction between structural or secular explanations and explanations centred on economic policy decisions in conjunctural circumstances. Sánchez then groups, among those that emphasise secular causes, those based on the relationship between factor endowments and institutions, and those based on the relationship between factor endowments, the distribution of wealth and distortionary trade

¹ “Rodrigazo” is the name by which the economic programme implemented by Celestino Rodrigo, Economy Minister during the government of Estela Martínez de Perón, is popularly known; it was characterised by a sharp increase in the exchange rate and in the prices under public-sector control, which gave rise to an unprecedented increase in the rate of inflation.

policies. Among the conjunctural explanations, he distinguishes between those that centre on channels operating through effects on saving and those that centre on channels mediated by macroeconomic instability. For his part, Míguez (2005) argues that both among the structural or deterministic explanations and among the conjunctural or “historical” ones, some are based on neoclassical theories and others on dependency/structuralist theories. The author uses these two divisions to survey the explanations based on the colonial legacy, institutions, the international division of labour, saving, the ruling classes, economic policies and the state.

Taylor (2018) also attempts a synthesis, empirically assessing the relative importance of different explanations. First, he acknowledges that there are narratives centred on “deep determinants” such as geography, colonial heritage and the legal system, which according to the author do not fit the Argentine case adequately. He then classifies the “proximate determinants” into those that affect trade and those that affect investment. Finally, Glaeser, Di Tella and Llach (2018) organise their review according to four categories: the perspectives that question whether there is an Argentine exceptionality, those that centre on the absence of growth-friendly institutions, those that focus on external shocks, and those that emphasise policy decisions.

For the purposes of this chapter, it is worth stressing that the explanations put forward to account for the transformations in each of these episodes make use, *explicitly or implicitly*, of theoretical models that make it possible to arrange the multiple factors in a causal chain. Identifying the economic theories underlying the different explanations is central to assessing their internal consistency, and to shedding light on the kind of theoretical assumptions involved in their narrative. It is therefore proposed to classify these explanations on the basis of diagram 1, which shows four possible theoretical configurations according to how the determination of the functional distribution of income and of the level of output in the long run is conceived (for further detail on these cleavages see Duménil & Lévy, 1999; Petri, 1989; and Vernengo, 2020). A similar classification can be found in Blecker and Setterfield (2019, p. 7).

Diagram 1. Classification of economic theories^a

		Level of output in the long run	
		Supply	Demand
Functional distribution of income	Endogenous	Conventional frameworks	Neo-Keynesians (Cambridge)
	Exogenous	Classical, Marxist	Classical-Keynesian framework

Source: Author’s elaboration.

^a The diagram contains *examples* of each category; it is not an exhaustive list.

The diagram seeks to illustrate that views on the determination of distribution can be schematised into two broad groups. On the one hand, there are the perspectives in which the

distributive variables (in particular, the wage and the rate of profit) are regarded as two prices like any other, determined by market mechanisms. In that sense, they could be said to be *endogenous* variables. This is the case of conventional economic theory, the marginalist or neoclassical one, in which the “normal” remunerations of capital and labour clear the markets for these factors. By contrast, distribution can be described as *exogenous* if its determination is not the result of a *necessary*² market mechanism within the corresponding theory. Without denying the influence that economic variables such as the level of employment may exert on the wage, for example, in the view of the classical economists the wage is determined by the cost of the basket required to reproduce the labour force, whose composition depends in turn on historical, political and institutional factors, which do not necessarily bear a direct relation to the situation of the labour market in a given period.

Views on how the quantities produced at the aggregate level are determined can also be classified into two groups. Some of them hold that the level of output is determined, ultimately, by the available *supply* of factors of production. Such is the case of neoclassical theory, where, thanks to the mechanisms of substitution in consumption and production, the quantities of labour and capital actually supplied tend to coincide with their respective endowments. In this framework, aggregate demand adjusts in the long run to supply, a proposition known as “Say’s Law”, and in particular to the full-employment position of the productive factors. Other positions, recovering the thought of John Maynard Keynes and Michał Kalecki, propose that, conversely, it is supply that adjusts to demand, a postulate known as the “principle of effective demand”. This framework does not deny that constraints associated with the supply of productive factors may exist, but postulates that economies tend to operate at levels of activity below the full utilisation of those factors. Once full utilisation is reached, the constraints associated with the scarcity of those resources would begin to operate. Moreover, the principle of effective demand is also compatible with the existence of obstacles to the expansion of activity other than the full utilisation of productive factors, such as technical dependence (Dvoskin & Feldman, 2018).

In this way, an explanation of the course of the economy could be classified into these four quadrants. At the theoretical level, the conventional frameworks are located in the upper left, with endogenous distribution and quantities governed by the supply of factors. In the upper right quadrant, where output is assumed to be determined by the principle of effective demand but distribution is determined endogenously, we can place, by way of example, the authors known as “neo-Keynesians”³, such as Nicholas Kaldor or Joan Robinson. These authors analyse the relationship between growth and distribution in the light of the “Cambridge equation”: $g = s/v$, where g is growth, given by demand, s the saving rate and v the capital-output ratio, which is assumed to be given. In this approach the saving rate must adjust to the other two

² On the distinction between necessary relations, of a general character within classical theory, see Garegnani’s (1984) notion of the theoretical “core”.

³ The term “neo-Keynesians” refers, following for example Lavoie (2015, p. 31), to authors of the “Cambridge school” of the 1950s, and must be distinguished from the so-called *New Keynesians*, whose school would be classified among the “conventional frameworks”.

variables. Since the capitalists' propensity to save is higher than that of workers, this adjustment takes place through the distribution of income. The latter is therefore endogenous, determined by the level of growth (Petri, 2002).

In the lower left quadrant we can place the classical authors such as Adam Smith and David Ricardo. In classical theory distribution is exogenous, with the wage given by the cost of reproduction of the labour force or by subsistence, as the case may be. But, having written before Keynes and Kalecki introduced the principle of effective demand, they maintain a supply-led growth: given wages and technique, the rate of profit is determined, and this — through saving and investment— determines the rate of growth (Salvadori, 2003). Some Marxist authors also belong to this category, in so far as they assume that the level of output is determined by saving. Such is the case of Duménil and Lévy (1999), who argue that in the long run investment converges towards the supply of saving. Whether Marx's own theory belongs in this quadrant is subject to controversy, in that it contains elements such as crises of overaccumulation or the reserve army which contradict Say's Law (which he himself rejected), but it does not offer a satisfactory alternative, such as the principle of effective demand (Petri, 2021, p. 60).

Finally, the classical-Keynesian framework adopted in this work and outlined in the next chapter is located in the lower right quadrant⁴, since it adopts an exogenous determination of the distribution of income (like the classical authors) and holds that the principle of effective demand (as in Keynes and Kalecki) applies in the long run. In this approach, the growth trend is determined by the autonomous components of effective demand, through the multiplier effect of expenditure and the accelerator effect of investment, jointly known as the Sraffian supermultiplier (Serrano, 1995). Distributive variables such as the level of wages and the rate of profit are regarded as exogenous, and although they do not alter the growth rate of output, they can affect its level in so far as they impact on aggregate demand (Freitas & Serrano, 2015). A model with these characteristics is fully developed in the Appendix (Dvoskin & Torchinsky Landau, 2023).

2.2. The academic discussion on Argentina's growth in the twentieth century

What follows is a systematisation of articles that discuss the divergence in Argentina's growth during the twentieth century, or that argue in favour of a change of trend at one of these breaks in particular. The survey seeks to place the explanations of the causes of the break in Argentina's growth trend within the quadrants of the diagram, setting aside for the moment

⁴ This is not the only theory that could be placed in this quadrant. For example, neo-Kaleckian models such as that of Bhaduri and Marglin (1990) are also characterised by an exogenous distribution and quantities determined by effective demand.

the methodological differences between the various works. To that end, in each case the explanations given by the authors for the divergence are surveyed, highlighting the main factors involved and the causal chain between them. The reading is intended to be analytical, not merely descriptive: the aim is to reconstruct, from an analysis of the mechanisms implicit in their argument, which position corresponds to each in the classification. This implies that the literal word of the articles will not necessarily be taken at face value, so there is a clear possibility that other readings, including those of the authors themselves, may disagree with the classification presented here. A final methodological consideration is that the survey was restricted to articles published from 2000 onwards, so that their authors would have a complete retrospective view of the century.

A first set of works emphasises the role of the endowments of productive factors. Such is the case of Engerman and Sokoloff (2002), who argue that the divergence among the countries of the Americas is explained by institutional structures inherited from colonisation, which in turn respond to the heterogeneity in the distribution of productive factors across the continent. According to the authors, the unequal distribution of productive factors translated into inequalities in wealth, human capital and political power, which generated institutional structures that favoured the elites to the detriment of growth. For the Argentine case, Gerchunoff and Llach (2003, 2009) focus on how the unequal distribution of factors (not only among individuals, but also across the territory) gave rulers an incentive to adopt protectionist policies that turned out *ex post* to be harmful. For these authors, the availability of natural resources guarantees a more equitable distribution of income than in other countries and a lower relative industrial productivity, owing to the relative scarcity of capital, which would condition growth during the twentieth century. Tafunell and Ducoing (2015) also start from productive factors to explain the divergence between Argentina and Australia, two “similar” countries of late colonisation (*settler economies*⁵), arguing that the abundance of minerals in Australia encouraged early investment in machinery, which translated into higher growth rates after the fall in the availability of external financing in 1913. In other words, in this view, growth rates during the century are explained by the fall in the saving rate caused by the reduction in international capital flows after the First World War.

Gerchunoff and Fajgelbaum (2006) also compare the country with Australia by focusing on their factor endowments, taking up aspects common to both countries that are highlighted in the literature: they are countries with a similar geography (similar distance from the Equator, low prevalence of tropical diseases and direct access to two oceans), which produced and exported raw materials, and with a similar configuration of social conflict. In the authors’ explanation, differences in factors (greater abundance of minerals in Australia, fertile land in Argentina) combined with differentiated movements in international prices (larger increases in minerals), on the one hand, and with a different processing and institutionalisation of social conflict, on the other, to give rise to divergent growth trajectories. Along similar lines, Brambilla and others (2018) argue that, given Argentina’s factor endowment, changes in

⁵ The *settler economies* are: Argentina, Australia, Canada, the United States, New Zealand and Uruguay.

international prices generated a political conflict around the protection/trade openness axis, which resulted in distortionary trade (and other) policies that affected growth.

Several works position distortionary economic policies as the ultimate cause of the slowdown in growth. For Buera and others (2011) the effect is direct: fiscal and monetary policies, particularly from 1974 onwards, led to a regime of macroeconomic instability in which incentives for unproductive behaviour are generated. Hopenhayn and Neumeyer (2004) add trade policies to monetary and fiscal ones (as do Ertola Navajas, 2020, and Gadea and Sáenz-Villarroya, 2020), and point, besides the direct effect on output, to an indirect effect through weaker incentives to invest. Finally, Cortes Conde (2005) argues that distortionary policies combined with social conflict, which was in turn aggravated by the impossibility of sustaining higher wages owing to the distortions, to give rise to institutions that negatively affected investment, and with it growth. For Marzetti and Spruk (2023), from 1943 onwards distortionary policies combined with institutional reforms that restricted economic freedom, damaging property rights and contracts, which affected productivity and therefore growth. In a similar argument, Spruk (2019) proposes that democracy (brought by the Sáenz Peña Law) encouraged the redistribution of income and wealth and the erosion of property rights, affecting capital accumulation and growth. In this view it is democracy itself that explains the divergence in economic growth.

Other studies seek to explain the determinants of policies and political institutions on the basis of exogenous factors. For both Taylor (2018) and Galiani and Somaini (2018), the international *shocks* that hampered international trade gave rise to the political conflict and the institutions that encouraged the adoption of policies harmful to growth. In a different argument, Míguez (2005) and Campante and Glaeser (2018) identify the availability and generation of human capital as the factor behind the institutions and, in the second article, the political stability that would have fostered higher growth. Lastly, there are also other explanations, which point to direct effects of adverse trade shocks or to lower saving and investment rates (Sáenz-Villarroya, 2005).

In all the analyses presented up to this point, the ultimate or immediate causes of growth are supply-side factors. These are interpretations that could be placed within conventional or neoclassical theory: the level of output and distribution are determined by the endowment of productive factors, at most mediated by distortions or institutional rigidities that move the economy away from its potential output by affecting the efficiency with which factors are employed or their pace of accumulation, giving rise to the *trade-off* between growth and distribution (consumption) characteristic of neoclassical theory. Other theories may stress the relevance of supply-side factors, such as the availability of fertile land and its implications for international integration, without this, in itself, implying the assumption of a conventional theory. The implicit neoclassical character of these explanations is given by the fact that there is some market mechanism which, in the absence of political or institutional factors, would lead the economy to a path of growth and distribution given by the supply of productive factors. What moves the economy away from this path are forces that are not strictly economic,

although perhaps also influenced by factor endowments. What allows them to be classified as theories of “supply-led” growth is the direct influence of factors on growth, so that denying the general validity of this theory does not imply denying that endowments may be relevant for other reasons.

The work of Gerchunoff and Rapetti (2016) is harder to categorise. In their view, the course of the Argentine economy during the twentieth century is explained by a structural distributive conflict arising from the discrepancy between workers’ wage aspirations and the economy’s capacity to satisfy those needs. This conflict leads to macroeconomic policies which, with the aim of satisfying social demands, generate recurrent balance-of-payments crises. For the purposes of this work, it should be noted that the authors make an exception to their explanation for the period 1983-2000, arguing that during this time the structural distributive conflict was suspended. Until 1991, the level of external indebtedness is said to have placed the exchange rate beyond its level of macroeconomic equilibrium, causing exceptionally low real wages. During Convertibility (1991-2001), real appreciation is said to have been accompanied by high levels of unemployment, preventing the satisfaction of social demands (or at least of all of them).

Beyond the exceptions, the difficulty in classifying this perspective lies in the fact that it combines expansions of activity caused by improvements in the real wage and demand policies with supply constraints (such as the low productivity of the tradable sector) that prevent wage aspirations from being met. One argument for placing it among the supply-side explanations is that its model assumes a labour demand curve with a negative slope that is sufficiently elastic for lower real wages to allow an increase in employment, through improvements in competitiveness that boost exports. On closer inspection, this relationship implies that the economy can only deviate from its full-employment level because of more or less transitory “rigidities”: institutional or political factors that do not allow market forces to function properly. If these factors were removed and the market could function freely, the economy would tend by itself towards full employment, as neoclassical theory assumes. Wages would fall until reaching the exchange rate of macroeconomic equilibrium, where competitiveness boosts exports until full employment is reached. The authors themselves characterise distributive conflict as a phenomenon that occurs at full employment (Gerchunoff and Rapetti, 2016, p. 258). As regards distribution, although it may vary according to exchange-rate policy, equilibrium is only reached when the wage equals the one compatible with macroeconomic stability⁶. That is, there is a tendency for the wage to gravitate around the one determined by market equilibrium, although the authorities may divert it from that level at the cost of destabilising the macroeconomy (a behaviour described as the “populist paradox” (Gerchunoff et al., 2020)).

Although less numerous, alternative proposals explain the Argentine divergence on the basis of an exogenous treatment of distribution in the sense set out above, that is, as determined by

⁶ For a critique of this perspective see Álvarez and Brondino (2024).

political and institutional factors, historically determined, and not amenable to characterisation by a general economic dynamic. In the interpretation of Basualdo (2006; 2001), for example, the military dictatorship of 1976 gave rise to a turn in economic policies and a profound change in institutions, persistently altering the balance of forces under a new “mode of accumulation”. This perspective belongs to a vast body of interpretations that attribute the country’s economic stagnation and political instability to the nature of its dominant classes (Castellani & Heredia, 2020; Gerchunoff, 2019; Nochteff, 1994; Sabato & Schvarzer, 1985). As Libman et al. (2022) point out, this interpretative framework comes from critical economics and economic sociology, and emphasises the emergence of a “new economic power” after the dictatorship (the authors refer, specifically, to Azpiazu et al., 1987; Basualdo, 2010; Castellani, 2006; Ortiz and Schorr, 2006; among others). Although they do not necessarily focus on economic growth over the whole twentieth century, there are several works that take up this interpretative framework to analyse different aspects of the period opened in 1976 (Azpiazu & Schorr, 2010; González, 2012; Kulfas & Schorr, 2003; Manzanelli et al., 2015; Pucciarelli, 2004). For example, González (2012) maintains that the change in the mode of accumulation meant, for workers, the beginning of a process of deterioration in their living and working conditions. Bona and Barrera (2021) and Ortiz and Schorr (2021) focus on financial aspects, highlighting that external debt constrained growth and how external creditors and the IMF (new actors emerging from financial liberalisation) prevented the choice of an alternative development model. For her part, Castellani (2021) analyses how trade liberalisation, the industrial crisis and the spheres of accumulation privileged by the dictatorship led to a concentration of capital in local economic groups (the new economic power), which attained veto power over economic policies. The lower growth rates, in the view of these authors, are explained by a mode of accumulation with financial predominance, established and defended by the elites, which favours deindustrialisation, the regressive redistribution of income, external indebtedness and capital flight. This rentier behaviour works like an endowment, which fundamentally conditions the trajectory of output without demand policies being able to reverse it.

Finally, an alternative interpretation of Marxist origin (Carrera, 1998; Kennedy, 2018) proposes that the specificity of the Argentine economy in the international division of labour, given by its abundance of fertile land, ties its economic cycles to the evolution of international commodity prices (and, therefore, of ground rent). This specificity entails the necessity of a conflictive process of accumulation, in which falls in ground rent weigh on the conditions of reproduction of the labour force. In the face of these contractions, the real wage falls, the distribution of income deteriorates, and the pace of accumulation slows. It should be noted that this perspective criticises the use of GDP at constant prices as an indicator of long-run growth. In Kennedy’s (2011) alternative indicator, the “new social wealth” performs significantly worse than GDP from 1976 onwards, with a fall in output (in levels, not in per capita terms) of 22% between 1975 and 2000⁷. Within this theoretical framework one could place the works of Kennedy (ed., 2018a), Kennedy and Aguila (2021), Graña (2015), Cazón et al. (2016) and

⁷ GDP at constant prices grew by 49% in the same period. Given the methodology, this difference is explained by the deterioration in Argentina’s productivity relative to the world frontier.

Grinberg (2022), among others. Grinberg (2022), for example, emphasises that agricultural prices entered a phase of decline that would last three decades, and which coincided with the incorporation of low-cost labour in East Asia. This forced accumulation to be sustained through external indebtedness, which in the long run would compete with industry in appropriating ground rent. In a complementary analysis, Graña and Kennedy (2017) focus on how advances in automation and telecommunications deepened the productive lag of Argentina, which had neither a domestic market attractive to foreign investment nor firms developing frontier technologies.

In so far as, in this perspective, the distribution of income is not determined by labour productivity, it is an explanation with exogenous distribution. On the other hand, larger inflows of foreign exchange are associated (automatically, or at least in a “socially necessary” way) with increases in domestic expenditure: a greater degree of external space is necessarily used by domestic absorption, so that it is demand that adjusts to supply, not vice versa. The increase in external space generated by a rise in ground rent is used—for example, by a growth of industry— without increases in effective demand necessarily having to intervene.

Table 1 summarises the account of the explanations for the Argentine divergence. In the first column they were classified according to whether the immediate determinants of growth are supply-side factors (factor endowments, the supply of saving and its distortions) or demand-side factors (components of autonomous demand or the distribution of income). In the second, according to whether the distribution of income is conceived as endogenous or exogenous. The last column sets out the causal chain that explains growth, and the legend can be consulted at the end of the table. Each arrow runs from a cause to its effect, and parentheses indicate that the variables act jointly to affect another and have common causes.

Table 1. Explanations of Argentina’s decline during the twentieth century

Level of output	Distribution	Study	Causality (Y: Growth)
Supply	Endogenous	Engerman and Sokoloff (2002)	$B \rightarrow E \rightarrow Y$
		Tafunell and Ducoing Ruiz (2015)	$B \rightarrow D \rightarrow Y$
		Gerchunoff and Fajgelbaum (2006)	$(B,F) \rightarrow Y; (B,E,I) \rightarrow Y$
		Cortes Conde (2005)	$(A,G,I) \rightarrow E \rightarrow D \rightarrow Y$
		Míguez (2005)	$C \rightarrow Y; C \rightarrow E \rightarrow Y$
		Galiani and Somaini (2013)	$F \rightarrow (E,I) \rightarrow G \rightarrow Y$
		Brambilla et al. (2013)	$(B,F) \rightarrow I \rightarrow (A,G) \rightarrow Y$
		Sánchez-Villarroya (2005)	$D \rightarrow Y; F \rightarrow Y$
		Gadea and Sanz-Villarroya (2020)	$(A,G) \rightarrow Y$
		Campante and Glaeser (2013)	$C \rightarrow Y; C \rightarrow (E,I) \rightarrow Y$

		Ertola-Navajas (2020)	$(A,G) \rightarrow Y$
		Taylor (2018)	$F \rightarrow (E,I) \rightarrow (A,G) \rightarrow D \rightarrow Y$
		Gerchunoff and Rapetti (2016)	$I \rightarrow (A,F) \rightarrow Y$
		Buera et al. (2011)	$A \rightarrow Y$
		Gerchunoff and Llach (2003; 2009)	$B \rightarrow (A,G) \rightarrow Y$
		Hopenhayn and Neumeyer (2004)	$(A,G) \rightarrow Y; (A,G) \rightarrow D \rightarrow Y$
		Spruk (2019)	$E \rightarrow (D,C) \rightarrow Y$
		Marzetti and Spruk (2023)	$(E,G,A) \rightarrow D \rightarrow Y$
	Exogenous	Kennedy (2018)	$(B,F) \rightarrow I \rightarrow H \rightarrow Y$
		Graña and Kennedy (2017)	$(B,F,H) \rightarrow (D,F) \rightarrow Y$
		Grinberg (2022)	$(B,F) \rightarrow (A,H,D,F) \rightarrow Y$
		Basualdo (2006; 2001)	$(A,E,G) \rightarrow H \rightarrow Y$
		González (2012)	$(A,G) \rightarrow (D,H) \rightarrow Y$
		Castellani (2021)	$(A,G) \rightarrow (A,D,F) \rightarrow Y$
		Ortiz and Schorr (2021)	$(A,E) \rightarrow (E,F,H) \rightarrow Y$
		Bona and Barrera (2021)	$(A,E,H) \rightarrow (D,F) \rightarrow Y$
Demand	Endogenous	No works on the "Argentine failure" were found	
	Exogenous	No works on the "Argentine failure" were found	

A	Fiscal and monetary policies	D	Saving and investment	G	Trade policies
B	Factor endowments	E	Institutions	H	Wages or distribution
C	Human capital	F	Foreign trade and balance of payments	I	Conflict

Source: Author's elaboration.

As a first conclusion of this review, then, the available explanations have tended to locate the causes of growth in supply-side factors, such as the supply of saving (capital) or the endowment of productive factors such as land and labour. A considerable group of authors accounts for the divergence on the basis of a *trade-off* between growth and distribution, which arises from the political will to apply measures that “distort” relative prices, generating inefficiencies that affect production. This is a *trade-off* characteristic of neoclassical theory which is not necessarily found within the classical approach, where prices (including the remunerations of labour and capital) and quantities bear no necessary relation to one another.

There are some articles which, without explicitly alluding to the “Argentine failure”, analyse Argentina's economic growth in the twentieth century from alternative views in which some distributive variable is exogenous. Broadly speaking, they can be grouped under the interpretations of Basualdo (2001), on the one hand, and Kennedy (2018), on the other. These alternative explanations are largely absent from the literature reviews available so far, such as

those of Sánchez (2018), Taylor (2018) or Glaeser, Di Tella and Llach (2018), although they are taken up by works that analyse the period, such as that of Libman and co-authors (2022).

None of the articles surveyed, including in particular those that propose explanations from views alternative to conventional economic theory, simultaneously adopts an exogenous distribution of income and demand-led growth in order to explain, specifically, economic growth. This implies that the positions adopted in the available works assume that the economy is governed by mechanisms that lack general validity (Garegnani, 1970, 1978, 1984; Petri, 2004), and this work seeks to remedy this shortcoming.

The absence of classical-Keynesian works on the breaks in Argentina's economic growth in the twentieth century does not imply that there are no studies of Argentine development from this perspective. Classical-Keynesian analyses have provided interpretations and empirical evidence in favour of the applicability of this perspective to the Argentine case, but with an emphasis on other research questions. Álvarez and Brondino (2024) analyse the scope of, and possible limits to, income redistribution in the Argentine case. The authors also identify two strategies for analysing the evolution of income distribution in Argentina in the literature, part of which coincides with that surveyed for this work. On the one hand, the strategy based on a model of two equilibria (social and macroeconomic), such as that of Gerchunoff and Rapetti (2015). On the other hand, the theory of factor proportions based on the Heckscher-Ohlin model, within which works such as those of Taylor (2018) or Gerchunoff and Llach (2009) are framed. The authors offer critiques of both views from a classical-Keynesian framework and propose an alternative reading of Argentine political economy. Along similar lines, Álvarez and Dvoskin (2023) study the dynamics of income distribution during the last military dictatorship, concluding that the government's class interests correspond only partially to the distributive outcomes.

Amico (2015) focuses on wages, covering much of the twentieth century and emphasising how the abandonment of the state-led industrialisation model entailed a deterioration in the power of the trade union movement and a break in the trend of real wages. Fiorito (2015) reconstructs a journey through Argentine economic history since 1930, presenting the main facts and their interpretation from a Sraffian perspective. In addition, he estimates autonomous demand, the supermultiplier and other parameters using cointegration techniques and time-series regressions. Along these lines, Médici (2010) carries out a cointegration analysis to assess whether Argentina's productive capacity is determined by effective demand, and finds evidence in favour of that hypothesis by validating a conditional model of output using the Johansen method. Similar evidence can be found in Dvoskin and Médici (2023), who estimate that growth depends on the growth rate of autonomous expenditures, and that productive capacity adjusts to demand. Other empirical analyses can be found in Amico and others (2011), who survey the available evidence on the stationarity of output and provide regressions for the Okun relation and Verdoorn's law, the endogeneity of labour supply and the investment accelerator, among others. Finally, Fiorito (2018) estimates impulse-response functions for

investment in the face of changes in autonomous expenditure using a VEC model, and finds empirical support for the main predictions of the supermultiplier model.

In a historical interpretation based on theoretical models, Álvarez (2020) covers the second half of the twentieth century, analysing distributive conflict and inflation also from a classical-Keynesian framework. To that end he employs an open-economy model, a monetary and financial theory of distribution, a demand-led growth model based on the supermultiplier, and the formalisation of the financial sector and capital account regulations. Amico and others (2011) also discuss how different elements of alternative growth theories are reflected in Argentine time series. The formulation of an explanation of the course of the Argentine economy with exogenous distribution and the principle of effective demand aspires to continue along the path set by these contributions.

Beyond their theoretical differences, the works surveyed also differ in their methodological approaches. Table 2 below summarises the empirical strategies used, the countries employed as a point of comparison and the breaks identified in each case.

Table 2. Studies on breaks in the trend of Argentine economic growth during the twentieth century

	Metodología	Trabajo	Países comparación	Quiebre
Comparativas	Narrativa	Engerman y Sokoloff (2002)	Americanos	Siglo XIX
		Tafunell y Ducoing Ruiz (2015)	Australia	1913
		Gerchunoff y Fajgelbaum (2006)	Australia	1930
		Cortes Conde (2005)	Latinoamericanos, Australia, otros	1945, 1974
		Míguez (2005)	Latinoamericanos	Siglo XX
	Narrativa-analítica	Galiani y Somaini (2013)	Australia	1930, 1945, 1975
		Brambilla, Galiani y Porto (2013)	Latinoamericanos y otros	Siglo XX
		Grinberg (2022)	Brasil, Colombia, México	1973
		Graña y Kennedy (2017)	EEUU	1976
	Turning points	Sánchez-Villarroya (2005)	Australia, Canadá	1896, 1899, 1913, 1975
	Calibración	Gadea y Sanz-Villarroya (2020)	Australia, Canadá	1945
		Campante y Glaeser (2013)	EEUU	1930
	Control sintético/DD	Ertola-Navajas (2020)	Suiza, Dinamarca, otros	1945
		Spruk (2019)	Chile, Alemania, EEUU, otros	1912, 1930, 1946, otros
		Marzetti y Spruk (2023)	México, Suiza, Eslovenia, otros	1853, 1943
No comparativas	Narrativa	Basualdo (2006; 2001)	-	1976
	Narrativa-analítica	Gerchunoff y Rapetti (2016)	-	1976
		Kennedy (2018)	-	1976
		González (2012)	-	1976
		Castellani (2021)	-	1976
		Ortiz y Schorr (2021)	-	1976
		Bona y Barrera (2021)	-	1976
		Gerchunoff y Llach (2003; 2009)	-	Siglo XX
	Contabilidad del crecimiento	Hopenhayn y Neumeyer (2004)	-	1975
	Calibración	Taylor (2018)	-	1929
		Buera, Navarro y Nicolini (2011)	-	1974

Source: Author's elaboration.

As can be seen, the literature has employed various methodologies to provide evidence for or against the presence of breaks in the growth trend at different moments, and for or against particular causal explanations of the divergence. While some authors have inquired into Argentine history in itself, which the table describes as “non-comparative” methodologies, others have employed comparative methodologies. Within each of these sets, some studies consist of a historical narrative of the main facts, actors and interpretations of the period. For example, Engerman and Sokoloff (2002) offer an interpretation of the course of the countries of the Americas, including Argentina, while Basualdo (2006; 2001) provides an interpretation of Argentine history since 1976. Other studies offer an “analytical narrative”, anchored in a specific theoretical framework (Kennedy, 2018; González, 2012; Gerchunoff and Llach, 2003; 2009) or in a formally specified economic model (for example, Galiani and Somaini, 2013).

In some cases, in addition to employing an explicit economic model, data are used to calibrate the main parameters or to contrast them with simulations. Such is the case, for example, of

Buera and others (2011), who simulate the trajectory of output under two parameterisations of the Solow model and look for which of them is compatible with its actual evolution, concluding that the differences between the observed and the simulated series are due, precisely, to the different parameters (in particular, from 1974 onwards the fit is better in a model with stagnant productivity). A similar approach is that of growth accounting, which starts from a theoretical model to derive an accounting decomposition of output, which is then calibrated with data (as in Hopenhayn and Neumeyer, 2004). There are also works that employ time-series econometrics to identify structural breaks (*turning points*) in series of per capita output and relative per capita output, among others (Sánchez-Villarroya, 2005).

In so far as the estimate of the magnitude of the divergence depends crucially on the point of comparison (the counterfactual), part of the discussion on the “failure” and its causes is also methodological. Which counterfactual is plausible for Argentina, had the factors that led to the divergence not intervened? Is it a country, or the rest of the world as a whole, summarised for example in an average growth rate? Is it reasonable to employ a single point of comparison throughout the century? If the reference is a country, is it one from the region, or one with an apparently similar economic structure, such as Australia? And, of course, there arises the meta-question of the criteria that legitimise the choice of one point of reference or another, the answer to which depends crucially on the economic theory used to explain the growth of countries. From this literature review it emerges that only three of the studies surveyed have used statistical techniques to establish an adequate point of comparison for Argentina (Ertola Navajas, 2020; Marzetti and Spruk, 2023; Spruk, 2019), by means of difference-in-differences (DD) or the synthetic control method, which will also be adopted for this work.

Two conclusions thus emerge from the literature review. On the one hand, studies of the “Argentine failure” from “heterodox” perspectives have been scarce, and none has adopted a framework with exogenous distribution and effective demand in the long run. The analyses that adopt a classical-Keynesian approach have addressed the question of growth in Argentina both theoretically and empirically, but they have not explicitly discussed the causes of the divergence between countries at particular historical moments, nor have they quantified it. On the other hand, empirical analyses of the Argentine divergence in the twentieth century have employed various methodologies, but only some have used statistical techniques to construct an adequate counterfactual, and they have done so with a focus on the 1945 break.

This work sets out to contribute simultaneously to both agendas by means of empirical evidence, starting from a classical-Keynesian framework. In particular, it sets out to study the 1976 break, offering an interpretation alternative to those available, and applying a synthetic control method. Unlike others, which capture relations of correlation, this strategy makes it possible to quantify the extent to which the country’s growth was affected by the policies implemented by the military dictatorship, identifying a causal effect. In addition, quantifying the divergence at each specific trend break would make it possible to assess the relative importance of each for Argentina’s performance during the century. Future research could advance in this comparison by inquiring into the other breaks.

As in any estimation, economic theory plays a central role in the synthetic control method. The theoretical framework will make it possible to identify the key variables in the determination of growth. These variables will then enter the estimation as “controls”, making it possible to distinguish the effect attributable to economic policies from that arising from changes in international factors and their (potentially heterogeneous) impacts on structurally different countries. In parallel, a narrative account of the dictatorship’s economic policies will make it possible to reconstruct which of these variables were affected, something that can also be corroborated in the data thanks to the synthetic control, producing evidence on the channels through which the policies constrained growth. In this way, the aim is not to validate economic theory (in particular, classical-Keynesian theory) on the basis of the evidence but, on the contrary, to use theory to inform the generation of evidence.

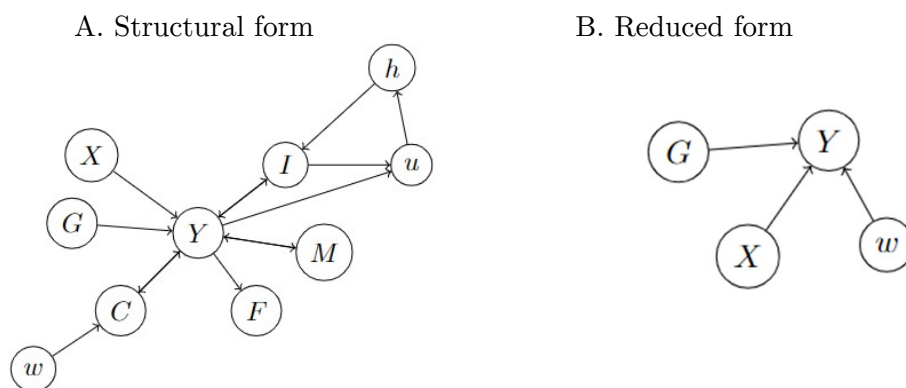
3. The structural transformations of 1976-1991

This chapter analyses the economic policies of the military regime and their consequences for growth during the period 1976-1991. The measures adopted by the dictatorship and their effects on structural features of the Argentine economy are what make it possible to hypothesise the existence of a break in economic growth from 1976 onwards, the magnitude of which is estimated in chapter 4. The aim is thus to complement the empirical estimation set out in the following chapter with a necessary conceptual understanding of the reasons for the divergence.

To that end we draw on the model for small open economies of Dvoskin and Torchinsky Landau (2023), which is presented in the appendix, with some slight adaptations. In terms of the interpretative framework of the previous chapter, this model belongs to the classical-Keynesian perspective, where the distribution of income is exogenous (that is, determined by political-institutional factors) and the level of activity is determined by the autonomous components of aggregate demand—that is, those that do not depend on the income generated during the period and that do not create productive capacity—.

By way of summary, figure 2 below presents the structural and reduced forms of the model. While panel A represents the set of causal relations involved in the determination of the level of activity (Y), panel B shows the exogenous variables, so that it could be interpreted as the directed acyclic graph (DAG) of the model. These relations are given by functional forms made explicit in the model together with their associated parameters. It can be seen there that, ultimately, the level of output depends on the level of public spending (G), the level of exports (X) and the wage (w)⁸. In the determination of the growth rate of output, the only exogenous determinants are the growth rate of public spending and that of exports.

Figure 2. Causal diagrams of the Dvoskin and Torchinsky Landau (2023) model^a



Source: Author's elaboration based on Dvoskin and Torchinsky Landau (2023).

^a Y : level of output. G : public spending. X : exports. w : wages. C : consumption. F : external asset formation.

M : imports. I : investment. u : utilisation of installed capacity. h : investment accelerator.

⁸ Therefore, the level of per capita output depends on the levels of public spending and exports per capita.

On the other hand, changes in the structural parameters of the model can affect the level of output. A fall in the wage share of income, a reduction in the employment or capital requirements per unit of output (v_l, v), an increase in the propensity to import in response to increases in income (m), a rise in the demand for external assets per unit of income (f) or an increase in the interest paid on external liabilities (i^e) would reduce the level of activity. The propensity to import, for example, besides reducing the multiplier by affecting the proportion of demand that is directed to domestic production, also determines a maximum level of output compatible with a sustainable level of external indebtedness. This is because the technical dependence assumed in this model implies that an increase in the level of activity will be associated with an increase in imports, deteriorating the balance of payments. Given a maximum level of external indebtedness, the balance-of-payments deficit is bounded and therefore determines a maximum level of output. A corollary of this aspect of the model is that a de-substitution of imports, caused for example by a process of deindustrialisation, reduces the maximum level of output.

Finally, it should be noted that, for a country with a negative net external position, an increase in the interest rate paid on external debt above the growth of export demand ($i^e > g_x$) would render the trajectory of external debt unsustainable. As discussed in this chapter, the changes in these parameters and in the determinants of growth caused by the dictatorship affected the economy's growth path persistently. That is, besides affecting the variables and parameters during the period of government, these transformations endured beyond 1983, producing structural changes in the productive structure (variables m, v_l, v, X), the wage (w) and the sustainability of external debt (f, i^e, D), which would limit growth and the space for boosting public spending (G). At the time of the estimation in chapter 4, these determinants of economic growth will have to be taken into account in the specification of the econometric model.

3.1. Structural changes during the last military dictatorship

“This is a change in the structure of financial institutions, a small revolution that goes much further than people can see. We are going to change them all and change the mentality, which is what matters”

Alfredo Martínez de Hoz⁹

The authorities, military and civilian, who took over the government after the civic-military coup had their own view of “the Argentine failure”, a view they sought to put into practice. This perspective acted as the guiding thread of their efforts and in particular of their economic policies, so that grasping it is a first central point for understanding the government’s motivations.

On 24 March 1976, the Military Junta took power with one central objective, which would end up subordinating other aspirations: the refounding of the nation. Ultimately, it sought to “correct” the —in its eyes— imperfect incorporation of the popular masses into politics (Sidicaro, 2004), a process said to have begun with Hipólito Yrigoyen (the first government after the secret ballot was made universal for men) and to have deepened from 1943 onwards with the emergence of Peronism. This view was shared by groups of intellectuals and liberal-conservative think tanks such as the Sociedad de Estudios y Acción Ciudadana (SEA) —to which the dictatorship’s economy minister, Alfredo Martínez de Hoz, belonged—, which advocated “reorganising” the nation after the Sáenz Peña Law of 1912 had frustrated the so-called “National Organisation Process” of the Argentine elites of the second half of the nineteenth century (Blaum & Román, 2022). From this reading comes the name “National Reorganisation Process”, with which the coup leaders christened their new project, whose priority was therefore the disappearance of populism as a national political culture (Semán, 2021, p. 198). This shared ideological substratum, liberal and conservative, made it possible to reconcile the heterogeneous interests of the different groups within the government beyond the basic agreement on the fight against “subversion” (Morresi, 2010). As the quotation that opens this section illustrates, the authorities of the economic team shared this objective of a comprehensive reform of society, and placed economic policies at its service.

This orientation of the regime had three implications worth highlighting. First, the success or failure of its project, if it is to be assessed on its own terms, must be analysed according to whether it managed to put an end to the development model in force until 1975 (Mesyngier, 1998). Second, the consensus within the civic-military coalition extended to the will to reform, but not necessarily to the shape of the model that would replace the previous one. This indeterminacy would be reflected in the lack of a development model for the subsequent period, with consequences for economic growth. The state-led industrialisation model would not be

⁹ Speech before the Legislative Advisory Commission, 1977.

recovered, but neither would it be replaced by a new one, beyond a new prominence of the financial sector which some authors characterised as “financial valorisation”. Lastly, for the purposes of our research objective it should be noted that economic growth appears as a secondary objective on the military government’s agenda. Although allusions to growth as an expected result of their programme can be found in officials’ statements, the promotion of growth is not at the centre of the objectives, but subordinated to the reform of the economy and to social disciplining¹⁰.

Without growth being among its highest priorities, the dictatorship thus sought a structural change. And various authors agree that this did indeed take place (Águila, 2023; E. M. Basualdo, 2001; Blaum & Román, 2022; Bona & Barrera, 2021; Castellani, 2021; Cuesta & Trupkin, 2022; Mesyngier, 1998; Müller, 2001; Rappoport, 2008; Zack & Pryluka, 2022). The policies and reforms that led to that structural transformation can be classified into three groups: (i) trade liberalisation, (ii) the structural reduction of the real wage, and (iii) financial liberalisation.

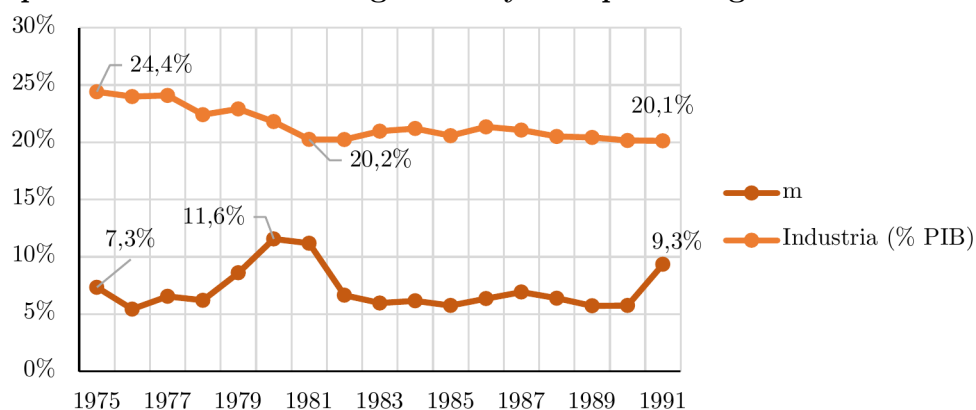
Trade liberalisation was one of the pillars of the dictatorship’s economic programme, with tariffs on imported products and their dispersion falling sharply until 1979 (Brambilla et al., 2018). Besides disciplining entrepreneurs through competition, in the economic team’s conception liberalisation would contribute to raising productivity and to reducing inflation through the convergence of domestic prices with external ones, an aspect that would be complemented by the introduction of the “tablita cambiaria”. This policy, based on the then novel Monetary Approach to the Balance of Payments introduced by the Chicago School (Blaum & Román, 2022; Heredia, 2004), set a series of pre-announced and decreasing devaluations with the aim of making domestic inflation expectations converge with international inflation. Since it was assumed that relative prices would be determined by “real” variables, while inflation would be determined by nominal variables, the advance announcement of a nominal price (the exchange rate) would cause both the money supply and the remaining prices to adjust their variation to that of this price. The gradual reduction of the rate of devaluation would allow domestic inflation to converge with international inflation¹¹.

In practice, the tablita did not manage to eliminate inflation, and price increases remained above the programmed rates of devaluation. This gave rise to a strong exchange-rate appreciation, which combined with the reduction of tariffs sharply reduced the competitiveness of the local economy. The joint effect of these measures was an increase in imports per unit of output (a de-substitution of imports, which in the model would be reflected in a rise in the parameter m) and a severe deindustrialisation, which affected the economy’s productive capabilities.

¹⁰ This is an example of Kalecki’s (1943) hypothesis that in certain historical contexts capitalists might prefer the disciplining of the working class to an increase in the mass of profits.

¹¹ For further details on the link between the “tablita” and the monetary approach to the balance of payments, see De Pablo (1983), Zack and Pryluka (2022), Blaum and Román (2022) or Heredia (2004).

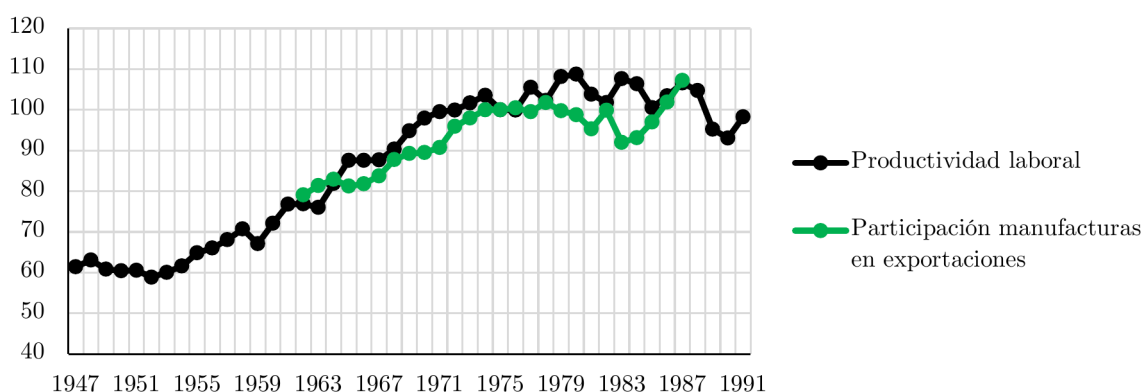
Figure 3. Imports and manufacturing industry as a percentage of GDP



Source: Author's elaboration based on Kidyba and Suárez (2020).

As a consequence, a period of marked productivity stagnation began (Buera et al., 2011, figure 4), driven by the shift of employment from sectors of high labour productivity to sectors of low productivity (Cortes Conde, 2005; Hopenhayn & Neumeyer, 2004). Although these effects were not those expected within the Monetary Approach to the Balance of Payments, they are an expected result according to the model described in the introduction to this chapter. The model is compatible with the Kaldor-Verdoorn law, which postulates that a reduction in economic growth has a negative impact on productivity growth. This fall in relative productivity would affect the external competitiveness of the economy, increasing the requirements of imported inputs and reducing exports (CEPAL-OIT, 2023). The export profile also deteriorated, as its concentration in primary products increased (Rappoport, 2008). Both the fall in productivity and the fall in the diversity of exports (in particular, towards products of lower technological content) generate, in terms of the model, a reduction in the income elasticity of exports, affecting both exports (X) and their growth rate (g_X).

Figure 4. Labour productivity and manufacturing exports^a (1975 = 100)

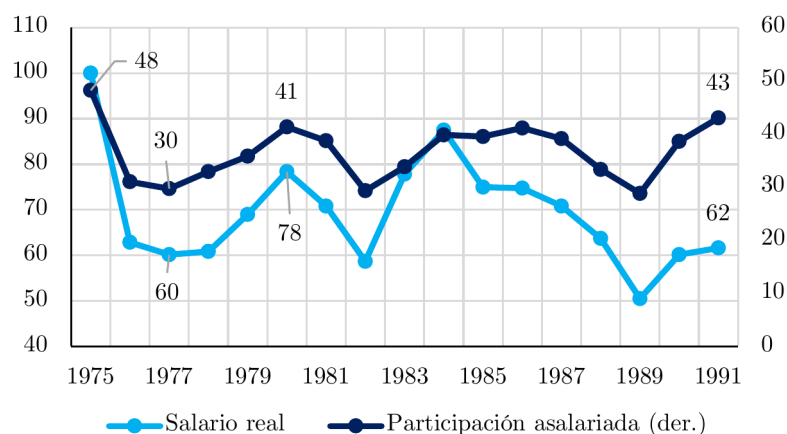


Source: Author's elaboration based on Graña and Kennedy (2008) and World Bank.

^a Evolution in percentage points, normalising the initial period to 100.

For its part, the structural reduction of the real wage (w in the model) found a first expression immediately after the coup. On 2 April 1976 nominal wages were frozen for 120 days, suspending every form of wage bargaining under threat of criminal prosecution. At the same time export duties were reduced (Brambilla et al., 2018), raising the domestic cost of food. In the month of the announcement inflation reached 33%, accumulating 54% after three months¹². In this way, the real wage fell by 36% in 1976, and by 1983 the share of the wage bill in income had fallen to 27% (from 45% in 1974, Rappoport, 2008). The peaks in the real wage and in the wage share of output (those of 1974) would not be recovered during the rest of the century (Graña & Kennedy, 2011). This marked volatility of real wages and their reaction to administrative acts illustrate the political-institutional character of their determination, reinforcing the relevance of the theoretical model developed in the introduction to this chapter for the study of this period. The fall in productivity would constrain the growth of the real wage under the new growth pattern (Graña & Kennedy, 2011). Again, within the framework of the model described, although the wage (w) is exogenously determined, productivity growth determines the maximum real wage that the economy can reach without entering a balance-of-payments crisis, through its impact on foreign trade (Cimoli & Porcile, 2013).

Figure 5. Wage share of output and real wage (1975 = 100)



Source: Author's elaboration based on Graña and Kennedy (2008).

In addition to the short-run changes, the multiple reforms of labour institutions and the systematic plan of repression and extermination deployed by the dictatorship converged in a curtailment of workers' bargaining power, which would leave scars for the subsequent period. A new ceiling on the real wage was established (Bona & Barrera, 2021), inaugurating a stage of structural deterioration (Kennedy, 2018b). The set of reforms to labour institutions included the prohibition of trade union activity or any political activity by workers' organisations, including the government takeover of the CGT (the General Confederation of Labour) and the suspension of the right to strike, and the modification of labour laws (collective wage bargaining, the Employment Contract Law, collective agreements, professional associations, union-run health funds, the law on trade union immunities, and the law allowing the dismissal

¹² Data calculated from the IPC-GBA (consumer price index for Greater Buenos Aires), INDEC.

of public employees) (Águila, 2023). By modifying the Basic Legal Regime of the Civil Service, the dictatorship also sought to dismantle opposition groups within the state bureaucracy (Mesyngier, 1998). Although these measures did not prevent trade union organisation from carrying decisive weight with the return of democracy, they undoubtedly affected workers' wage bargaining power. All these institutional changes, then, contributed to structurally reducing the real wage (w) (Amico, 2015), which according to the model would limit the level of per capita output.

The repression of the political cadres of workers' organisations through state terrorism also acted in this direction, resulting in the forced disappearance of some and the demobilisation or exile of others. The disappearance of a generation of activist and trade union cadres led to a scant capacity for activism (Grimson, 2020). During the period after 1983, the annihilation of "organic intellectuals of the popular camp" prevented the formation of a leadership class (Basualdo, 2001). The "fight against subversion" was particularly effective in putting an end to what was called "the age of youth", a period of intense political mobilisation by young students, whose alliance with industrial workers had been decisive for the fall of the previous dictatorship, headed by Onganía, especially in the so-called "Cordobazo" of May 1969 (Manzano, 2017, p. 21).

As a result of these measures, but also of the restructuring of the productive fabric and the slowdown in economic growth, the labour market underwent profound transformations. The social structure thus began a stage of fragmentation, driven by a deterioration of industrial employment, whose workforce was absorbed by sectors of lower productivity, such as the tertiary sector and construction (Castellani, 2021). Unemployment, underemployment, informality and self-employment increased, worsening working conditions (Carregal et al., 2015). Besides the migration of wage earners towards informality and self-employment, low-quality outsourcing increased (Gerchunoff & de León, 2018). Although the dictatorship sought to prevent unemployment from growing, employment fell and hidden unemployment rose (Rappoport, 2008). A stage of growing heterogenisation of the working class thus opened, which among other consequences entailed a fragmentation of its interests, weakening its capacity to organise.

Lastly, as a result of the changes at the productive and institutional levels, industrial entrepreneurs and their relationship with workers were also affected. Economic policy functioned as a disciplining device for both groups, whose association had found, during the previous stage of industrialisation, a confluence of interests in the expansion of the domestic market (Canitrot, 1980). Breaking the material basis for that "multi-class alliance" was a central objective of trade liberalisation, and it entailed a hardening of the political conditions for raising the real wage from then on. Taken together, the transformations eroded the economic power of trade unions and of firms oriented towards the domestic market, and gave greater weight to the banking sector, traditional exporters and large conglomerates in the determination of economic policies (Mesyngier, 1998). Within local business, the large groups gained weight, above all those linked to the military-industrial complex, increasing the

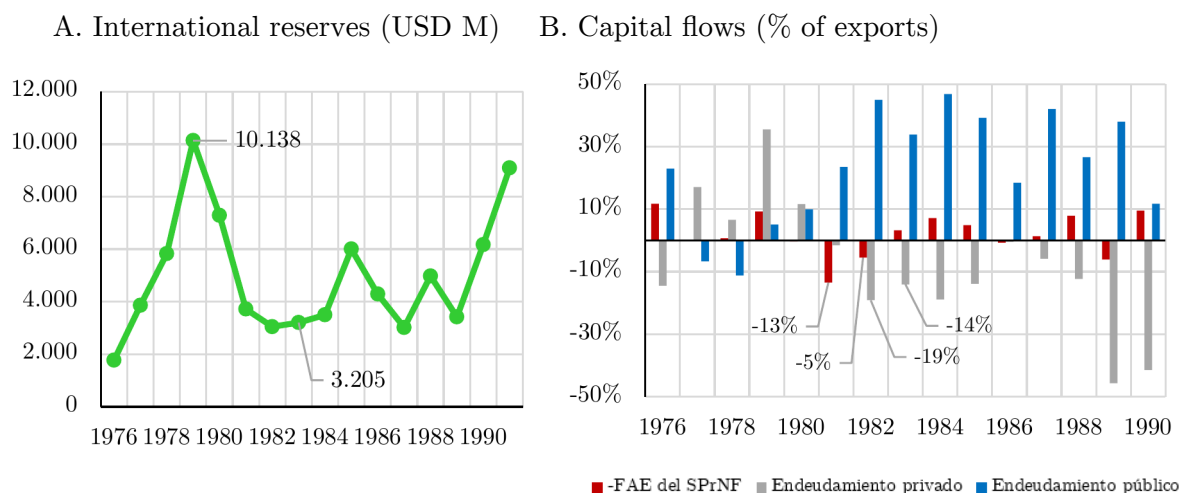
concentration of economic power (Castellani, 2004). The strengthening of this corporate sector —what came to be called the “*patria contratista*”, the state-contractor establishment— and its market power may have favoured an increase in the rate of profit, to the detriment of the real wage. Likewise, the capture of the state and the veto power of these new actors conditioned the orientation and level of public spending from then on, as detailed in the following section. In short, the set of actions and institutional changes concerning the position of the working class tended to erode its political power, which in turn affected the real wage. Although the wage followed a non-monotonic dynamic, in which the distribution of income corresponded only partially to the dictatorship’s class interests, a new structural level for w was consolidated, markedly lower than that of the previous period (Alvarez & Dvoskin, 2023).

The last pillar of the dictatorship’s policies and reforms, and probably the most relevant for long-run growth, was financial liberalisation. The cornerstone of the financial reform was the enactment of the Financial Institutions Law in 1977, which transformed the institutional framework of the financial sector in a way that proved difficult to reverse (Mesyngier, 1998). The law removed every significant restriction on capital flows and the regulations on domestic interest rates, and introduced other modifications such as the reduction of bank reserve requirements. A first effect was the increase in the economic weight of the banking sector (Canitrot, 1981). The financial sector as a whole became more important than in the past, and international financial capital emerged as a new actor in Argentine political economy, analogously to the way in which the so-called “international bourgeoisie” had gained weight in the capital goods sector during the developmentalist period (Álvarez, 2020, p. 70). External creditors thus consolidated themselves as a new actor in the distributive struggle (Gerchunoff and Hora, 2021).

Second, the resulting increase in the domestic interest rate, combined with a high degree of international liquidity and the exchange-rate cover implied by the pre-announced devaluations of the “*tablita*”, encouraged the inflow of short-term speculative capital. The inflow of portfolio funds sought to take advantage of the high returns on local assets, far above the pre-set devaluations (Zack & Pryluka, 2022), and was made possible by the free outflow of capital (financed in turn by the Central Bank, with the aim of sustaining the exchange-rate target). The reversal of capital flows was triggered by a change in expectations about the sustainability of the *tablita* after the increase in US interest rates in 1981, which would give way to a “lost decade” not only in Argentina but in the region (D’Amato & Katz, 2018). Both trade liberalisation and financial deregulation had eroded the state’s capacity to control the exchange rate, raising devaluation expectations and inflation (Alvarez & Dvoskin, 2023). Inflation in turn encouraged the formation of external assets by residents, by weakening the capacity of the domestic currency to fulfil the function of store of value (Corso, 2021). This increase in the proportion of income allocated to external asset formation would be reflected, in the model, in a rise in the parameter f , and it reached very significant levels: in 1981, 13% of exports. International reserves fell rapidly, as figure 6 shows, both because of the payment of liabilities contracted with non-residents (which in 1982 reached 20% of exports) and because of the

formation of external assets by residents. The fall in reserves could only be halted through public borrowing.

Figure 6. International reserves and capital flows

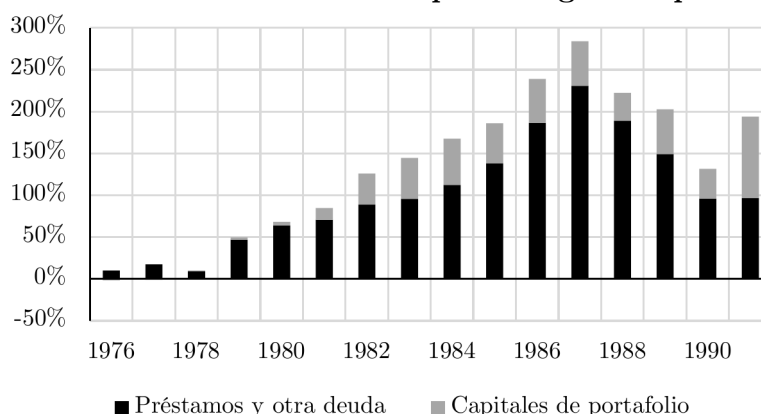


Source: Author's elaboration based on IMF.

Beyond the short run, financial liberalisation imposed a new constraint on interest-rate policy, in that local returns would now have a new floor, given by international interest rates and devaluation expectations. Given the inverse relationship—for a given exchange rate—between the interest rate and the wage, this new constraint may have contributed to the structural reduction of the real wage (Álvarez, 2020, p. 71).

More importantly still, the turn in the global financial cycle left behind an extremely vulnerable external sector, in that external liabilities had grown while, owing to capital outflows, reserve assets had diminished. The military government sought to counteract the financial fragility of the private sector by means of exchange-rate insurance schemes, *de facto* nationalising private liabilities, a debt that would condition the Argentine economy for decades (D'Amato & Katz, 2018). The public sector would have to face an external debt that could not be refinanced in global markets and that reached an unprecedented magnitude: the dictatorship accumulated liabilities equivalent to 150% of exports (figure 7). This increase in principal maturities and interest payments (reflected in the term $i \cdot D$ of the model) persistently conditioned the balance of payments (Gerchunoff & Rapetti, 2016) and entailed a need to achieve trade surpluses in order to cope with the new levels of indebtedness. The margins for growth during the following decade would thus be severely restricted, in that this required reducing imports and consequently investment (Gerchunoff & Llach, 2019).

Figure 7. Accumulated external debt^a as a percentage of exports



Source: Author's elaboration based on IMF.

^a Owing to data availability, the figure shows the debt accumulated since 1976, not the total outstanding stock.

3.2. Consequences for economic growth, 1983-1991

The consequences for the first democratic government were dramatic, severely limiting its capacity to stabilise the economy, boost activity and raise wages. Economic policy would be constrained by the need to preserve democracy (Gerchunoff & Llach, 2019, p. 453) in a context of permanent threats of a new coup d'état and of the marked restrictions inherited from the de facto government on meeting society's demands and aspirations (see Torre, 2021). The impossibility of resolving this dilemma would lead to the hyperinflation of 1989, which would not be fully subdued until after a second hyperinflationary episode in 1991.

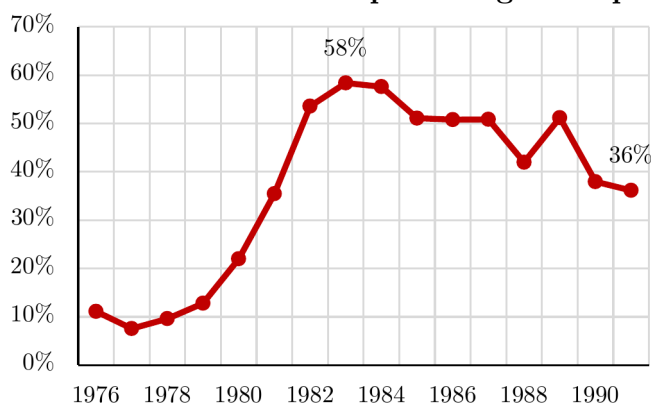
As regards wages, their dynamics were shaped by the stabilisation attempts that characterised the period, given the high inflation rates inherited. Wage aspirations, until then contained by dint of unprecedented repression, resurfaced with democracy, as did other sectoral demands (Libman et al., 2022). The government's first economy minister, Bernardo Grinspun, sought to satisfy these aspirations through favourable incomes policies and expansionary fiscal policies. But he came up against the new veto power of the new local economic groups, now under the name of the "group of 8" and later the "group of 20" (Castellani, 2021; Ortiz & Schorr, 2021). As mentioned in the previous section, with the multi-class alliance around the expansion of the domestic market broken, these actors no longer had incentives to support the expansion of wages or of public spending, unless the latter was channelled directly towards their corporate interests. In addition to these political constraints on the growth of wages and public spending, the need to achieve trade surpluses in order to meet debt maturities imposed additional "economic" limits. Indeed, one of the reasons behind the return of inflation after the initial success of the Austral Plan was the increase in import demand brought about by the recovery of economic activity.

Another factor that limited growth during the first democratic government was the state's inability to guarantee exchange-rate stability, which gained weight in the determination of

inflation as the indexation of contracts initiated by the dictatorship expanded and deepened (Álvarez, 2020, p. 118). The capacity to control the exchange rate was conditioned by two processes inherited from the dictatorship: very high inflation and external indebtedness. The persistence of inflation fed the demand for external assets as a store of value, in a context of lax monetary policies, which would only be tightened after José Luis Machinea took office at the Central Bank (Gerchunoff & Llach, 2019, pp. 465-467). This implied greater pressure on the international reserves of the Central Bank, which established exchange-rate regimes with multiple exchange rates, whose management proved difficult.

But the main constraints came from external indebtedness. In 1983, interest payments on the external debt reached 58% of exports (Libman et al., 2022 and figure 8), and principal maturities had to be faced throughout the period. In the absence of external financing — explained by the explosive trajectory of the debt and the high interest rates prevailing internationally—, the current account balance had to be high enough to meet the payments, which limited fiscal and/or wage expansion, in open tension with the political needs of the democratic restoration.

Figure 8. Interest on external debt as a percentage of exports



Source: Author's elaboration based on IMF.

As a result of the dictatorship's economic programme, then, the first democratic government was severely constrained in restoring wages, boosting public spending, promoting exports, controlling foreign-exchange outflows and carrying out other structural changes that could have boosted growth. In terms of the model, the impacts during both the democratic and the pre-democratic period would be reflected at three levels: the multiplier of autonomous demand (and, consequently, the long-run *level* of per capita income Y^i), the rate of change of public spending (g_G) and the constraints on the growth of exports (g_X).

The conjunctural reduction of, and the new structural ceiling on, the real wage (w) lowered the level of per capita income. Deindustrialisation operated in the same direction: by affecting the composition of output in favour of labour-intensive sectors, it reduced the capital requirements per unit of output (v). The consequent de-substitution of imports raised imports per unit of output (m), reducing the external space for the expansion of demand. Finally, exchange-rate

volatility raised devaluation expectations, so that the dollar returns on local-currency assets fell significantly, giving rise to a marked tendency towards the dollarisation of portfolios. The increase in the propensity to form external assets resulting from financial liberalisation, in a context of high inflation and exchange-rate volatility, thus also operated to reduce the long-run level of output.

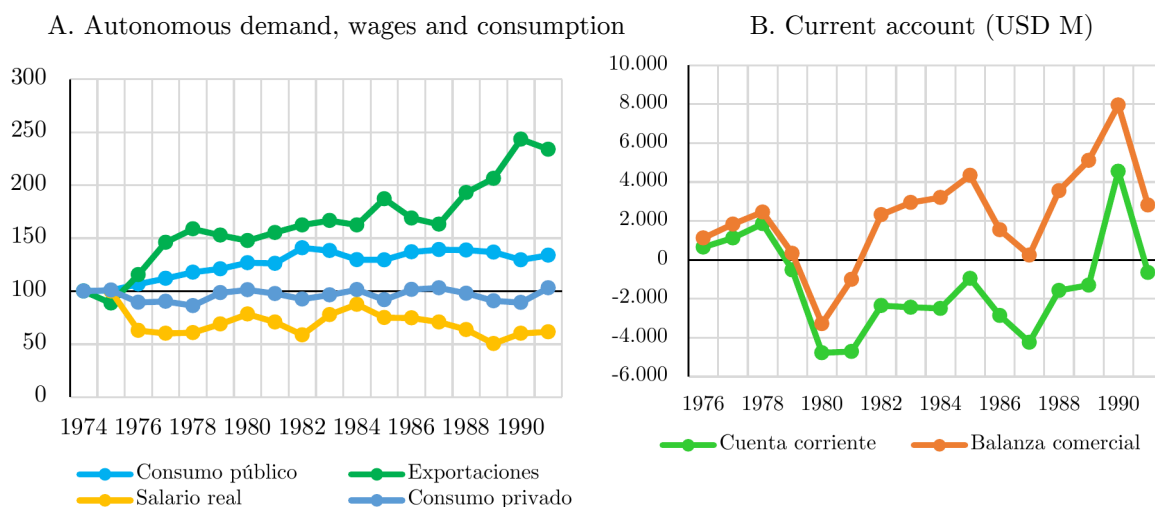
With regard to the impulse to public demand (g_G), it was discussed how the democratic authorities faced considerable political and external constraints. International creditors, strengthened after the dictatorship, and especially the International Monetary Fund, watched over fiscal austerity in order to guarantee the repayment of the debt. As argued, the logic of fiscal austerity operates through its impact on economic activity (according to the model, though not according to the IMF), which would induce an increase in the trade balance and in the availability of foreign exchange. Consequently, the successive stabilisation plans included a component of fiscal adjustment, which also responded to the need to increase the trade surplus in order to service the external debt. The economic authorities of the dictatorship, although they sought to limit public spending and dismantle part of the state apparatus, also responded to demands from sectors of the army, giving rise to a change in the composition of spending rather than to an absolute cut. During the democratic government, trade union and social pressures over the purchasing power of public employees' wages operated in the same direction, in that this expenditure represented a significant proportion of the total. Torre (2021, pp. 80-86) recounts how the IMF's fiscal adjustment proposals at the beginning of the government would have required a 30% fall in public-sector wages, a fall that the economic team and the President judged to be socially and politically unacceptable.

Finally, the external constraints were the most acute. The deterioration of the productive apparatus and the reprimarisation of the export basket affected the elasticity of exports, affecting export dynamics (g_X). But, above all, external indebtedness worsened the balance of payments via the increase in interest payments ($i^i D$). Interest rose because of the increase in the external debt contracted by the dictatorship (D), but also because of the increase in the interest rates paid on Argentine debt in the new international context (i^i), already at a high level that fed the perception that the country's liabilities were uncollectable. The debt thus entered an explosive dynamic in which the interest burden far exceeded the growth of exports ($i^i > g_X$). Refinancing during the Alfonsín government, while it could be sustained, only increased the debt burden to more than 250% of exports.

Figure 9A summarises the dynamics of the main sources of demand over the whole period. During the dictatorship, public spending expanded significantly, only to stagnate from 1982 onwards. Despite the initial fiscal expansion, consumption (and consequently GDP, in that in 1975 consumption represented 71% of Argentine output) contracted, anchored by the drastic fall in real wages and the consequent reduction in the multiplier. Export volumes expanded strongly in the first two years, but their effect on demand did not manage to offset the fall in the wage, while the external space gained was soon reversed, as a result of growing interest payments on the debt (panel B). Exports remained practically stagnant from 1978 onwards,

with real wages oscillating around their new level constrained by the balance of payments. Thus, by 1990 the level of output was 3% below that reached in 1974.

Figure 9. Main sources of demand at constant prices and current account



Source: Author's elaboration based on Kidyba and Suárez (2020), Graña and Kennedy (2008) and IMF.

3.3. The international and regional context

The policies described and their consequences cannot be analysed in isolation from the international context in which they took place. Especially bearing in mind that changes at the global level also affect the growth of countries, so that identifying an effect of the dictatorship's policies requires distinguishing their specific effects from global changes. As we shall see, many of the dynamics described in this chapter are shared by other countries.

There are multiple conceptualisations of the transformations that took place internationally from the mid-1970s onwards, which Boyer (2000) classifies into five hypotheses. According to the first, the exhaustion of the Fordist growth regime based on mass consumption and mass production ran into limits to the increase of productivity, giving rise to a regime based on product differentiation. According to the second, based on analyses of industrial revolutions in the long run, the diffusion of information and communication technologies gave rise to a new productive paradigm. A third hypothesis suggests that a new knowledge-based paradigm was established, which allowed a better articulation between scientific knowledge and production. The fourth hypothesis set out by Boyer emphasises the development of services and tertiarisation, while the fifth focuses on the shift towards greater openness in trade, finance and migration. Finally, there is the fifth and last hypothesis—to which the author subscribes—of a new financialised regime. Boyer's review serves to illustrate, beyond the characterisation of the new regime, the multiplicity of structural transformations that became evident at the global level from the mid-1970s onwards, both in productive terms and in the structure of the financial sector and its relationship with the rest of the economy.

As regards the changes in the international financial architecture, the institutions established in the post-war period by the Bretton Woods agreements were abandoned, giving way to a global system based on flexible exchange rates (Eichengreen, 2008). The post-war international monetary system had been based on the fixing of exchange rates against the US dollar, which would have a fixed parity with the price of gold. This scheme was sustained by interventions in the foreign-exchange market, at times supported by IMF assistance, and by regulations that limited international capital flows. The growing mobility of these flows eroded the system, making exchange-rate arrangements harder to sustain, until the United States abandoned the parity in 1971 and other countries with weight in international capital markets, such as Japan, followed in its footsteps.

In the context of the stagflation produced by the oil shocks of the 1970s, existing ideas and institutions seemed unable to respond to the problems of the time, and were replaced by new ones. At the institutional level, “embedded liberalism” (the commitment of states to full employment and welfare policies characteristic of the post-war period) began to be resisted by business sectors, and was gradually replaced by a set of institutions that in retrospect would be characterised as neoliberal (Blyth, 2002, pp. 3-8). In the realm of ideas, the growing influence of intellectuals and *think tanks* clustered around the Mont Pèlerin Society stands out (see Mirowski and Plehwe, 2015). Monetarism, rational expectations theory and *public choice* theory gained ground; besides dominating academic discussions within the field of economics, they served as the basis for the turn in economic policies¹³. The very functioning of the Bretton Woods multilateral organisations shifted, with the IMF promoting trade liberalisation, financial liberalisation and other “structural adjustment” policies through the conditionalities of its loans, and the World Bank leading the intellectual dispute over development policies (Polanyi-Levitt, 2013).

How did these global changes manifest themselves in Latin America? In the productive sphere, these transformations gave rise to a new international division of labour, in which countries of South-East Asia specialised increasingly in industrial activities, to the detriment of Latin American industry (Graña & Terranova, 2022). The tensions emerging from this global process were superimposed on criticisms of the state-led industrialisation model prevailing in the region, both from economic orthodoxy and from the political left (Bértola & Ocampo, 2013, p. 243). In the first case, criticising macroeconomic instability and state interventionism. In the second, warning about the model’s limitations in overcoming external dependence and transforming the social structure, in the context of struggles for emancipation from oppressions structured around gender, race and the global North-South divide (Fraser, 2020). But the former were more successful in reaching state power. In a systematic and repressive response to the initiatives of emancipation, the region witnessed a wave of right-wing military regimes which in some cases shared with the Argentine one ideological traits such as economic liberalism (Hobsbawm, 1994, pp. 440-441).

¹³ Mark Blyth (2002, pp. 164-247) gives an account of this process for the cases of the United States and Sweden.

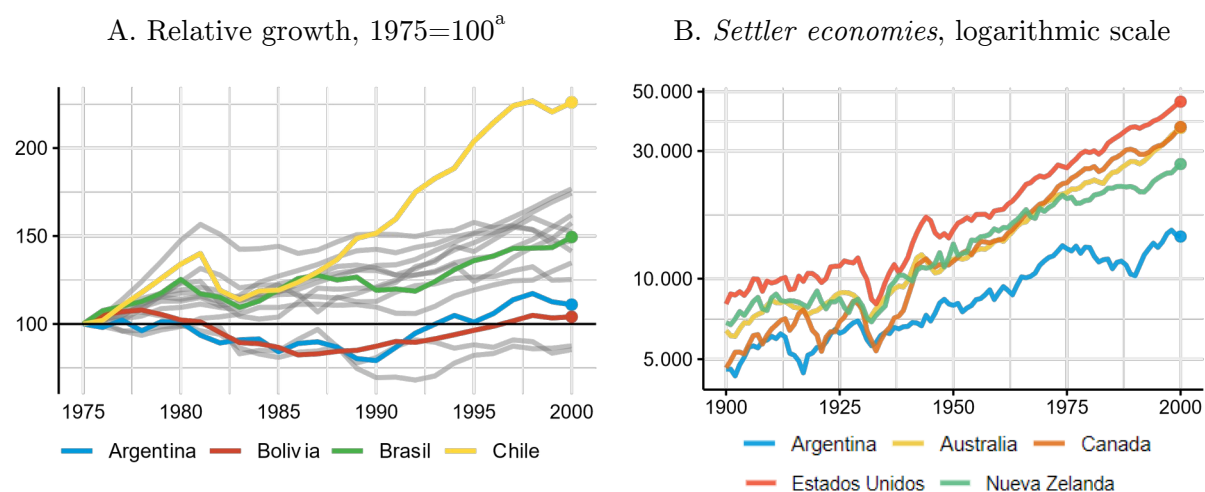
In the financial sphere, the oil price increases of 1973 and 1979 favoured a rise in global liquidity that translated into more attractive interest rates for debtors. The countries of Latin America, affected by the oil shocks and driven by the context of the growing dismantling of restrictions on capital flows, contracted external debts with international creditors, while experiencing rapid increases in current account deficits and, in some cases, strong exchange-rate appreciations (Frenkel & Rapetti, 2010). But those flows were reversed with the change in the monetary policy cycle of the US Federal Reserve, which under Paul Volcker raised interest rates sharply in 1979 and 1981. Exchange-rate tensions were added to the high costs of oil, feeding inflation and deepening economic stagnation (Zack & Pryluka, 2022). The countries of Latin America faced renegotiations of the debt inherited from the crisis, with international banks and the IMF, and had to make substantial payments (Frenkel & Rapetti, 2010). As a result of these processes, Latin America's economic stagnation lasted a decade, christened the "lost decade".

From a medium-term perspective, "peripheral financialisation" emerged in Latin America during the 1970s, hand in hand with the deregulation of national financial systems and the growing international mobility of capital (Abeles et al., 2018). Since then, the financial sector has grown faster than the rest of the economy at the global level, a discrepancy that was even greater in Latin America (Beck et al., 2014). The financial transformations are far from unrelated to the processes of deindustrialisation mentioned above: in Latin America, investment and growth rates were affected, and never recovered (CEPAL, 2018).

As shown in this section, in the period analysed in this thesis other countries —besides Argentina— also suffered the effects of global transformations, with heterogeneous results. Any assessment of a country's economic growth during this period must therefore take into account that part of economic growth is explained by these changes common to all countries, and not by factors of its own. Are these differences explained by the dictatorship's policies? The review in this chapter of the secondary sources that have analysed the period suggests that they are, but in order to produce a divergence the measures adopted would have to have been different from those of other countries, whether in qualitative terms (different kinds of measures) or quantitative terms (different intensity of the same measures). Müller (2011), for example, puts forward the hypothesis that the main factor behind the break in growth was the increase in external debt, but that the historical importance of the dictatorship's policies is qualified by the fact that external indebtedness was not a peculiarity of Argentina, but a manifestation of international financial conditions. A contribution to this debate may therefore come from the empirical evidence.

As panel A of figure 10 below shows, Argentina's performance during the period that began in 1976 was even weaker than that of other countries in the region, such as Brazil and above all Chile, and more in line with that of Bolivia. Panel B shows how the divergence from some core countries also becomes more pronounced from that year onwards. This explains why a certain consensus is found in empirical works regarding the existence of a break in 1975/1976 (Sánchez, 2018), even though this fact is sometimes overlooked when discussing the causes of the "Argentine failure".

Figure 10. Per capita GDP of selected countries



Source: Author's elaboration based on Maddison.

^a The grey lines show the growth of other South American countries, Mexico, the United States, Canada, New Zealand and Australia.

But this observation is not enough to reach a consensus on its causes. As far as the evidence is concerned, observing a divergence in the per capita output series is not sufficient: growth trends could differ by country, according to their different structural characteristics and independently of the dictatorship's policies. Second, given its structural characteristics, did twentieth-century Argentina find a good point of comparison in countries such as Bolivia, or in countries such as Brazil? If Bolivia were an adequate point of comparison, it could not be said that the Argentine military dictatorship caused the decline¹⁴, in that both countries followed a similar trend without Bolivia having had an "Argentine dictatorship". Lastly, even if there were an effect of the dictatorship on growth from 1976 onwards, can it explain Argentina's divergence during the twentieth century? In the following chapter we set out to answer these questions empirically.

¹⁴ Provided it is assumed that the impacts of the Argentine dictatorship on Bolivia's growth are negligible, as they can be expected to be. Within the framework of the model described, for this assumption not to hold, Bolivian exports to Argentina would have to represent a significant fraction of Bolivia's autonomous demand.

4. Estimating the impact on growth

This chapter quantifies the impact of the policies described in the previous chapter on economic growth during the last quarter of the century. The first section describes the synthetic control methodology, the second presents its application to the estimation of the impact of the Argentine military dictatorship’s policies on economic growth, and the third sets out the results obtained. The fourth section inquires into the channels through which the policies affected growth, and the last presents the robustness exercises.

4.1. Methodology: the synthetic control

In order to estimate the impact of the military dictatorship on long-run growth, a synthetic control research design is employed (Abadie & Gardeazabal, 2003; Abadie et al., 2010). This is a strategy for the quantification and empirical identification of causal effects in comparative case studies, in which the absence of a control group to act as a counterfactual for the group under study is addressed by constructing (“synthesising”) that counterfactual. This method is particularly suitable for studying the effects of an intervention at the aggregate level (such as a macroeconomic policy or, in our case, a dictatorship), because it allows estimation and inference *even* if the number of units affected by the intervention (the “treatment group”) is reduced to 1 unit.

In such cases, as shown in chapter 2, many studies use other untreated units as a control group, on the basis of bilateral comparisons. That is, comparative studies establish a counterfactual by comparing Argentina with another country. But it may happen that none of the untreated units constitutes a good reference, in that each is subject to idiosyncratic aspects of its own¹⁵. In our case, it may be that none of the countries is by itself a good counterfactual for Argentina, or at least not the best one. A synthetic control seeks to construct a more adequate counterfactual than these bilateral comparisons, on the basis of a weighted average of the untreated units, that is, those that act as “candidates” for a possible point of comparison. Thus, for our estimation a “synthetic Argentina” is constructed, which arises from an average of other comparable countries, and which seeks to be more representative of Argentina’s specificities than any one of these countries. A more detailed explanation of how synthetic control methods work can be found in Abadie (2021), whom we follow below in the formal description of the method¹⁶.

Suppose that information is available on $J+1$ units, where 1 of them is the unit affected by the intervention to be studied (in our example, Argentina) and the remaining J , unaffected, are called the “donor pool” (other a priori comparable countries). Data are available for T periods, T_0 being the periods prior to the intervention (periods up to and including 1975). Each unit is characterised by k predictor variables of the outcome variable Y , which in our case is the level

¹⁵ Indeed, there are no two countries in the world whose growth trajectory, for example, is identical.

¹⁶ The estimates were carried out using the ‘synth’ package for the R programming language, developed by the pioneering authors in the use of synthetic controls (Abadie et al., 2011).

of per capita GDP, such as characteristics of the productive structure, the level of development, the level of schooling, trade openness, etc. These variables constitute a vector of characteristics of the treated unit, X_1 , and a matrix for the remaining units $X_0 = [X_2, \dots, X_{J+1}]$, which will be relevant when defining the weights of the synthetic control and, in addition, make it possible to assess the similarity between the latter and the treated unit¹⁷. Finally, we define the causal effect of the intervention τ_{1t} as the difference between the level of Y that the treated unit has after the intervention Y_{1t}^I and the level it would have in the absence of the intervention Y_{1t}^N . The effect is indexed in time because the effects might not be instantaneous, and might accumulate or be reversed over time.

$$\tau_{1t} = Y_{1t}^I - Y_{1t}^N$$

The aim of the synthetic control is to estimate Y_{1t}^N , the level of per capita GDP that the country would have in the absence of the intervention, which is unobservable. The estimator of the effect is then the difference between the observed level and the synthetic control, $\hat{\tau}_{1t} = Y_{1t}^I - \hat{Y}_{1t}^N$. The estimator $\hat{Y}_{1t}^N$ is an average of the Y_{jt}^N , the per capita outputs of the untreated countries, whose weights $W = (w_2, \dots, w_{J+1})$ are usually chosen so as to make the synthetic control resemble the treated unit as closely as possible during T_0 . The similarity between the synthetic control and the treated unit is measured by the predictor variables X . Formally, the vector $W^{\hat{\cdot}}$ is chosen that minimises:

$$\|X_1 - X_0 W\| = \left(\sum_{h=1}^k v_h (X_{h1} - w_2 X_{h2} - \dots - w_{J+1} X_{hJ+1})^2 \right)^{1/2}$$

subject to the weights w being positive and summing to 1. That is, the Euclidean distance is minimised, but weighting each variable within X by a positive scalar $v_h \in V$. Each choice of the weights V is associated with a different synthetic control. Abadie et al. (2010) suggest choosing the V that minimise the mean squared prediction error (MSPE) of the variable of interest Y during the periods prior to the intervention T_0 . Finally, then, given the results of both minimisations the weights $W^{\hat{\cdot}}$ are obtained, and the estimator of the causal effect is determined by:

$$\hat{\tau}_{1t} = Y_{1t} - \sum_{j=2}^{J+1} w_j^{\hat{\cdot}} Y_{jt}$$

The aim of both minimisations is to find a set of weights that make it possible to replicate the trajectory that output would have followed in the absence of an intervention, an identification assumption whose plausibility can be put into context by comparing the evolution of the synthetic control and of the variable of interest before the intervention (that is, the periods in which Y_{1t}^N is observable). By construction, both the weights of the units (W) and those of the

¹⁷ It should be noted that X may include values or averages of the variables of interest Y in periods prior to the intervention, in so far as these are not affected by the intervention itself.

variables (V) are chosen to minimise that (observed) difference¹⁸, which is a reason not to interpret the possible similarity between the synthetic control and the actual pre-intervention evolution as conclusive evidence that the identification assumption holds¹⁹. In any case, even with the optimal weights, differences of considerable magnitude might prevail between the observed series and the control, which would weaken the plausibility of the estimates or, at least, of their precision. In addition, Abadie et al. (2010) argue that this *matching* of pre-intervention outcomes helps to control for unobservable factors and for the heterogeneity of the effects of those factors (and of the observable ones) on the variable of interest.

Abadie and his co-authors (2010) show that one of the cases in which the synthetic control estimator is asymptotically unbiased is when the variable of interest is determined by a factor model. Factor models are a class of models in which the determinants of the variable of interest can be expressed as:

$$Y_{it} = f_t' \gamma_i + u_{it}$$

Where Y_{it} is the variable of interest for unit i in period t , f_t is an unobservable vector of “factors” and u_{it} an also unobservable vector of “factor loadings”. The first of these vectors could be thought of as a set of time fixed effects, while the second is similar to a set of country fixed effects, but in this model the two terms interact. When assessing the level of output of different countries, f_t represents the set of shocks common to all countries in a given period, while γ_i captures the structural (time-invariant) characteristics of the country that determine how those common shocks affect output. This model thus captures the fact that the same shock can have heterogeneous effects across countries, according to their structural attributes.

It turns out that the model described in chapter 3 can be rewritten as a factor model of growth, which implies that the synthetic control estimator would be unbiased for estimating the effects of an idiosyncratic shock in a given country. To see this, recall that, according to the model of Dvoskin and Torchinsky Landau (2023) developed in the appendix, growth is determined by:

$$g_{it} = g_{Zit} = \sigma_i g_{Git} + (1 - \sigma_i) g_{Xit}$$

where σ_i is the share of public spending and exports in autonomous demand, which we assume to be invariant over time. Let us suppose that exports are determined by external demand, with an income elasticity of exports that we take as given (ε_i):

$$X_{it} = (Y_t^i)^{\varepsilon_i} \Rightarrow g_{Xit} = \varepsilon_i g_{Y^i_t}$$

¹⁸ Strictly speaking, the weights of the units (W) arise from minimising the distance of the predictor variables (X), not of the variable of interest (Y). But that measure of distance is mediated by the weights of the variables (V), so it depends indirectly on Y . In addition, it is usual, and recommended by the literature, to include among the X an average of Y for some period, which reinforces the fact that the differences between the synthetic control and the observed Y are small by construction.

¹⁹ A second reason, among others, is that similarity before the intervention is neither a necessary nor a sufficient condition for it to hold afterwards, something that has already been pointed out for tests of pre-existing trends in difference-in-differences studies (Roth, 2022).

On the other hand, let us suppose that public spending is subject to political and institutional determinants that may operate on a global scale —with differential impacts according to the country’s structural political conditions θ_i —:

$$g_{Git} = \theta_i g_{G^i_t}$$

where $g_{G,t}$ is the growth rate of public spending in country i , $g_{G^i_t}$ the rate at which the common determinants of public spending grow at the global level and θ_i a parameter reflecting how those global conditions affect domestic spending. This equation seeks to generalise the determination of public spending, admitting that it may have global determinants. In the particular case where $\theta_i = 0 \forall i$, each country would determine its level of public spending independently of the behaviour of the others, and we would observe zero correlation between changes in fiscal policy. Other values of θ capture the fact that global historical processes may differentially affect the capacity or willingness of domestic authorities to expand or contract public spending, which would induce a cross-country correlation in the stance of fiscal policy. For example, we might conjecture that part of the synchronisation in the expansion of fiscal policy during the “golden years” of post-war capitalism is explained by the threat posed by communism and the USSR. Financial globalisation would operate in the opposite direction.

In this way, growth is determined by:

$$g_i = \sigma_i \theta_i g_{G^i_t} + (1 - \sigma_i) \varepsilon_i g_{Y^i_t}$$

Re-expressing in vector terms:

$$g_i = (g_{G^i_t} \quad g_{Y^i_t}) \begin{pmatrix} \sigma_i \theta_i \\ (1 - \sigma_i) \varepsilon_i \end{pmatrix} = f'_t \gamma_i$$

This expression captures possible determinants of the components of autonomous demand that are common to the different countries. Considering that, in a given country, the local government has discretion to boost public spending, this expression could be modified to allow for possible idiosyncratic shocks to growth. An additional term could thus be included:

$$g_i = (g_{G^i_t} \quad g_{Y^i_t}) \begin{pmatrix} \sigma_i \theta_i \\ (1 - \sigma_i) \varepsilon_i \end{pmatrix} + u_i = f'_t \gamma_i + u_i$$

Where u_i captures the discretionary changes in g_G , while $g_{G^i_t} \sigma_i \theta_i$ captures possible changes driven by determinants common to the different countries such as, for example, a tightening of international financial conditions that restricts countries’ capacity to carry out expansionary fiscal policies or, more indirectly, a rise of discourses unfavourable to state expansion at the global level.

Note that the synthetic control estimator $\hat{Y}_i = \sum w_j Y_{jt}$ could be re-expressed as:

$$\hat{Y}_i = \sum w_j f'_t \gamma_j = f'_t \sum w_j \gamma_j$$

This implies that, if the estimator manages to approximate adequately the “*factor loadings*” of the treated unit, that is, if it holds that $w_j Y_j \approx Y_1$, the synthetic control will have a growth similar to that which the treated unit would have had in the absence of the idiosyncratic shock. Shocks common to all countries will have an impact on the synthetic control similar to the one they have on the treated unit, because the degree of exposure to those shocks, determined by Y_j , is similar. This illustrates the importance of $w_j Y_j \approx Y_1$, that is, of the structural characteristics of the synthetic control resembling those of the treated unit. If they are similar, the bias of the estimate is bounded by the scale of the idiosyncratic shocks u_i and the sample size (length of the pre-treatment period, T_0).

In short, the synthetic control seeks to estimate country weights that make it possible to reproduce the trajectory of Argentine per capita output before 1976. A *critical identification assumption* is then employed: that Argentina’s per capita output would have followed the trajectory of that weighted average of countries had there been no dictatorship. Under this assumption, the average per capita output (the “synthetic Argentina”) is used as a counterfactual to estimate the long-run effects of the dictatorship’s policies. This synthetic country shares with Argentina structural characteristics and a similar economic trajectory at least until 1976, but it did not have a dictatorship or, if it did, it was not a dictatorship whose policies had the same characteristics (whether in qualitative or quantitative terms). Therefore, any divergence from 1976 onwards is explained by the policies of the Argentine dictatorship itself. The robustness tests seek to lend plausibility to the identification assumption, but it cannot be proved, nor can the effect of the dictatorship be distinguished from the effect of another possible shock specific to Argentina that occurred in 1976. In this context, the discussion based on historical arguments and on the economic model of the determination of growth, presented conceptually in chapter 3 and developed in the Appendix, becomes relevant. The hypothesis that the break occurred in 1976 arises from that discussion; it is not merely extracted from the data by empirical methods²⁰. For that reason, the results must be placed within (i) a broader analysis of other possible breaks in the growth trend, and (ii) of other possible causes of the break that might be contemporaneous with the dictatorship. After presenting the results in this chapter, chapter 5 seeks to offer a view on these two discussions.

4.2. A synthetic control for twentieth-century Argentina

Turning now to our estimation, the donor pool must be made up of countries that could, a priori, be considered similar to the Argentina of the last quarter of the century. At the methodological level, it is recommended that this pool include units comparable to the case of interest (Abadie et al., 2015), avoiding, for example, the inclusion of countries with a much higher level of development, or with dissimilar historical trajectories. Including too many units

²⁰ For example, even if a break from 1976 onwards could be observed empirically, it could always be attributed to events that took place earlier, whose effects only became manifest from 1976.

in the donor pool runs the risk of generating overfitting and interpolation bias, giving rise to spurious results.

Of course, the extent to which a country deserves to belong to this pool is open to discussion. In this case, the choice of countries within the donor pool was made on the basis of the historical literature that has already carried out comparative studies between Argentina and other countries and argued for their similarity, surveyed in section 2. In particular, only countries with which this historical literature has compared Argentina were included in the donor pool. This pool is made up of the rest of South America, Mexico, and some English-speaking countries outside the region, which together with Argentina and Uruguay make up what is known in the literature as the “*settler economies*”: the United States, Canada, New Zealand and Australia. These are countries that share part of their history, cultural traits or institutional heritage, and that are peripheral countries specialised in the export of raw materials, or at least were so at the beginning of the century. Many of them had dictatorships or hyperinflations, as Argentina did. In the case of Australia, New Zealand, Canada and the United States, they are countries of “recent” colonisation, endowed with natural resources (some of them similar, such as fertile grasslands and a temperate climate) and which at the beginning of the century had relatively similar per capita outputs. All had a marked expansion of the primary export sector at the end of the nineteenth century that attracted waves of European migrants, similarities that were pointed out even by observers at the time (Ladeuix & Schiaffino, 2022). Table 3 sets out the complete list of donor countries, although it should be stressed that not all of them enter into the construction of the counterfactual, but only those that turn out to have a positive weight.

Table 3. Donor pool for the synthetic control.

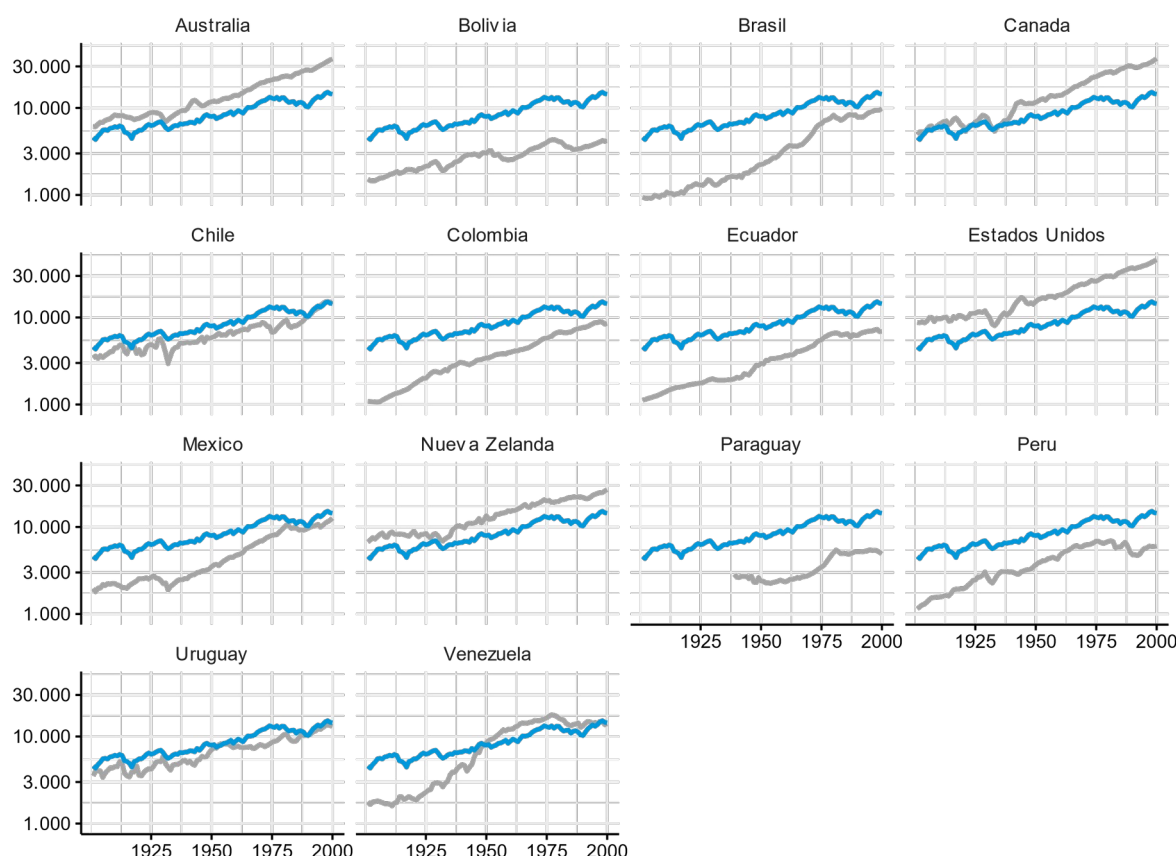
Donor pool			
	Bolivia	Brazil	Chile
Colombia	Ecuador	Mexico	Paraguay
Peru	Uruguay	Venezuela	Australia
USA	Canada	New Zealand	

Source: Author’s elaboration.

The choice of these countries as the donor pool requires restricting the optimisation period of the synthetic control to the years after 1939, given that sufficient data are available only from that year onwards. It is a reasonable cut-off year, in that official per capita output data for Argentina are available only from 1935. The estimates for earlier years have serious methodological shortcomings, which are discussed in section 4.5.7, where a robustness exercise is also carried out using alternative measurements for years after 1935. In any case, section 4.5.3 presents specifications in which the countries with missing data are excluded in order to extend the optimisation period back to the beginning of the century.

As figure 11 below shows, none of these countries is, by itself, a good counterfactual. Throughout the century, Argentina's level and growth rate differed considerably from those of the countries that have been used as a point of comparison. This does not mean that any comparative exercise between Argentina and one of these countries is in vain, but it does mean that noting the divergence between them is not enough to establish that, at some particular moment, the actions of a government caused a deterioration in growth. To establish a more adequate point of reference, the synthetic control method can be used.

Figure 11. Per capita GDP of Argentina and the countries in the donor pool



Source: Author's elaboration based on Maddison.

The synthetic control constructed consists of a weighted average of these countries, which seeks to establish a “synthetic Argentina” similar to the real Argentina in different dimensions, given by the predictor variables X . The counterfactual estimated for the periods after the onset of the military dictatorship consists of the average of the per capita incomes of the countries in the donor pool, weighted by the coefficients W^i of the synthetic control. In this way, an international shock that simultaneously affected both Argentina and the countries within the control would not introduce bias into the estimate. This is precisely the kind of shock that one seeks to isolate by establishing a point of comparison.

As for the predictor variables X , proxy variables for the structural characteristics of the economy were used. As shown in the model employed, long-run growth and the long-run level

of per capita GDP depend on the foreign trade elasticities and on other structural parameters such as the labour requirements per unit of output. The main specification included, in addition to the average per capita GDP recommended in the literature, the shares of imports and of industry in output, and the share of primary products in exports. That is, the synthetic control will have, in the five years immediately preceding the onset of the dictatorship in Argentina, a degree of industrialisation, a level of import substitution, a diversification of the export basket and a level of development similar to those of Argentina. The use of averages of the variables X is a recommendation of the methodological literature, to avoid biases from outliers in a particular year, and periods close to the year of the shock are used precisely so that the economies are as similar as possible at the time of the shock.

As will be seen below, these similarities are sufficient for no significant divergences to be recorded between Argentina and the control from 1939 to 1976. Until 1976, although the various international shocks may have affected Argentina differently from other countries, they did not move it significantly away from the control. The importance of these variables lies, then, in the fact that they affect the way in which the economy's growth responds to global changes (common to the different countries), so that two economies with the same characteristics along these dimensions should not diverge other than temporarily, except in the face of an idiosyncratic shock that persistently affected one of them.

Table 4. Predictor variables and weights

Average variable	v
Imports (% of GDP, 1970-1975)	55%
Industry (% of GDP, 1970-1975)	0.4%
Primary products (% of exports, 1970-1975)	1.2%
Per capita GDP (1965-1975)	43%

Source: Author's elaboration.

The methodology also requires the determination of weights (v) for these variables, set out in table 4. These weights are determined by an optimisation that minimises the differences between Argentina's observed trajectory and that of the control, as is common in empirical studies that use this method. In this case, that technique notably favours the weighting of per capita GDP and imports to the detriment of the other variables, but as will be seen, the values of these variables in comparison with the synthetic control are nevertheless similar. In any case, section 4.5.3 shows specifications in which these weights are set without resorting to computational methods, or in which the predictor variables are altered, showing that the results are not drastically affected.

Finally, it is necessary to dwell on the applicability of the usual synthetic control estimator in this context. As anticipated, the estimator consists of calculating the difference between the

observed series of Argentine per capita income, where the Argentine military dictatorship broke in, and that of the synthetic control, which is an average of countries in which there was no such dictatorship. The identification of a causal effect requires that the *shock*, the military dictatorship, was orthogonal (i.e. uncorrelated) to the trajectory that Argentine growth would have followed in the absence of the dictatorship. It is worth noting that the dictators themselves might perhaps have disputed this assumption, in that their project was founded on the claim that Argentina was in a debacle, which was precisely what justified their action. Their view assumes that the country was destined for a fall in growth, and that it is precisely for that reason that there was a dictatorship. The estimates below start from the opposite assumption: that Argentina was not destined for failure, and that the dictatorship's motivations were not associated with an expectation of deteriorating growth, but with other factors, already discussed in the previous chapter. Moreover, the dictators also regarded as justified the political and economic repression that took place at the same time in other countries of the region, so that this trajectory was not necessarily specific to Argentina. It is argued here that although the dictatorships in the region were motivated by a regional context common to all countries (including Argentina), the specific characteristics of the Argentine dictatorship do not respond univocally to an economy that was predestined for relative decline. The estimated effect is attributable to the idiosyncratic component of the Argentine dictatorship vis-à-vis the course of the countries within the control, regardless of whether these are countries under democracy or under other dictatorships.

4.3. Results

Table 5 shows how the synthetic control is composed under the main specification, described in the previous section. In this specification, the synthetic Argentina is a combination of only two countries: Brazil —with a weight of approximately two thirds— and the United States. Section 4.5.4 presents analyses of robustness to the removal of some of the countries in the sample, to ensure that the results do not depend on any particular donor.

Table 5. Country weights of the synthetic control

Country	w
Bolivia	0%
Brazil	63%
Chile	0%
Colombia	0%
Ecuador	0%
Mexico	0%
Paraguay	0%
Peru	0%
Uruguay	0%
Venezuela	0%
Australia	0%
New Zealand	0%
USA	37%
Canada	0%

Source: Author's elaboration.

Table 6 compares the synthetic Argentina with the real one, and for reference the simple mean of the donor pool and the countries with a positive weight are included. As can be seen, the synthetic Argentina has a partially industrialised productive structure (a 36% share of GDP), an export basket heavily concentrated in primary products (60% of total exports), a low import coefficient (9%) and a level of per capita output (on average over the decade prior to the intervention) similar to that of the observed Argentina. The synthetic control is thus closer to Argentina, in its structural characteristics, than any of the countries taken separately, or than the average of the whole sample.

Table 6. Comparison between Argentina and the synthetic counterfactual

Average variable	Brazil	USA	Sample average	Argentina	Synthetic
Imports (% of GDP, 1970-1975)	10	6.6	17	6	9
Industry (% of GDP, 1970-1975)	39	31	35	43	36
Primary products (% of X, 1970-1975)	77	30	68	79	60
Per capita GDP (1965-1975)	5,012	24,586	10,423	11,533	12,086

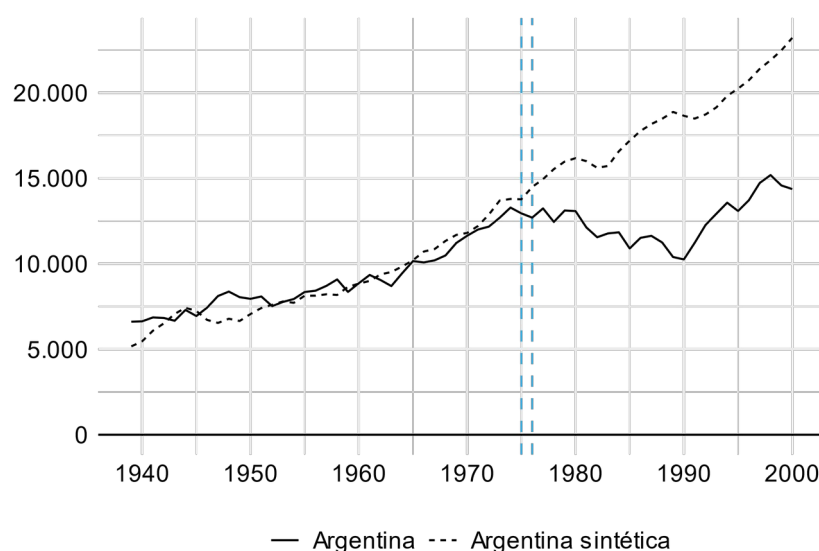
Source: Author's elaboration based on Maddison, COMTRADE, UNCTAD and World Bank.

It is interesting that Brazil obtained a high weight. It is another country in the region with a semi-industrialised productive structure, which had implications for the shaping of its social structure and, therefore, for the web of political alliances throughout its history. The country also had a populist regime (Vargas) that promoted material reforms for the urban masses, sought to accelerate the growth of national industry and rejected liberal principles (Anderson,

1988). The pressures for income redistribution would in both cases provoke a reaction from the armed forces, including coups d'état and the establishment of a new dictatorial regime. Brazil, like Argentina, would go through a regime of high inflation and finally a hyperinflation in 1989-1990. But in Argentina, unlike Brazil, the dictatorship would embody a project of a neoliberal character, as discussed in the previous chapter: the last Argentine dictatorship distinguished itself from the earlier ones by its aspiration to eliminate not only the political representation of populism (Peronism), but the whole set of economic, social and institutional structures that gave it material support. By contrast, the Brazilian dictatorship gave continuity to the developmentalist strategy it had adopted at the time of the coup, in 1964. This specificity of the Argentine dictatorship is one of the first idiosyncratic aspects to point out when reflecting on the reasons for the divergence between the country and the synthetic control, a discussion that is postponed until the following chapter.

This synthetic control, besides attaining similar predictor variables, adequately reproduces the evolution of Argentine output until 1976, as figure 12 shows. Before the onset of the military dictatorship, both economies perform very similarly. What stands out here, for example, is the similarity between the two even during Perón's second government, characterised by tensions in the external sector that would be repeated in later decades and identified by other works as a breaking point in growth. From 1976 onwards, by contrast, the synthetic Argentina continued its upward trajectory, while the country entered a marked deterioration that could not be reversed even with the expansions of the 1990s. During the "lost decade", in which the growth of Latin American countries slowed markedly, the synthetic Argentina records a temporary deterioration, while the real one deepened its divergence with a fall in per capita output.

Figure 12. Counterfactual for Argentina's growth, 1975-2000^a



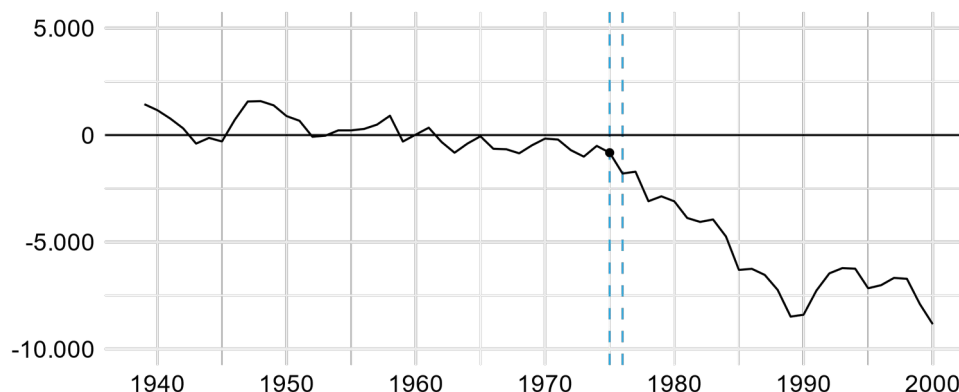
Source: Author's elaboration.

^a The dotted lines mark the years 1975 and 1976.

The effects are of a very considerable magnitude. Figure 13 and table 7 show the gaps between the two series, netting out the difference in 1975 in order to avoid incorporating that year's

recession into the estimates. While the fit is adequate before 1976, from that period onwards effects greater than 6% of per capita GDP are recorded. Even netting out the initial difference of 6 percentage points, the gap is large for all periods, reaching 27.1% at the end of the dictatorship (1983) and a maximum of 75.5% during the hyperinflation (1989-1990). That is, our estimates suggest that, in the absence of the structural transformations introduced by the dictatorship, Argentina's per capita income in 1990 could have been 75.5% higher than that observed, or 55.2% higher at the end of the century.

Figure 13 and table 7. Gaps between Argentine output and its counterfactual^a



Year	1976	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989
Effect (% of per capita GDP)	-7.8	-6.6	-18.5	-15.5	-17.3	-25.6	-28.8	-27.1	-33.7	-51.5	-47.9	-49.8	-58.0	-75.3

1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000
-75.5	-58.4	-46.3	-41.7	-39.6	-48.3	-44.8	-39.0	-37.9	-47.8	-55.2

Source: Author's elaboration.

^a The dotted lines mark the years 1975 and 1976.

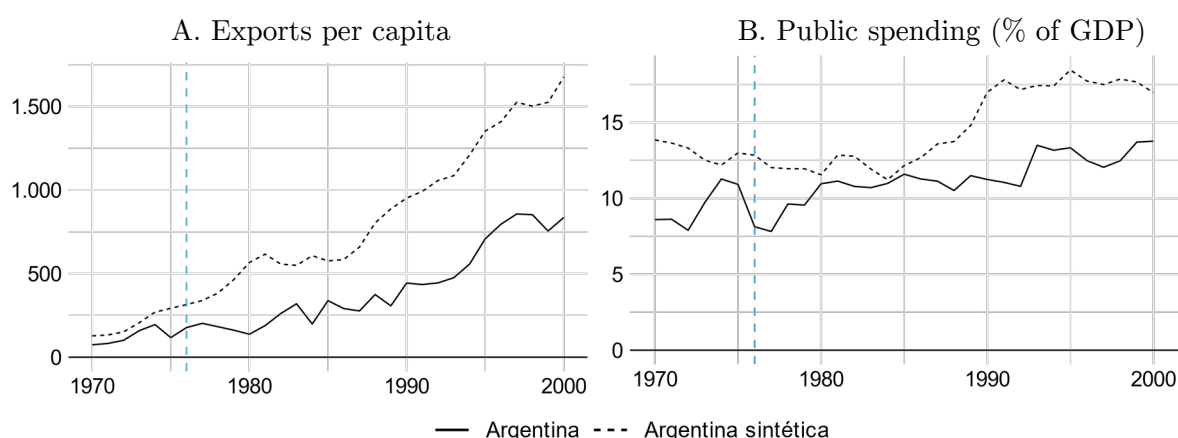
The magnitude of the effects implies that, in the absence of the dictatorship, by the end of the century Argentina would have reached a per capita GDP similar to that of New Zealand, well below that of Australia but far above that of countries such as Chile.

4.4. Channels

Given that the synthetic Argentina is represented by an average of other countries, all the counterfactual macroeconomic variables could be estimated from these weights. This exercise makes it possible to inquire into the channels through which the dictatorship's policies could have affected growth. However, this exercise does not constitute an identification of the causal effect of the dictatorship on these variables, which would require re-estimating the synthetic control taking each variable of interest as the target variable. The evidence emerging from the analysis in this section should therefore be interpreted as indicative—but not conclusive—of the routes through which the dictatorship's policies affected growth.

First, as figure 14 shows, the dictatorship's policies appear to have significantly affected the level of exports in the long run. In 1975 there was a fall in Argentine exports that cannot be attributed to the dictatorship, but the previous level was not recovered until the early 1980s, and the discrepancy with respect to the counterfactual grew during almost all the rest of the century. In the case of public spending, a gap in spending as a percentage of GDP appears immediately after the coup, closes around 1980 and reappears during the first democratic government. Given that in this period the per capita GDP of the counterfactual Argentina grew, the data imply that spending in real terms also grew, while in Argentina it remained in line with a falling GDP. After 1984, in the counterfactual spending grew even faster than GDP, reaching a new level of spending as a percentage of GDP. This evolution is consistent with the hypothesis that the structural current account deficit limited the external space for fiscal policy, and also with the possibility that the Alfonsín government, in the context of the stabilisation plans, limited public spending in the conviction that this would help to reduce inflation. After the hyperinflation, public spending recovered part of the gap, but did not reach the counterfactual level, as was also the case for per capita output.

Figure 14. Autonomous demand in the real and the synthetic Argentina^a

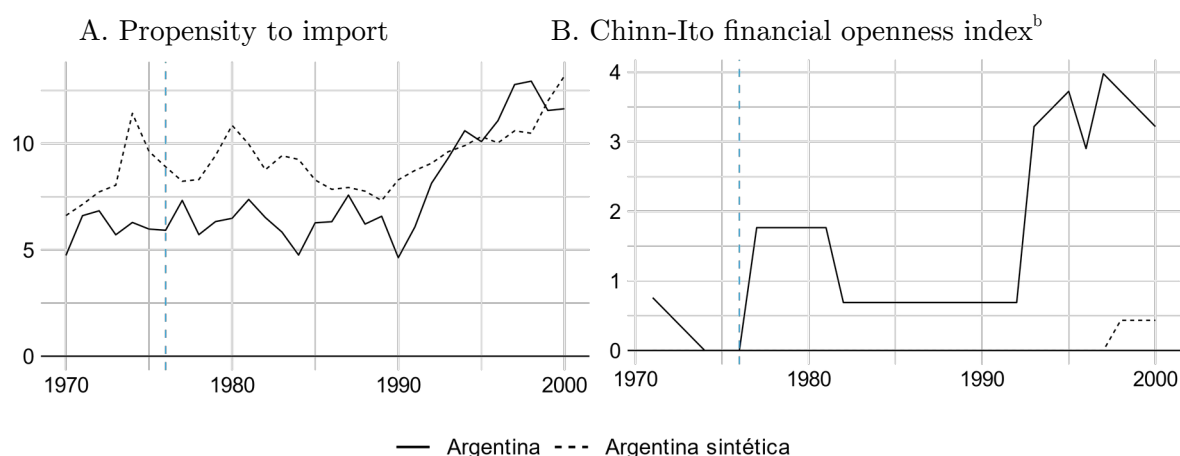


Source: Author's elaboration based on World Bank, Maddison and UNCTADStat.

^a The dotted line indicates the year 1976.

Although the de-substitution of imports appears in historical studies as one of the consequences of the policies, during the period under analysis there was no considerable divergence in import requirements per unit of output (figure 15.A). While the synthetic Argentina deepened import substitution during the 1980s, the increase in Argentina's propensity to import and its convergence with the counterfactual took place in the 1990s, with the trade liberalisation of the government of Carlos Menem.

Figure 15. Trade and financial openness in Argentina and its counterfactual



Source: Author's elaboration based on World Bank.

^a The dotted line indicates the year 1976.

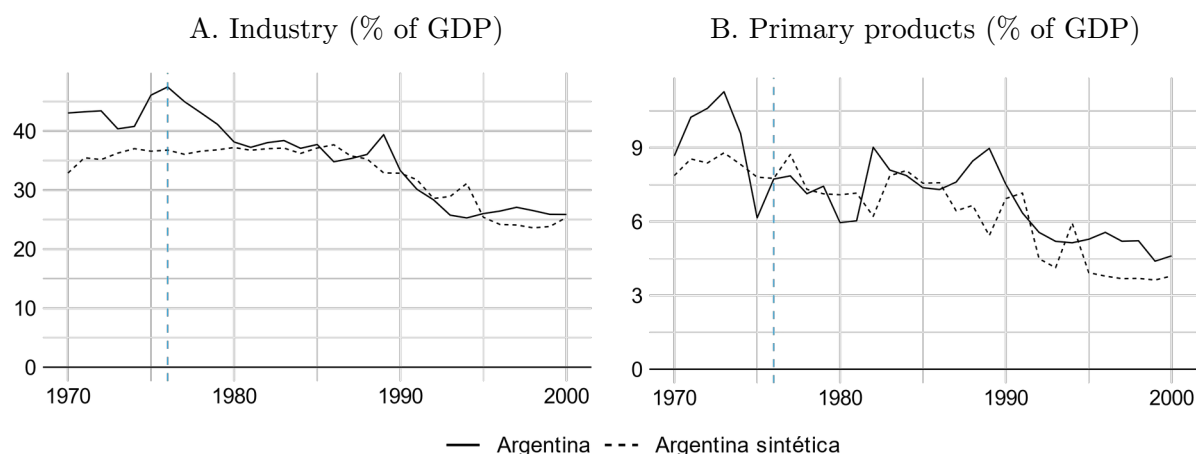
^b The difference with respect to 1976 is shown.

A significant change is observed, however, in financial openness, as captured by the Chinn and Ito indicator (2008). This is an index of *de jure* financial openness, that is, one constructed on the basis of a survey of measures regulating or liberalising the financial account, and not from *de facto* financial indicators such as the magnitude of capital flows or stocks²¹. Panel B of figure 15 shows the difference in this indicator with respect to 1976 for Argentina and its counterfactual. As is evident, while in the counterfactual *de jure* financial openness remained constant until 1998, Argentina went through a significant financial opening that was only partially reversed with the macroeconomic crisis of 1982, and that deepened significantly after 1991. This gap suggests that part of the Argentine divergence from the dictatorship onwards could be explained by the magnitude and speed of the deregulation of the financial account.

In terms of the productive structure, figure 16 shows a downward trend in the industrialisation coefficient in both the real and the synthetic Argentina. Argentina started from a higher initial level, and with the dictatorship it converged to the counterfactual level. The share of the primary sector in output, by contrast, was not especially affected. This suggests that, given the inability of the primary sector to increase production and absorb the labour expelled from a declining industry, there was a migration of production and employment from the industrial sector —of higher productivity— towards the services sector —of lower productivity—, in line with what Castellani (2021) points out.

²¹ In the period analysed the index ranges between -2 and 2.3.

Figure 16. Effects on the productive structure



Source: Author's elaboration based on UNCTADStat and COMTRADE.

^a The dotted line indicates the year 1976.

As shown, the evolution of exports was affected by the dictatorship's policies. One of the routes is through changes in the composition of the export basket, which underwent considerable transformations. From 1976 onwards, in the counterfactual Argentina the diversification of the basket deepened (figure 17), with the share of industrial goods rising and that of food falling²². In Argentina, by contrast, the slight tendency towards a falling share of food and a growing share of manufactures observed in the previous decades came to a halt. The gap in the share of manufactures, which seemed to be closing before 1975, tended to widen after the coup d'état.

Figure 17. Effects on the composition of the export basket



Source: Author's elaboration based on World Bank.

^a The dotted line indicates the year 1976.

²² The lack of data prevents the export basket from being compared much beyond the variables mentioned in this paragraph. It may be noted that the share of agricultural raw materials and of fuels remained at similar levels in Argentina and its counterfactual, at least until 1990.

In conclusion, evidence is found of long-run effects of the dictatorship’s policies on both components of autonomous demand, financial openness, the degree of industrialisation and the export profile. Although the methodology employed does not make it possible to estimate the relative contribution of each of these transformations, it does suggest that they could explain Argentina’s divergence in terms of growth after the military dictatorship. The rapid liberalisation of the capital account allowed a dizzying rise in external debt, which in the long run (during the Alfonsín government) restricted the capacity to expand fiscal policy. At the same time the evolution of exports was substantially affected, further reducing financing capacity and weakening demand. In parallel, deindustrialisation at a faster pace than in the counterfactual worsened the conditions of the domestic market and the country’s integration into international markets. Possibly, the trend decline in wages also had an impact on the gap with respect to the counterfactual, but the available data do not make it possible to test this hypothesis.

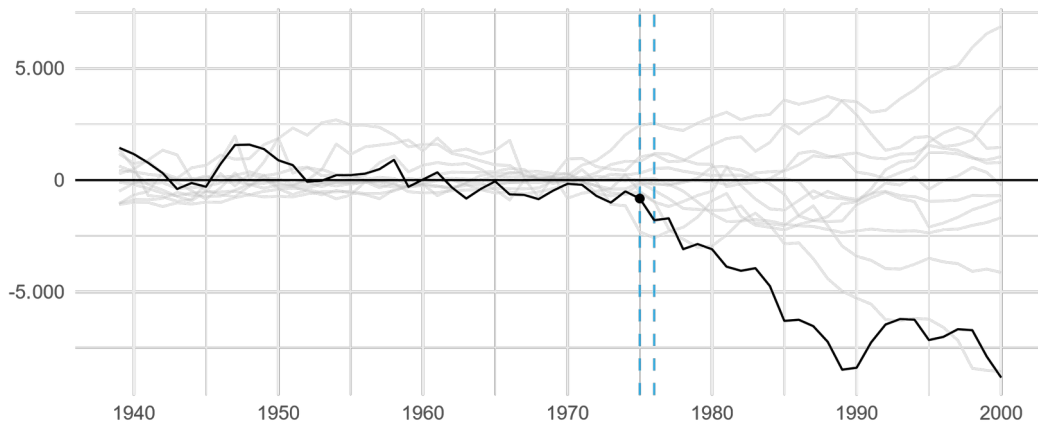
4.5. Robustness

4.5.1. Permutation

To assess the statistical significance of this estimate it is not possible to construct confidence intervals, since the sample consists of 1 observation within each “group” compared. However, following Abadie et al. (2010) it is possible to carry out inference without parametric hypothesis tests, by means of a permutation analysis of placebo effects. The test consists of repeating the estimation of the gap between the actual evolution of GDP and a control synthesised from the same donor pool for each country within that pool. It is a simulation in which the procedure is repeated as if each country had received the shock of the Argentine military dictatorship in 1976, when in truth none of the other countries had that impact (hence the name “placebo” effects). This makes it possible to generate an observed distribution of discrepancies between the observed country and the synthetic one, which are the product of chance. The plausibility of the effect in Argentina increases to the extent that its magnitude is greater than a larger number of placebo effects. As figure 18 shows, Argentina’s effect is the largest among all the permutations between 1978 and 1992²³, which makes it possible to reject the hypothesis of no effect with a p-value of 0.07. Given the sample size (deliberately small to avoid introducing bias into the estimate by comparing with non-comparable countries), this is a low value, and one that falls when more countries are incorporated into the sample (see section 4.5.4). For the remaining periods after 1976, the p-value is 0.14, while in the periods before 1976 the gap is around 0, in line with the placebo effects.

²³ In line with the recommendations of the literature, the countries for which it was not possible to achieve an adequate fit in the periods before 1976 (Venezuela and the United States) are excluded. In particular, these are countries for which the root mean squared error in the pre-intervention period is more than 5 times larger than that of Argentina, as recommended by Abadie and co-authors (2015).

Figure 18. Permutation test (in-space placebo effects)



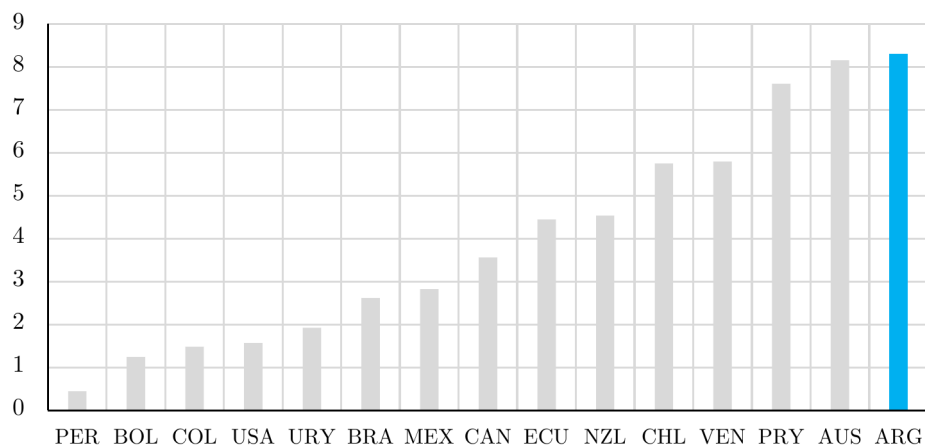
Source: Author's elaboration.

A second method for assessing robustness to in-space placebos is the ratio between the root mean squared error (RMSE) of the observed series with respect to its counterfactual, measured post- and pre-intervention. That is, if $R_j(t, t')$ is the RMSE between the series and its counterfactual between periods t and t' for country j , the ratio assessed is:

$$r_j = \frac{R_j(T_0 + 1, T)}{R_j(1, T_0)}$$

Where T_0 is the period of the intervention. The higher the ratio, the greater the divergence between the observed and the counterfactual trajectory after the intervention, making it more credible that the break occurred in the period under analysis. Figure 19 below shows that this metric reaches its maximum in the case of Argentina.

Figure 19. Ratio of the root mean squared error before and after 1975

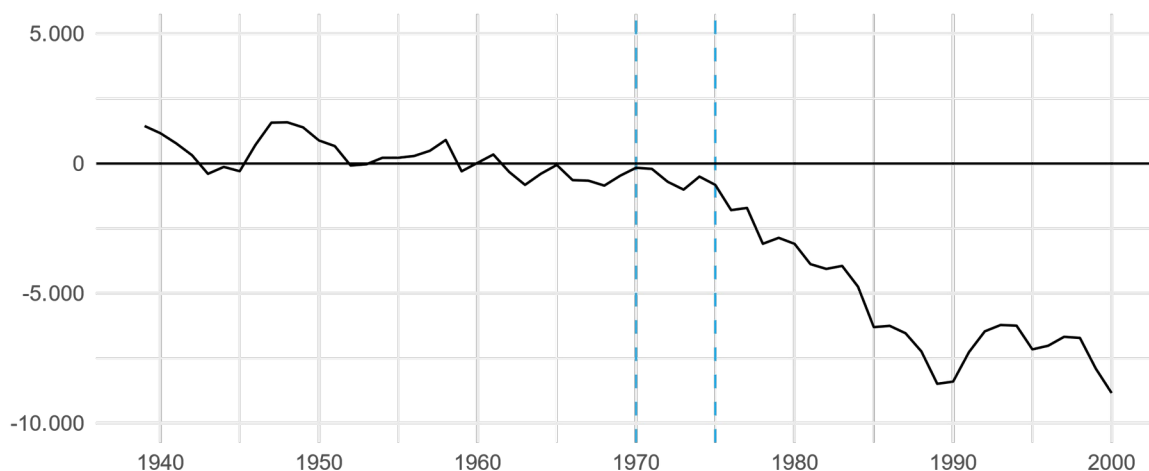


Source: Author's elaboration.

4.5.2. Backdating

A second robustness test consists of estimating “in-time” placebo effects. The estimation is repeated simulating that the shock had occurred in some earlier period, for example 1970. That is, all the weights are recalculated without using any information after 1970, and it is checked whether the effect begins to be observed only from the year of the shock (1976). Indeed, as figure 20 below shows, the significant divergence from the counterfactual begins to be observed only after 1975 even if the estimation is fitted up to 1970 (the first year with sufficient information available).

Figure 20. Backdating test (in-time placebo)^a

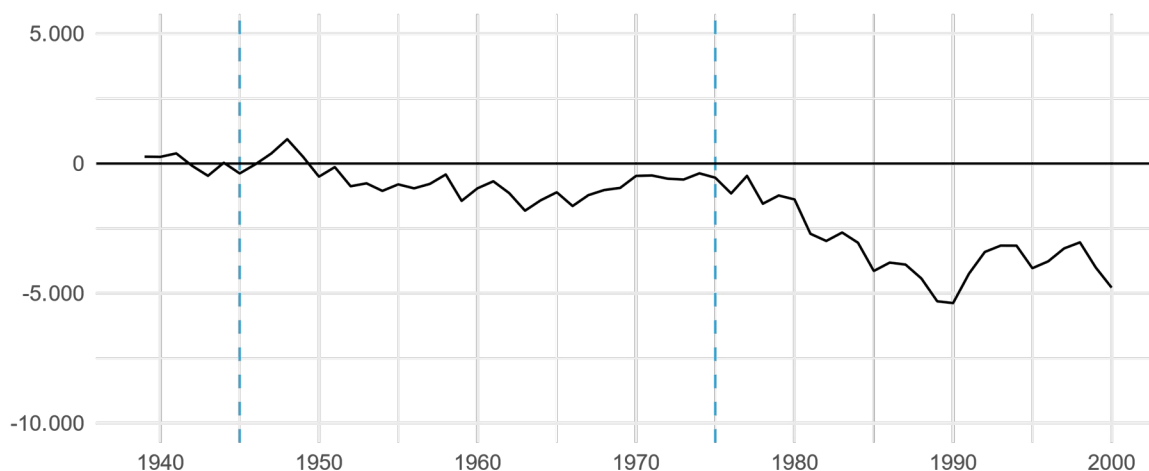


Source: Author’s elaboration.

^a The dotted lines indicate the years 1970 and 1975.

Data availability prevents going back in time beyond 1970, but the same exercise could be replicated with an alternative specification that uses variables available for a longer period. Figure 21 presents the per capita output gap if the synthetic Argentina is estimated using only per capita output and population for the years 1939 to 1945 as predictor variables. Once again, the divergence is observed only from 1976 onwards.

Figure 21. Backdating to 1945 under an alternative specification (in-time placebo)^a



Source: Author's elaboration.

^a The dotted lines indicate the years 1945 and 1975.

This result contrasts with the studies that find a divergence from 1945 onwards (such as Ertola Navajas, 2020). According to this evidence, between 1945 and 1975 the country grew at the pace of an average of the countries of the region with similar population and level of per capita output, and only departed from that path from 1976 onwards. The factors that explain the differences with respect to the results of Ertola Navajas are discussed in section 5.2.2.

4.5.3. Sensitivity to changes in specification

Changes in the specification of the predictor variables (X), their weights (v) or the optimisation period may affect the choice of country weights (w), and with it the estimates. Table 7 shows the results for different specifications of the synthetic control. Model 1 was chosen as the preferred one because it has a better fit in the pre-dictatorship period and, in addition, includes structural variables among the X , which makes it possible to mitigate the bias in the face of shocks with heterogeneous effects. The optimisation period was restricted to the years 1939 to 1975 because of missing data for some of the countries in the donor pool. The remaining models seek to quantify the sensitivity to these methodological decisions. The effects are calculated netting out the difference between the observed value and the counterfactual in 1975, in order to net out that year's recession.

Table 7. Different specifications for the synthetic control

Modelo	1	2	3	4	5	6
Periodo de optimización	1939-1975	1900-1975	1939-1975	1900-1975	1900-1975	1900-1975
Países (w)	BRA (0,63)	AUS (0,39)	COL (0,53)	AUS (0,37)	AUS (0,6)	AUS (0,56)
	USA (0,37)	MEX (0,39)	AUS (0,33)	CHL (0,32)	BRA (0,4)	BRA (0,43)
		URY (0,22)	NZL (0,14)	URY (0,22)		
			BRA (<0,01)	Otros (0,09)		
X (v)	Importaciones (0,5)	Poblacion (0,03)	Gasto público (0,01)	Importaciones (0)	Importaciones (0,1)	Importaciones (0,2)
	Industria (<0,01)	Apertura (<0,01)	Exportaciones (0)	Industria (<0,01)	Industria (0,1)	Industria (0,04)
	Prod. Primarios (0,01)	PIB 50-60 (0,96)	Importaciones (<0,01)	Prod. Primarios (<0,01)	Prod. Primarios (0,1)	Prod. Primarios (0,62)
	PIB 65-75 (0,4)	PIB 60-70 (0)	Industria (0)	PIB 65-75 (0,09)	PIB 65-75 (0,1)	PIB 00-10 (0,13)
			Otras (0,99) ^a	Otras (0,88) ^b	Otras (0,6) ^c	
v optimizado/manual	Optimizado	Optimizado	Optimizado	Optimizado	Manual	Optimizado
Efectos						
1980	-17%	-14%	-8%	-14%	-13%	-13%
1983	-27%	-22%	-20%	-18%	-22%	-22%
1989	-76%	-59%	-57%	-60%	-73%	-70%
1991	-59%	-47%	-43%	-49%	-54%	-52%
2000	-56%	-52%	-43%	-57%	-62%	-60%

^a Primary products (0.04), food (0.04), manufactures (<0.01), exports per capita (0), GDP 50-60 (0.24), GDP 60-70 (0.67).

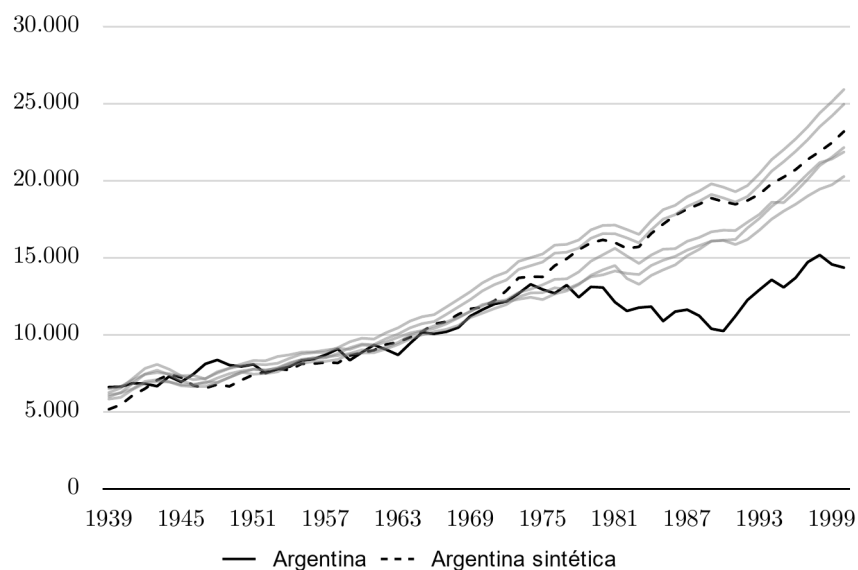
^b GDP 55-65 (0.23), GDP 45-55 (0.19), GDP 35-45 (0.25), GDP 25-35 (0.06), GDP 15-25 (0.01), GDP 00-15 (0.17).

^c GDP 55-65 (0.1), GDP 45-55 (0.1), GDP 35-45 (0.1), GDP 25-35 (0.1), GDP 15-25 (0.1), GDP 00-15 (0.1).

For the sensitivity to changes in the X , model 2 used predictor variables frequently employed in synthetic control studies (Abadie, 2021; Ertola Navajas, 2020), such as the level of population and trade openness approximated by the magnitude of foreign trade (exports and imports as a proportion of output). Model 3 used proxy variables for all the variables and parameters which, according to the model developed in the appendix, affect growth. To extend the estimation to the whole century, model 4 used the three structural variables employed in the preferred specification (model 1), plus the average per capita GDP of each decade of the century up to 1975. Model 5 used these same parameters, but without employing computational methods to choose the variable weights (v) optimally, instead assigning a fixed and uniform weight of 10% to each variable. Finally, in model 6 the estimation was carried out replacing the variable for average GDP in the decade before the dictatorship (1965-1975) with average GDP in the first decade of the century, in order to prevent a “forced” fit just before the intervention from giving rise to spurious effects immediately after the intervention.

As figure 22 shows, negative effects on per capita GDP are found in all specifications, for years during the dictatorship (for example, 1980 and 1983), the Alfonsín government (1989) and the end of the century (the year 2000). Considering all the models, the range of effects in 1980 runs from a minimum of 8% to a maximum of 18%; in 1989 the effects range from 57% to 76%; and at the end of the century from 43% to 62%. The preferred specification, however, shows a better fit in the period before the dictatorship, while also incorporating structural variables into the specification.

Figure 22. Counterfactual growth under different specifications^a



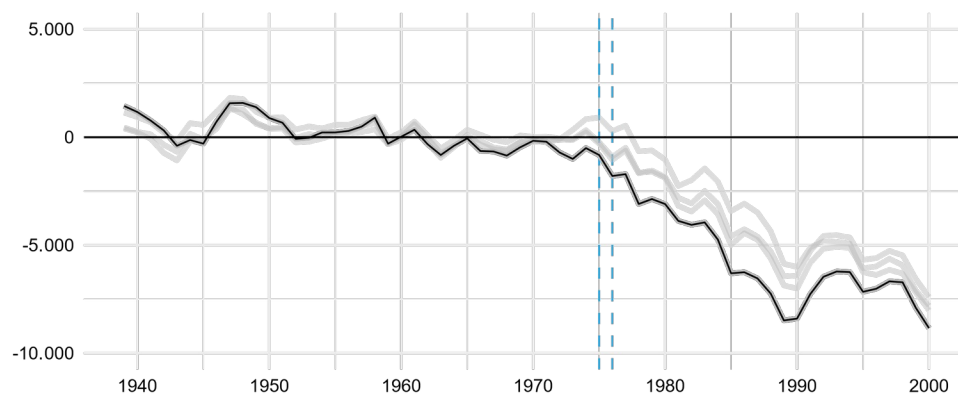
Source: Author's elaboration.

^a The black lines show the observed trajectory and the counterfactual one under the main specification, and the grey lines show estimates under alternative specifications

4.5.4. Robustness to changes in the donor pool

Another robustness test common in the literature concerns the sensitivity to changes in the donor pool (figure 23). First, the control is re-estimated iteratively, excluding one country at a time from the donor pool (a test known as *leave-one-out*). Most of the resulting estimates coincide with the main one (which is why they cannot be distinguished in the figure), while some reduce the magnitude of the effects, but without reversing the divergence.

Figure 23. Robustness to the exclusion of countries from the donor *pool* (leave-one-out)

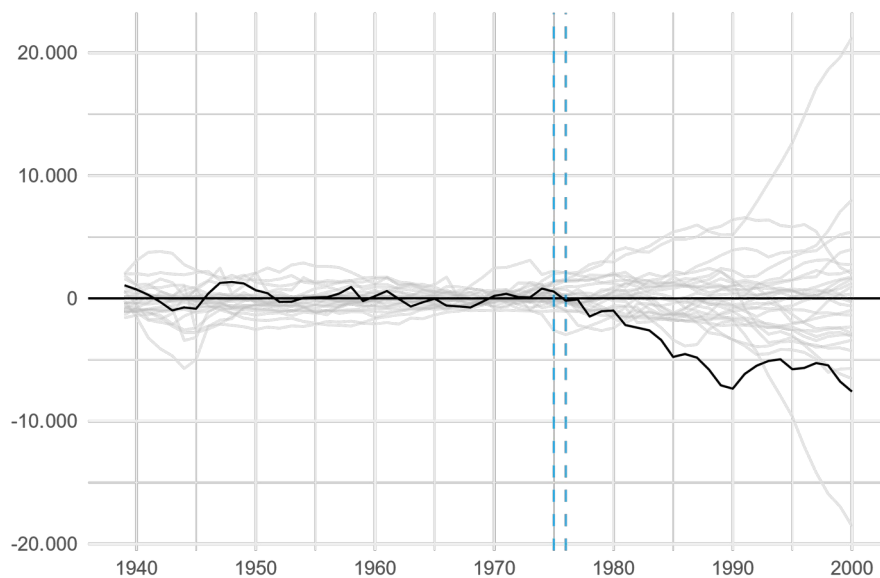


Source: Author's elaboration.

A second possibility for assessing robustness to changes in the donor pool would be to incorporate countries previously excluded. The inclusion was assessed of all the countries with available information that had a positive weight in any of the specifications of Ertola Navajas

(2020), namely: Australia, Belgium, Denmark, Finland, France, Greece, India, Italy, Japan, the Netherlands, Norway, Portugal, Spain, Sri Lanka, Sweden and the United Kingdom (Germany and Switzerland were excluded for lack of data). The estimation yielded a vector of weights whose highest values are those of India (52.4%) and the United States (42.5%), and effects similar to the base specification of model 1. The results are shown in figure 24, where each grey line is the placebo effect on one of the countries in the donor pool, so that the plausibility of the hypothesis of an effect caused by the dictatorship increases to the extent that the black line lies below the grey ones. Thanks to the inclusion of a larger number of countries, the p-value of the “in-space placebo” test fell to 3.8% for the periods before 1993 and to 7.7% for the later ones.

Figure 24. Robustness to the inclusion of new countries in the donor pool



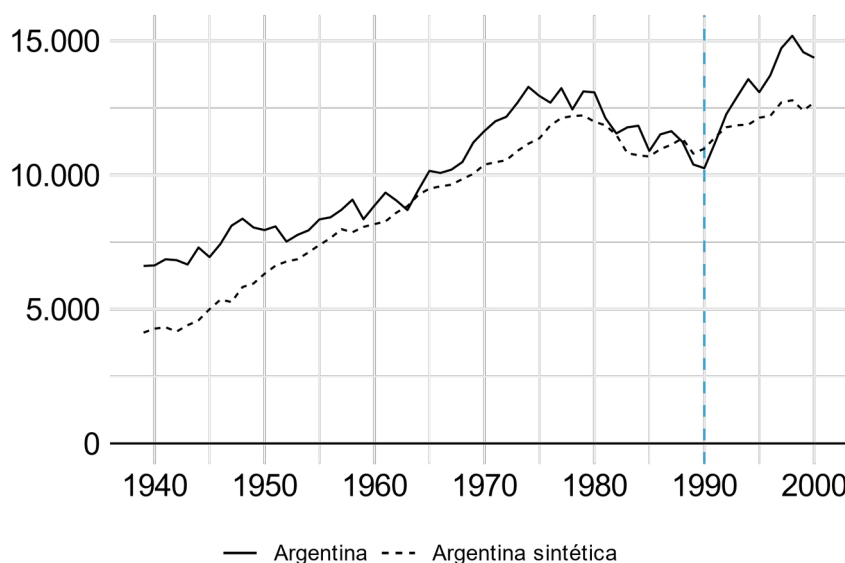
Source: Author's elaboration.

4.5.5. Effects after 1983

The fact that the effects of certain shocks on growth may be persistent implies that they can accumulate over time. If a second shock occurred after the shock under analysis, the synthetic control estimator would capture the joint effect of both causes, so that it would be incorrect to attribute the estimated effects to the first of them. Chapter 5 develops the implications of this point for the discussion on the “Argentine failure”, while this section presents a robustness test.

The methodology itself makes it possible to assess whether, in a given period, a shock occurred that moved Argentine output away from its counterfactual. In this robustness test a placebo effect is simulated in periods after 1983, in this case from 1990, with the same specification of the model. As figure 25 shows, the fit for the previous periods is not adequate, which prevents the presence of a causal effect from being identified. No significant divergence appears to have occurred from 1990 onwards.

Figure 25. Absence of effects after the dictatorship^a



Source: Author's elaboration.

^a The synthetic control was constructed with the same predictor variables as the base specification, but changing the fitting period of average per capita output to 1985-1990. The resulting synthetic control has a positive weight on all the countries in the donor pool, the highest being Venezuela (40%), Bolivia (17%), Brazil (6%) and Peru (5%).

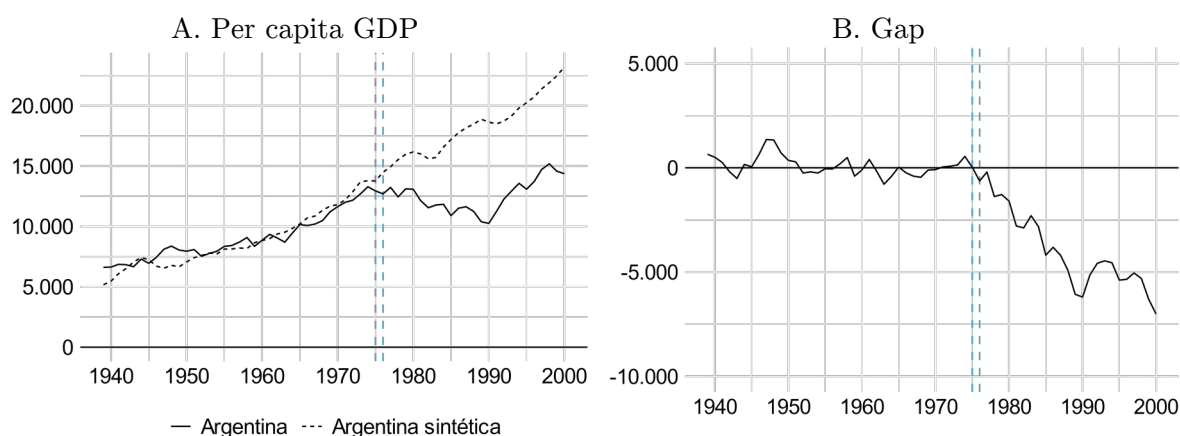
In addition, from 1990 onwards Argentina shows slightly *higher* growth than the synthetic control. This suggests that, if there were an effect, it would be positive, with the implication that, once it is netted out, the estimated effect would be even larger than under the base specification. The results are similar for other years after 1983.

4.5.6. Imperfections in the fit

Ferman and Pinto (2021) point out that an imperfect fit in the pre-intervention period could introduce a bias into the estimate if the predictor variables are not a good approximation of the determinants of the variable of interest. The inclusion of structural variables, as in the base specification, addresses this problem, in that these seek to capture how the economy would react to shocks, at the risk of achieving a worse fit in the pre-intervention period. An alternative is to look for a specification with a better fit based on variables that are not necessarily structural. For example, by resorting to the standard variables in the literature (Abadie, 2021; Ertola Navajas, 2020): per capita output, the level of population and trade openness²⁴. As figure 26 shows, in this specification a significant divergence from the counterfactual after the dictatorship is again observed, similar to the base specification.

²⁴ This specification differs from the one presented in figure 21 in three respects: the inclusion of trade openness as a predictor variable, the periods used for measuring the variables and the span used for optimising the parameters $\mathbf{v}$.

Figure 26. Per capita GDP, counterfactual and gap in a standard specification



Source: Author's elaboration.

4.5.7. Measurement error

Up to this point, the data of the Maddison Project have been used, in line with other works that analyse long-run growth from a comparative angle. Such is the case of the three studies that use the synthetic control method for the case of Argentina, so that the results of the main specification are comparable. However, these data are subject to controversy, especially in the case of Argentina. The use of different sources could significantly alter some results of the synthetic control, above all for periods before 1940, which were excluded from the main specification of this thesis but which are used by the other three studies that employ this methodology for the Argentine case. For example, a Maddison Project working paper shows that Argentina is an outlier in the change in per capita income estimates between the methodology used by the 2020 edition and an alternative estimate based on multiple PPP benchmarks (Bolt & van Zanden, 2020). In particular, its per capita income in 1913 is approximately three times higher under the current methodology than under the alternative methodology, a very significant difference that illustrates how sensitive Argentine GDP data for the beginning of the century can be to methodological decisions.

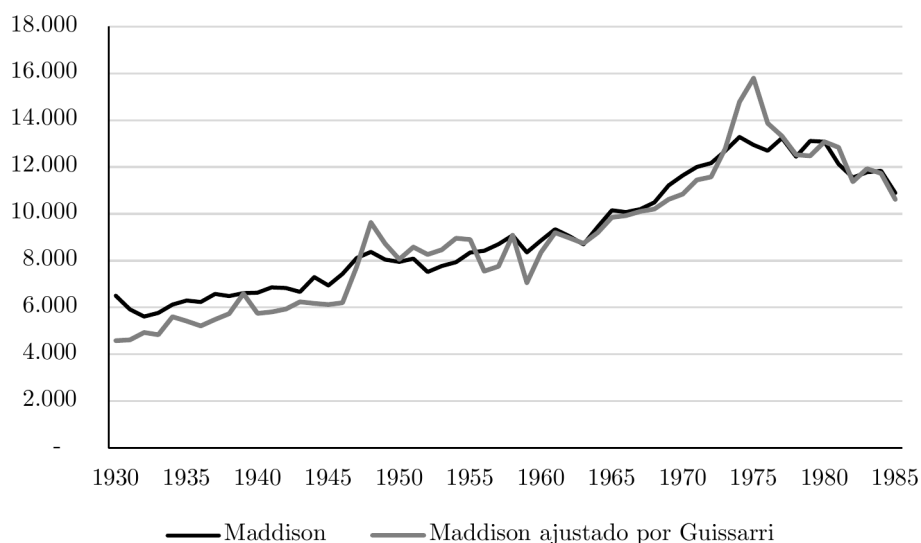
The differences between the various estimates of Argentine per capita output are manifold. As Guissarri (1989, pp. 70-75) shows, the official GDP sources (available since 1935) do not adequately capture the output of the informal sector of the economy, producing a bias in the estimation of growth that grew over time. By the second half of the century this informal sector represents between 30% and 40% of total output. Given that Maddison's estimates extrapolate PPP per capita GDP backwards using the growth rates of GDP at constant prices estimated by the countries, the use of official data (without the informal sector) leads to a considerable overestimation of per capita output, an underestimation that reaches its greatest magnitude during the 1930s (the first decade with available data on the informal sector). Maddison's estimates suggest that Argentina's per capita GDP in 1930 represented 61% of that of the United States, whereas extrapolating from growth rates that take the informal sector into account, that proportion reaches only 43%. Given that by 1980 the proportion stood at around

44%, the original estimates would suggest a process of divergence between Argentina and the United States that is not observed in the adjusted data. The drawbacks of splicing PPP measurements by backward extrapolation, the methodology used by the Maddison Project, are not limited to the case of Argentina and its informal sector, but tend to bias per capita GDP estimates upwards, especially for countries undergoing structural change (Prados De La Escosura, 2016).

On the other hand, for the period 1900-1950 the Maddison database uses estimates made by ECLAC, which have been criticised for using a weighting structure based on 1950 prices, and for lacking industrial production series between 1914 and 1935 (Sánchez, 2018). Some studies seek to remedy these limitations by using the alternative series of Cortés Conde and Harriague (1994) for periods before 1935. In a recent review of the sources of historical data in Argentina, Harriague and Rayes (2018) argue that these estimates would be preferable to those available in the Maddison database. However, after an exhaustive review of their methodology, Francis (2013, pp. 46-52) concludes that Cortés Conde's estimates are not very transparent or reliable, and that they could significantly overestimate industrial growth by imputing to the wine sector a growth rate of 64% per year, when the area under vines grew by only 5% per year. Other series surveyed do not manage to overcome these limitations, in that they are based on the same original sources. For example, those of Ferreres (2010) make some corrections but are based on the ECLAC estimates, and do not differ significantly from Maddison's (at least in the case of per capita GDP in PPP).

Unfortunately, then, there do not seem to be reliable sources for per capita output before 1935, while after 1935 the Maddison data might require a correction that takes into account the growth of the informal sector. Guissarri's estimates are available only up to 1985, so they allow a correction for the period 1935-1985, as is done below. Figure 27 shows Argentina's per capita output in the Maddison database and the one adjusted according to Guissarri's estimates.

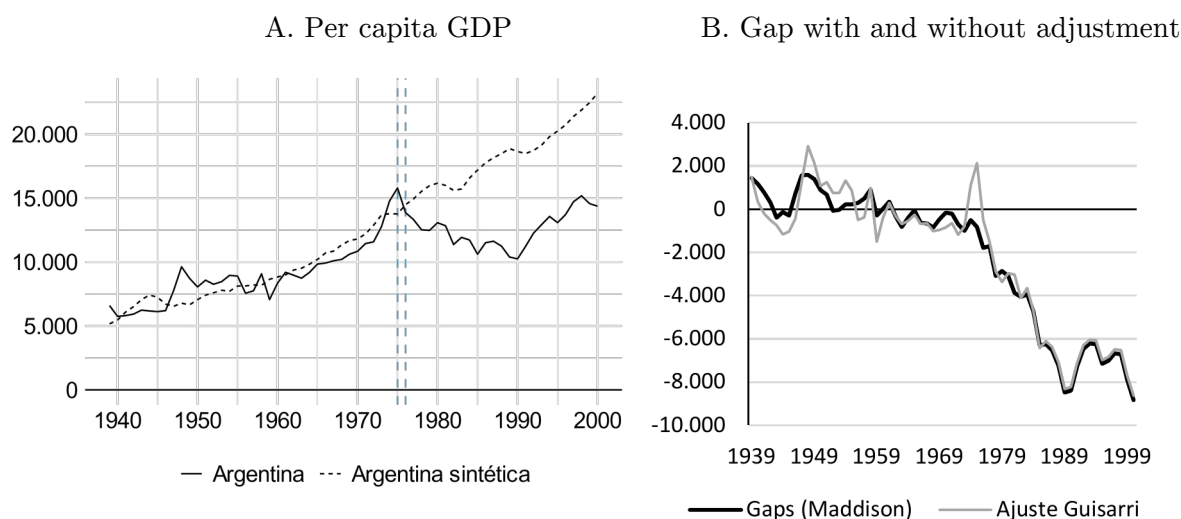
Figure 27. Comparison between the Maddison data and those adjusted for the informal sector



Source: Author's elaboration based on Maddison and Guissarri (1989).

Figure 28 shows that the conclusions of this work are not drastically affected when the synthetic control is re-estimated using these data. In fact, the break from 1976 onwards appears to be reinforced, although this is partly explained by a very marked increase in the informal sector in the early 1970s. The magnitude of the effects is even similar to that of the main specification.

Figure 28. Re-estimation with data adjusted according to Guissarri (1989)

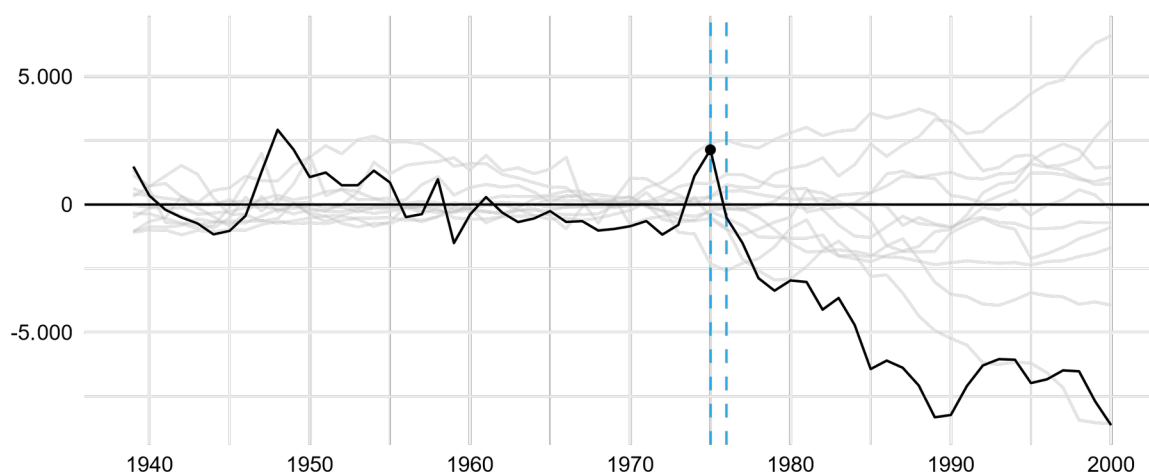


Source: Author's elaboration based on Maddison and Guissarri (1989).

The placebo test, shown in figure 29, also yields results favourable to the existence of a significant effect from 1978 onwards. As a preliminary conclusion of this chapter, the results of the empirical estimation provide evidence in favour of a causal effect that is statistically significant and economically relevant for explaining the divergence in Argentina's economic

growth. These results are robust to a wide range of sensitivity analyses and do not require the use of sources with serious measurement flaws.

Figure 29. In-space placebo test with data adjusted according to Guissarri (1989)



Source: Author's elaboration based on Maddison and Guissarri (1989).

5. Discussion

This chapter seeks to place the results obtained within the discussion on the “Argentine failure”. In particular, it seeks to contribute to two controversies. First, whether the economic deterioration of the last quarter of the twentieth century should be attributed to the dictatorship’s policies or, instead, to the entry into a regime of high inflation after the “Rodrigazo”. Second, to what extent the effect of the dictatorship’s policies can explain the divergence in Argentina’s growth during the twentieth century.

5.1. The “Rodrigazo” or the dictatorship’s policies?

The occurrence of practically simultaneous events can bias the synthetic control estimate. When assessing the impacts of the military dictatorship’s policies, one highly disruptive event that cannot be set aside is the “Rodrigazo”. This was the economic programme implemented in 1975 by the Economy Minister Celestino Rodrigo, characterised by a sharp increase in the exchange rate and in the prices regulated by the public sector, together with a ceiling on wage increases. The resistance of workers, including a 48-hour general strike, resulted in the removal of the ceiling on collective wage bargaining and the resignation of the economic team. As a result, the exchange rate, utility rates and wages all recorded nominal increases of more than 100%, with real wages falling by only 2% in 1975 despite the objectives of the programme (Kacef et al., 2022). This package of measures gave rise to an unprecedented increase in the rate of inflation, and was followed by 15 years of a high-inflation regime. As set out in chapter 3, some authors attribute to this episode the deterioration of Argentina’s economic performance during that period.

It is clear that macroeconomic instability during the period cannot have been neutral for growth, if only because of its indirect effects. But it would also be difficult to deny that inflation interacted with the economic policies of the period, without which the results would have been different. For example, the intensification of capital outflows cannot be understood without the high levels of inflation, but neither can it be understood without the financial liberalisation reforms. Although it is complex to attribute to each variable a contribution to the deterioration of growth, there are theoretical reasons, and evidence emerging from the estimation in chapter 4, for favouring the dictatorship’s policies as the main cause.

From a theoretical point of view, inflation in itself, in so far as it does not affect any relative price, should not have an impact on the level of output. For example, if the theoretical model employed for this work (and developed in the appendix) is correct, the level of output and growth depend on exports and public spending, which make up autonomous demand. Inflation appears as a result, not as a cause, of cycles of exchange-rate depreciation and nominal wage increases resulting from an incompatibility between the real wage demanded by workers and external equilibrium (Dvoskin & Torchinsky Landau, 2023). In so far as exogenous nominal wages keep pace with the increases in other prices, the real wage would not be affected in the

long run and therefore neither would the level of output. It is clear that in practice nominal wages need not follow inflation, but that relationship is mediated precisely by variables that are not determined by the inflationary regime, such as workers' bargaining power. If the power to negotiate wage increases in line with inflation fell with the dictatorship, the consequences this had for output must be attributed to the dictatorship, not to the inflationary regime.

Of course, it can always be argued that it is not inflation in itself that caused the decline in growth, but the way in which macroeconomic volatility was historically processed. Even within the classical-Keynesian framework used here, it could be argued that both inflation and the dictatorship's policies respond to the distributive conflict between workers and the dominant classes. Although in this argument inflation may be a necessary condition, the sufficient conditions for the deterioration must be sought in the subsequent process, and cannot be attributed solely to the Rodrigazo. The question to be asked in this case is: were there no other ways in which inflation or the conflict could have been processed, without resulting in a long-run deterioration of the economy?

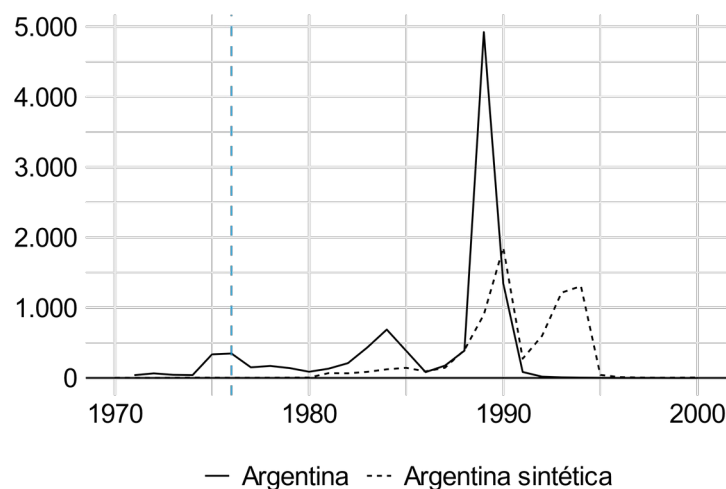
It cannot be maintained that the dictatorial government had its "hands tied" by the inflation and conflict inherited from the democratic government. The dictatorship enjoyed exceptional conditions for taking the path it considered the right one, the most "favourable" conditions imaginable for carrying out its project: a totalitarian regime. Moreover, despite the internal tensions, the economic team enjoyed wide room for manoeuvre during the first 5 years of the regime, leveraged on the decisive support of Videla (Águila, 2023, pp. 111-115). It is clear that the direction taken by the Argentine economy is, at least in part, the product of its action, and not a mere expression of underlying processes or inherited conditions. It is the way in which the dictatorship sought to "solve" the conflict, and not the conflict or inflation in themselves, that explains the subsequent economic stagnation.

From the empirical point of view, moreover, Argentina is not the only country in the region to have suffered a high-inflation regime or a hyperinflation, and yet its divergence from these countries is considerable. On this point the methodology adopted in chapter 4 can shed light on the magnitude of the impacts of inflation. In the base specification of the synthetic control, Brazil has a positive and high weight (63%). As it happens, it is a country that also entered a high-inflation regime that led it to a hyperinflation. It was also governed by a military dictatorship, but one that had not a neoliberal but a developmentalist programme (Bona & Páez, 2020; Crespo & Chibaudi, 2013).

As a consequence of the inclusion of Brazil in the synthetic Argentina, the point of comparison for establishing the counterfactual in chapter 4 is a country that also went through a high-inflation regime and a hyperinflation. As figure 30 shows, the synthetic Argentina reached inflation of approximately 1,000% in 1989 and 2,000% in 1990, and then again exceeded 1,000% inflation in 1993 and 1994. This means that the observed divergence is computed with respect to a point of comparison that suffered the effects of inflation on growth. Put differently, although inflation may have caused a deterioration in growth, the estimate in chapter 4 already

incorporates that effect by including Brazil in the counterfactual, isolating the effects of the Argentine dictatorship's policies.

Figure 30. Inflation in the synthetic Argentina



Source: Author's elaboration.

To conclude this section, it has been argued that the effects estimated in chapter 4 are attributable to the dictatorship and not to the Rodrigazo. The impacts of inflation on growth are mediated by the dictatorship's economic policy decisions. Other measures to address inflation or the distributive conflict would have led to other results. The estimated synthetic control provides evidence in favour of this hypothesis and gives indications of the kind of path Argentina could have followed: one in which, even with a hyperinflation, a moderate pace of growth could have been sustained.

5.2. The “Argentine failure”

What can be said about the divergence in Argentina's economic growth during the twentieth century —what in chapter 3 was called the “Argentine failure”— on the basis of the empirical results of this work? To answer this question some methodological considerations are necessary.

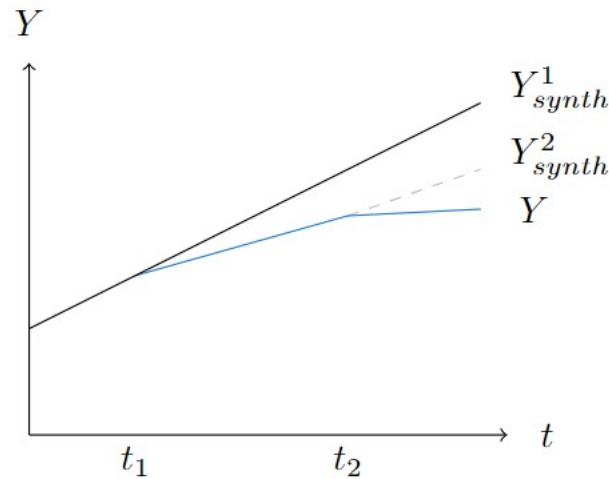
5.2.1. A framework for interpreting synthetic controls on the Argentine divergence

To speak of a “failure” it would be necessary to argue that the trajectory of the Argentine economy could have been different from the one observed, which requires formulating a counterfactual, a point of comparison. The synthetic control employed in this and other works on the “failure” is a statistical method for identifying an adequate counterfactual. But when interpreting the results of different studies it is necessary to bear in mind that the effect

captured by the synthetic control estimator is a cumulative effect²⁵. A country is exposed to idiosyncratic historical events (not shared with other countries) that alter its long-run growth trend. Since the synthetic control method is used to assess effects on a single unit²⁶, every idiosyncratic shock to which that unit is exposed accumulates on top of the previous ones as long as these have not dissipated. The synthetic control estimator will capture the effects of both events, something that must be taken into consideration when interpreting the results.

To analyse more formally which effect the synthetic control estimator captures, it is assumed that a country is subject to shocks that affect its long-run trend, producing structural breaks in periods t_1 and t_2 . Let D_1 and D_2 denote the variables that cause these breaks, which can be interpreted as economic policies or other changes. Let us also suppose that the effects of each of these changes can be identified by means of a synthetic control, which adequately estimates the counterfactual trajectory after each break, denoted Y_{synth}^1 and Y_{synth}^2 respectively, as diagram 2 shows.

Diagram 2. Effects and counterfactuals with cumulative impacts



Source: Author's elaboration.

If the break in t_2 is attributable to the intervention that occurred in t_1 , the total effect of this intervention on Y is, indeed, the difference between the synthetic control constructed in t_1 and the observed series:

$$\tau = Y - Y_{synth}^1$$

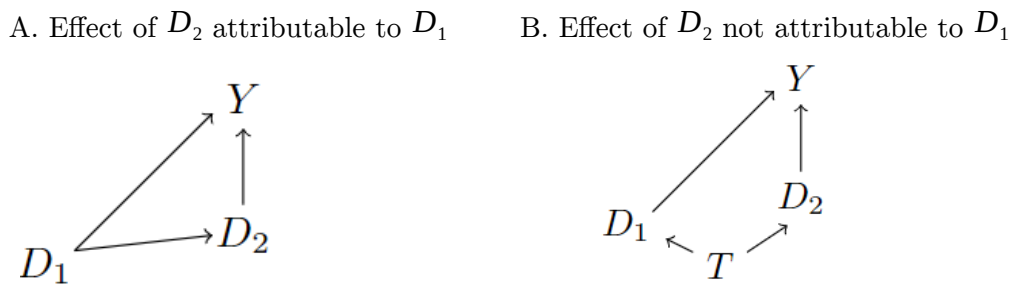
This total effect can be decomposed into a partial (direct) effect, given by $\tau_{direct} = Y_{synth}^2 - Y_{synth}^1$, and a cumulative (indirect) one given by $\tau_{indirect} = Y - Y_{synth}^2$. It holds that $\tau = \tau_{direct} + \tau_{indirect}$.

²⁵ As far as we have been able to ascertain, this distinction has not been pointed out in previous articles on the synthetic control method.

²⁶ When comparing groups of units with one another, and in so far as shocks tend, on average, to offset each other within each group, these changes do not pose major difficulties for the econometric estimation of differences between groups.

By contrast, if the second break (in t_2) is not endogenous to the previous one (in t_1), but responds to other causes (or is exogenous), a bias is introduced into the estimate. The cumulative effect, captured by the estimator $\tau = Y - Y_{synth}^1$, is not, from t_2 onwards, a causal effect of the intervention in t_1 , but includes a component that corresponds to the shock that occurred in t_2 . Put differently, the synthetic control estimator assumes that the correlation between events D_1 and D_2 reflects a causal effect from D_1 to D_2 . But that causal relationship is an assumption, and it could be argued that D_2 is not a consequence of D_1 . We can depict the two situations as shown by the DAGs in diagram 3. In the left-hand diagram, D_2 is attributable to D_1 , in which case the synthetic control estimator captures the total effect of D_1 on Y . D_2 acts as a mediator of the effect of D_1 on Y , giving rise to an indirect effect.

Diagram 3. DAGs in the presence of cumulative structural breaks



Source: Author's elaboration.

In the second case, by contrast, D_2 is a *confounder*, that is, a variable correlated with the cause and with the outcome variable, which biases the estimate of the effect. The correlation in this case does not reflect a causal effect, but is spurious, generated by the fact that D_1 and D_2 (“having been affected by event 1 or 2”) are non-stationary variables. The mere passage of time generates a statistical association between D_1 and D_2 (something that would not bias the estimate only if their effects were transitory). The causal effect of D_1 can still be isolated if one controls for D_2 , something that can be computed by subtracting from the cumulative effect τ the effect of D_2 ($\tau_{indirec}$). Therefore, in estimating the long-run effects of an intervention, it is indispensable to assess to what extent the synthetic control estimator may be capturing deviations from the trend produced *after* the intervention (t_1).

In the context of the estimation of effects on growth, it has been noted that part of the literature identifies breaks in the trend of Argentine output prior to 1976, for example in 1914. If one aspired to estimate the effect of a shock that occurred in 1914 beyond 1976 (that is, to estimate τ_{62} or later ones) by means of a synthetic control, it is necessary to:

1. assume that the structural break produced by the dictatorship is attributable to the 1914 shock, or
2. net out the effect of the 1976 shock from that year onwards.

Strictly speaking, this argument applies to any period after the moment of the shock, not only to 1976. That is, shocks that affected the long-run trend in the intervening years would also have to be attributed to the earlier intervention or netted out. To the extent that no structural break is recorded in a given year, the indirect effects in those periods are nil and do not need to be netted out. But if there were an event that produced a break, for example in 1930, it would have to be estimated and netted out in order to find the causal effect of the 1914 event. Given that changes in the long-run trend of output are not directly observable, in principle every year after the shock would have to be tested up to the period for which the effect is to be estimated. Given the complexity of this task, at the very least the years for which the historical literature has identified possible turning points should be tested (in the example, those after 1914 would be 1930, 1945 and 1976).

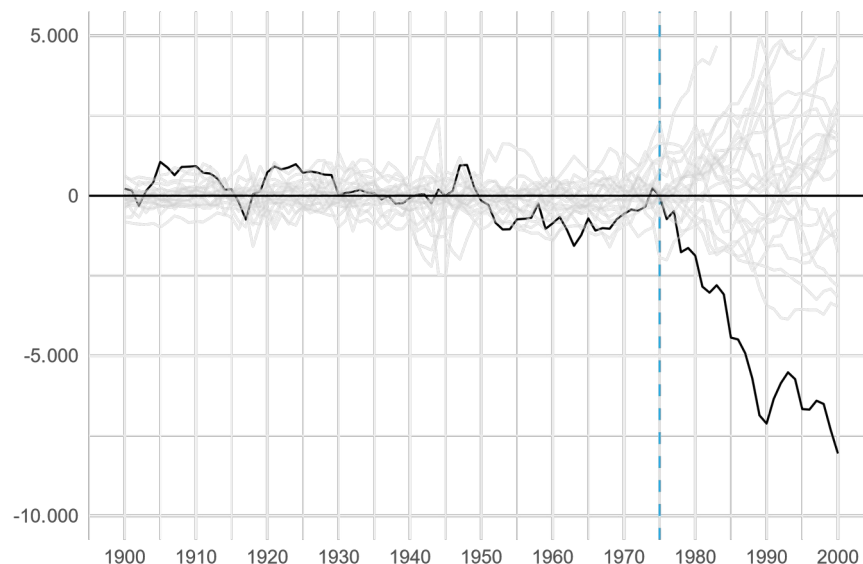
5.2.2. Implications for the interpretation of the available evidence

Far from being a methodological detail, the distinction between the direct effect and the cumulative effect suggests a framework for the discussion on the “Argentine failure”. In short, if one wishes to attribute the divergence over the whole century to one event, it is necessary to attribute all subsequent breaks to that event. This is an implicit assumption in the available works on the subject that use synthetic controls. Both Ertola Navajas (2020) and Marzetti and Spruk (2023) attribute the gap in the level of per capita output estimated at the end of the century to the rise of Peronism in 1945, thereby assuming that every subsequent break —in particular, the one which according to our estimates was produced by the military dictatorship — is caused by Peronism. If one were not prepared to maintain that the last military dictatorship is in its entirety a consequence of Peronism, the estimates would have to be revised. In the case of Spruk (2019), the set of “institutional breakdowns” is assessed, considering the Sáenz Peña Law (1912), the military coup of 1930, Peronism (1946), the military coup of 1955, the Frondizi government (1958), the dictatorship that styled itself the “Argentine Revolution” (1966), the third Peronist government (1973), the last military dictatorship (1976) and the democratic transition (1983). In this case, the divergence is attributed to the set of breaks, which would make it possible not to take a position on the causal relations between events, but prevents the relative contribution of each to the total divergence from being assessed. In the conclusions of that work, however, it is stated that the path of institutional deterioration began as a result of the first of the events. This assumes, for example, that the 1966 dictatorship is a consequence of the Sáenz Peña Law.

If one did not accept that the last dictatorship was a mere consequence of the earlier shocks, how significant is the bias introduced by the accumulation of effects? Taking the work of Ertola Navajas (2020), the effect of the dictatorship was estimated using the same specification employed there to estimate the effect of Peronism from 1945. Figure 31 shows the per capita output gap with respect to the counterfactual using the same donor *pool* and the same predictors to estimate the effect from 1976. The optimisation period of v was extended to 1975, and, as in the original model, the per capita output of the year before the intervention (in this

case, 1975) was included among the predictor variables. The p-values associated with the in-space placebo test are 8.3% for the first two years and 4.1% thereafter.

Figure 31. Gap with respect to the counterfactual following the specification of Ertola Navajas (2020) but applied to 1976



Source: Author's elaboration based on Maddison and Ertola Navajas (2020).

Two conclusions can be drawn from these results. First, following the same methodology as the original article it is possible to identify an effect from 1976 onwards. Unless it is maintained that the last military dictatorship is a consequence of the rise of Peronism in 1945, this effect would have to be netted out of the one originally estimated. The *gap* estimated by that work reaches approximately USD 23,000 at the end of the century, while this estimation yields a gap of USD 8,053, so that at least 35% of the estimated gap must be attributed to the dictatorship rather than to Peronism.

Second, and more importantly, this estimation yields a counterfactual for Argentina with respect to which no significant divergence is observed before 1976, as illustrated by the fact that the gap oscillates around 0 during the first three quarters of the century. This implies that, using the same methodology as the original work, no evidence is found in favour of an effect of Peronism on growth. Instead, until 1976 Argentina followed a trajectory similar to that of its counterfactual, until the dictatorship moved it away from that growth path. This conclusion is consistent with the findings of this thesis, and in particular with the in-time placebo test of section 4.5.2, where it was shown that even without using information beyond 1945, the Argentine divergence is observed only from 1976 onwards. Taking this second conclusion into consideration, the dictatorship explains not 35% of the divergence (as suggested in the previous paragraph), but 100%. Taking as valid the Maddison Project data for the period (questioned in section 4.5.7) and the methodology of Ertola Navajas (2020), no evidence is found in favour of an effect on the growth trend before 1976.

5.2.3. The “Argentine failure” and the estimates of this work

How is the possibility of breaks after 1983 addressed in this work? In our estimation, too, it is necessary to take one of the two routes: to argue that the divergence after the dictatorship is a consequence of its policies, or to re-estimate at the later dates and, should an effect be found, net it out. For the years of the dictatorship (1976 to 1983) the first route is taken, arguing that during this period it is the dictatorship itself, and not some other contemporaneous idiosyncratic shock, that is the cause of the break in the trend. To defend this position, the policies and reforms during the dictatorship that affected growth were reconstructed in the light of a theoretical model (chapter 3). Section 5.1 discussed in particular the hypothesis that the “Rodrigazo” might have caused the effects attributed to the dictatorship. Section 4.4 also set out indications of the channels through which the dictatorship’s policies affected growth. Despite these arguments, it should be stressed that the empirical evidence cannot settle this discussion, in that it can always be argued that the breaks observed in a given period are due to actions that occurred in earlier periods. For example, although the in-time placebo test shows that no divergence occurs until 1976, whether that break is attributable to the dictatorship or to earlier events is open to discussion.

Extending the estimation beyond the return of democracy requires additional discussion. It could be argued that the presidency of Raúl Alfonsín constitutes an exogenous event that altered the course of history, maintaining the trajectory of output against a counterfactual in which it would have fallen (or risen) more than it did. In this work we rely, on the one hand, on earlier scholarly work, which does not identify breaks after 1975/1976, at least until the end of the century, but mainly continuities in the historical stage opened by the dictatorship. On the other hand, chapter 3 gave arguments for the view that at least part of the course of the Argentine economy during the first democratic government is attributable to the persistent constraints imposed by the dictatorship. The analytical reconstruction seeks to reinforce this assumption explicitly. In this way, should there be effects on the long-run trend caused by the political action of the first democratic government, these are attributed to the military dictatorship. This hypothesis was tested empirically in section 4.5.5, where the synthetic control was re-estimated for periods after 1983. No evidence was found in favour of a break in the trend for these periods, leading to the conclusion that the estimate in this work is robust to the criticism levelled at other synthetic controls in the previous section (5.2.2).

6. Conclusions

The discussion on the determinants of Argentina's economic growth throughout the twentieth century cannot be settled definitively on the basis of econometric estimates alone. Nor is it possible to pinpoint *one* characteristic that qualitatively distinguishes Argentina from other countries, or one policy that explains its inability to sustain growth rates similar to those of the core countries. Rather, the divergence appears attributable to a succession of policies, reforms and effects of an international order, with diverse impacts according to the specific structural factors of each country; that is, a historical process.

In such cases, where the aim is to assess the macroeconomic impacts of multiple overlapping policies which coincide in time with exogenous shocks of great magnitude, attributing breaks in economic growth to particular historical events requires theoretical assumptions about how the economy works. Chapter 2 reviewed the different explanations offered by economic historiography for the course of the Argentine economy, classifying them according to their assumptions about the determinants of the distribution of income and of the level of output. What emerged from this review was the absence of an explanation based on a classical-Keynesian framework, in which the distribution of income is not fundamentally determined by market mechanisms and the level of activity in the long run responds to the principle of effective demand. This led to a detailed account, in chapter 3, of how the last military dictatorship produced structural changes that affected growth, not only during the seven years that the *de facto* government lasted, but also during those that followed. It was then proposed, in line with other historical works, that the divergence is due mainly to the economic deterioration of the last quarter of the century, but starting from an interpretation that focuses on the negative effects of the transformations on the long-run evolution of demand.

Chapter 4 provided empirical evidence for this hypothesis. Establishing a causal identification in this case requires establishing a point of reference for the comparison, which is neither more nor less than formulating a counterfactual for Argentina: the Argentina that would have been had events unfolded differently. This question about the distance between what was and what could have been may be of little interest to some historians, but it is indispensable if one seeks to assess the economic impacts of a programme such as that of the military dictatorship. The synthetic control method employed in this work allows this discussion to be conducted transparently, bringing out the conceptual hypotheses on which the analysis rests. It also makes it possible to formulate a clear point of comparison, which does not stray far from a methodology common in studies of comparative economic development: contrasting the growth trajectory of two countries considered similar. The synthetic control refines the counterfactual so that the countries to be compared are as similar as possible.

Different combinations of the growth paths of countries with which Argentina shares structural characteristics make it possible to reproduce the observed evolution of Argentine per capita output until 1975. For example, the country grew at a pace similar to a weighted average of Brazil and the United States, or of Australia, Uruguay and others. This implies that, until the

end of the century, the divergence between Argentine growth and that of other countries is due not to idiosyncratic policies or institutions, but to its structural characteristics. But from 1976 onwards there is a break in growth which, as has been argued, is attributable to the policies and reforms introduced by the last military dictatorship. These policies should not be thought of in a vacuum; rather, they interacted with a changing international context that was adverse for an economy like Argentina's, giving rise to a meagre economic performance. Despite this interrelation, the results of this work show that the economic deterioration is not explained by the external context alone, but must be attributed to the actions of the military government: an economic programme centred on financial liberalisation, deindustrialisation and the repression of the working class. Without the profound changes brought about by the military dictatorship, Argentina could have reached 1991 with a per capita output between 43% and 59% higher, greater capacity to carry out expansionary fiscal policies and a more diversified productive structure. Although the figures may seem high, it would have sufficed for the Argentine economy to grow at an average annual rate of between 1.4% and 1.9%, in line with countries such as Australia, Brazil, Mexico or Uruguay. These results are robust to a wide range of sensitivity analyses, including changes in the specification of the model, exclusions from or inclusions in the pool of donor countries and placebo effects in time and space.

As discussed in chapter 5, the economic deterioration during the last quarter of the century is not explained solely by the entry into a high-inflation regime after the "Rodrigazo". Beyond the fact that it is hardly plausible that inflation by itself could cause the estimated effects, the estimated counterfactual also went through a process of high inflation and a hyperinflation. The difference would therefore seem to lie in the way in which the dictatorship dealt with macroeconomic volatility and political conflict, and in the constraints that this handling imposed on the first democratic government (mainly, the external debt). During the first three quarters of the twentieth century, the estimates suggest that the country did not undergo a break in its growth trend, but that its structural characteristics led it to grow at a slower pace than other countries that are not strictly comparable to Argentina (such as the European ones). Thus, the impacts of the policies of the last military dictatorship are the main factor behind the divergence in Argentina's economic growth during the twentieth century.

7. References

- Abadie, A. (2021). Using Synthetic Controls: Feasibility, Data Requirements, and Methodological Aspects. *Journal of Economic Literature*, 59(2). <https://doi.org/10.1257/jel.20191450>
- Abadie, A., Diamond, A., & Hainmueller, J. (2010). Synthetic Control Methods for Comparative Case Studies: Estimating the Effect of California's Tobacco Control Program. *Journal of the American Statistical Association*, 105(490), 493-505. <http://dx.doi.org/10.1198/jasa.2009.ap08746>
- Abadie, A., Diamond, A., & Hainmueller, J. (2011). Synth: An R Package for Synthetic Control Methods in Comparative Case Studies. *Journal of Statistical Software*, 42(13).
- Abadie, A., Diamond, A., & Hainmueller, J. (2015). Comparative Politics and the Synthetic Control Method. *American Journal of Political Science*, 59(2), 495-510. <https://doi.org/10.1111/ajps.12116>
- Abadie, A., & Gardeazabal, J. (2003). The Economic Costs of Conflict: A Case Study of the Basque Country. *American Economic Review*, 93(1), 113-132.
- Abeles, M., Pérez Caldentey, E., & Valdecantos, S. (2018). *Estudios sobre financierización en América Latina*. Comisión Económica para América Latina y el Caribe (CEPAL).
- Águila, G. (2023). *Historia de la última dictadura militar. Argentina, 1976-1983*. Siglo XXI Editores Argentina.
- Álvarez, R. E. (2020). *Essays on the Argentine political economy through the lens of the classical-keynesian approach* [Doctoral, Universities of Florence, Siena and Pisa join PhD Program in Economics]. <https://usiena-air.unisi.it/handle/11365/1105887>
- Álvarez, R. E., & Brondino, G. (2024). The limits to redistribution in small open economies: The case of Argentina. *Review of Keynesian Economics*, 12(1), 53-73.
- Alvarez, R. E., & Dvoskin, A. (2023). On Income Distribution Dynamics in Argentina During the 1976–1983 Dictatorship: A Classical-Structuralist Interpretation. *Review of Political Economy*, 35(3), 738-761. <https://doi.org/10.1080/09538259.2022.2149921>
- Amico, F. (2015). Los salarios reales en el largo plazo: Surgimiento de un nuevo piso estructural de las remuneraciones en Argentina. *Documentos de Trabajo del CEFIDAR*, 67, 90.
- Amico, F., Fiorito, A., & Hang, G. (2011). Producto potencial y demanda en el largo plazo: Hechos estilizados y reflexiones sobre el caso argentino reciente. *Documentos de Trabajo del CEFIDAR*, 35, 90.
- Anderson, P. (1988). Democracia y dictadura en América Latina en la década del 70. *Revista de Ciencias Sociales*, n. 3, pp. 15-23. https://www.colibri.udelar.edu.uy/jspui/bitstream/20.500.12008/26385/1/RCS_Anderson_1988n3.pdf
- Azpiazu, D., Basualdo, E., & Khavisse, M. (1987). *El nuevo poder económico en la Argentina*. Legasa.
- Azpiazu, D., & Schorr, M. (2010). *Hecho en Argentina. Industria y Economía, 1976-2007*. Siglo XXI Editores.
- Basualdo, E. (2006). *Estudios de historia económica argentina*. Siglo XXI Editores. <https://sigloxxieditores.com.ar/libro/estudios-de-historia-economica-argentina/>
- Basualdo, E. (2010). *Estudios de historia económica argentina. Desde mediados del siglo XX a la actualidad*. Siglo XXI.

- Basualdo, E. M. (2001). *Sistema político y modelo de acumulación en la Argentina. Notas sobre el transformismo argentino durante la valorización financiera (1976-2001)*. Universidad Nacional de Quilmes Ediciones.
- Beck, T., Degryse, H., & Kneer, C. (2014). Is more finance better? Disentangling intermediation and size effects of financial systems. *Journal of Financial Stability*, 10, 50-64. <https://doi.org/10.1016/j.jfs.2013.03.005>
- Bértola, L., & Ocampo, J. A. (2013). *El desarrollo económico de América Latina desde la independencia*. Fondo de Cultura Económica.
- Bhaduri, A., & Marglin, S. (1990). Unemployment and the real wage: The economic basis for contesting political ideologies. *Cambridge Journal of Economics*, 14(4), 375-393.
- Bhering, G., Freitas, F., & Serrano, F. (2019). Thirlwall's law, external debt sustainability, and the balance-of-payments-constrained level and growth rates of output. *Review of Keynesian Economics*, 7(4), 486-497.
- Blaum, L., & Román, V. (2022). Disciplinamiento social y vaivenes de política económica. Las ideas liberal-conservadoras en Argentina 1976-1980. In P. Gerchunoff, D. Heymann, & A. Jáuregui (Eds.), *Medio siglo entre tormentas. Fluctuaciones, crisis y políticas macroeconómicas en la Argentina (1948-2022)*. Eudeba.
- Blecker, R., & Setterfield, M. (2019). *Heterodox macroeconomics: Models of demand, distribution and growth*. Edward Elgar Publishing.
- Blyth, M. (2002). *Great Transformations. Economic Ideas and Institutional Change in the Twentieth Century*. Cambridge University Press. <https://doi.org/10.1017/CBO9781139087230>
- Bolt, J., & van Zanden, J. L. (2020). Maddison style estimates of the evolution of the world economy. A new 2020 update. *The Maddison Project*.
- Bona, L. M., & Barrera, M. A. (2021). El endeudamiento como motor de la economía. In A. Wainer, *¿Por qué siempre faltan dólares? Las causas estructurales de la restricción externa en la economía argentina del siglo XXI* (p. 232). Siglo XXI Editores Argentina.
- Bona, L. M., & Páez, S. M. (2020). Fases, similitudes y diferencias entre los casos de las dictaduras y economía política en Argentina, 1966-1973 y 1976-1983, y Brasil, 1964-1985. *América Latina en la historia económica*, 27(2). https://www.scielo.org.mx/scielo.php?pid=S1405-22532020000200006&script=sci_arttext
- Boyer, R. (2000). Is a Finance-led growth regime a viable alternative to Fordism? A preliminary analysis. *Economy and Society*, 29(1), 111-145. <https://doi.org/10.1080/030851400360587>
- Brambilla, I., Galiani, S., & Porto, G. (2018). Argentine Trade Policies in the XX Century: 60 Years of Solitude. *Latin American Economic Review*, 27(4). <https://doi.org/10.1007/s40503-017-0050-9>
- Buera, F., Navarro, G., & Nicolini, J. P. (2011). The Argentine Economy After Two Centuries. *Latin American Journal of Economics*, 48(2), 133-156.
- Burdisso, T., & Corso, E. A. (2011). *Incertidumbre y dolarización de cartera. El caso argentino en el último medio siglo*.
- Campante, F., & Glaeser, E. (2018). Yet Another Tale of Two Cities: Buenos Aires and Chicago. *Latin American Economic Review*, 27(2). <https://doi.org/10.1007/s40503-017-0052-7>

- Canitrot, A. (1980). Discipline as the central objective of economic policy: An essay on the economic programme of the Argentine government since 1976. *World Development*, 8(11), 923-928.
- Canitrot, A. (1981). Teoría y práctica del liberalismo. Política antiinflacionaria y apertura económica en la Argentina. *Desarrollo Económico*, 21(82), 131-189.
- Carregal, C., Coatz, D., & Sarabia, M. (2015). La matriz productiva argentina: Una mirada en retrospectiva para repensar la política industrial. In OIT, *Un enfoque productivo para el trabajo decente: Desarrollo industrial, entramado institucional y empleo de calidad en Argentina*.
- Castellani, A. (2004). Gestión económica liberal-corporativa y transformaciones en el interior de los grandes agentes económicos de la Argentina durante la última dictadura militar. In A. Pucciarelli, *Empresarios, tecnócratas y militares. La trama corporativa de la última dictadura militar* (p. 385). Siglo XXI Editores.
- Castellani, A. (2006). Los ganadores de la «década perdida». La consolidación de las grandes empresas privadas privilegiadas por el accionar estatal Argentina 1984-1988. In A. Pucciarelli, *Los años de Alfonsín. ¿El poder de la democracia o la democracia del poder?* Siglo XXI.
- Castellani, A. (2021). La consolidación de la «patria contratista» durante la última dictadura cívico-militar. In M. Schorr, *El viejo y el nuevo poder económico en la Argentina*. Siglo XXI Editores Argentina.
- Castellani, A., & Heredia, M. (2020). La reproducción fallida de las elites. Inestabilidad y transformaciones de las elites empresariales argentinas entre 1976 y 2015. *Revista Española de Sociología*, 29(3), 467-486. <https://doi.org/10.22325/fes/res.2020.30>
- Cazón, F., Kennedy, D., & Lastra, F. (2016). Las condiciones de reproducción de fuerza de trabajo como forma de la especificidad de la acumulación de capital en Argentina: Evidencias concretas desde mediados de los 70. *Trabajo y sociedad*, 27, 305-327.
- CEPAL. (2018). *La ineficiencia de la desigualdad: 2018*. Naciones Unidas, CEPAL.
- CEPAL, & OIT. (2023). Coyuntura laboral en la Argentina. Productividad y salarios: Una mirada de largo plazo. *Coyuntura laboral en la Argentina*, 2(1).
- Chinn, M. D., & Ito, H. (2008). A New Measure of Financial Openness. *Journal of Comparative Policy Analysis: Research and Practice*, 10(3), 309-322. <https://doi.org/10.1080/13876980802231123>
- Cimoli, M., & Porcile, G. (2013). Tecnología, heterogeneidad y crecimiento. Una caja de herramientas estructuralistas. *CEPAL - Serie Desarrollo Productivo*, 194.
- Corso, E. A. (2021). Dolarización financiera en Argentina: Un análisis histórico de una restricción vigente. *Documentos de Trabajo del BCRA*, 95.
- Cortes Conde, R. (2005). *La Economía Política de la Argentina en el siglo XX*. Edhasa.
- Cortés Conde, R., & Harriague, M. (1994). *Estimaciones del Producto Bruto Interno de Argentina, 1875-1935*. UDESA.
- Crespo, E., & Chibaudi, J. (2013). Las contradicciones del capitalismo brasileño y el mito de la burguesía nacional. *Entrelíneas de la Política Económica*, 6. <https://sedici.unlp.edu.ar/handle/10915/30260>
- Cuesta, M., & Trupkin, D. (2022). Deuda, guerra y crisis. La economía argentina entre 1981 y 1983. In P. Gerchunoff, D. Heymann, & A. Jáuregui (Eds.), *Medio siglo entre*

- tormentas. *Fluctuaciones, crisis y políticas macroeconómicas en la Argentina (1948-2022)*. Eudeba.
- D'Amato, L., & Katz, S. (2018). Una constante en la evolución macroeconómica argentina: Dinero, deuda y crisis (1945-2015). In R. Cortes Conde & G. Della Paolera, *Nueva Historia Económica de la Argentina*. Edhasa.
- de Pablo, J. C. (1983). El Enfoque Monetario De La Balanza De Pagos En La Argentina: Análisis Del Programa Del 20 De Diciembre De 1978. *El Trimestre Económico*, 50(198(2)), 641-669.
- Di Tella, G., & Zymelman, M. (1967). *Las etapas del desarrollo económico argentino*. Eudeba.
- Duménil, G., & Lévy, D. (1999). Ser keynesiano en el corto plazo y clásico en el largo plazo. *Análisis Económico*, XIV(30), 55-93.
- Dvoskin, A., & Feldman, G. D. (2018). Income distribution and the balance of payments: A formal reconstruction of some Argentinian structuralist contributions Part I: Technical dependency: Part I: Technical dependency*. *Review of Keynesian Economics*, 6(3), 352-368. <https://doi.org/10.4337/roke.2018.03.04>
- Dvoskin, A., & Médici, F. (2023). Evaluación Empírica de Dos Hipótesis Testeables del Supermultiplicador Sraffiano para Argentina 1993-2021. *Documentos de Trabajo del BCRA*, 112.
- Dvoskin, A., & Torchinsky Landau, M. (2023). Income distribution and economic cycles in an open-economy supermultiplier model. *Structural Change and Economic Dynamics*, 64, 273-291. <https://doi.org/10.1016/j.strueco.2022.12.012>
- Eichengreen, B. J. (2008). *Globalizing capital: A history of the international monetary system* (2nd ed). Princeton University Press.
- Engerman, S. L., & Sokoloff, K. L. (2002). Factor Endowments, Inequality, and Paths of Development among New World Economies. *NBER Working Paper Series*, 9259. <http://www.nber.org/papers/w9259>
- Ertola Navajas, M. G. (2020). Peron: A Synthetic Control Analysis. In M. G. Ertola Navajas, *Essays in applied economics*. Universidad de San Andrés. Doctoral thesis.
- Fagundes, L., & Freitas, F. (2019). The Role of Autonomous Non-Capacity Creating Expenditures in Recent Kaleckian Growth Models: An Assessment from the Perspective of the Sraffian Supermultiplier Model. *XII Encontro Internacional da AKB*. <https://associacaokeynesianabrasileira.org/wp-content/uploads/2017/08/379904d7ae8e4197ba711.pdf>
- Ferman, B., & Pinto, C. (2021). Synthetic controls with imperfect pretreatment fit. *Quantitative Economics*, 12(4), 1197-1221. <https://doi.org/10.3982/QE1596>
- Ferreres, O. J. (2010). *Dos siglos de economía argentina: Edición bicentenario 1810-2010; historia argentina en cifras*. Ed. El Ateneo.
- Fiorito, A. (2015). Patrones de desarrollo y distribución del ingreso en la Argentina. *Documentos de Trabajo del CEFIDAR*, 70.
- Fiorito, A. (2018). Supermultiplicador Clásico y Crecimiento Económico Dirigido por la Demanda en la Argentina. *Master's thesis, Facultad de Ciencias Económicas, Universidad de Buenos Aires*.

- Francis, J. (2013). *The Terms of Trade and the Rise of Argentina in the Long Nineteenth Century*. Department of Economic History of the London School of Economics and Political Science.
- Fraser, N. (2020). Las contradicciones del capital y de los cuidados. In N. Fraser, *Los talleres ocultos del capital. Un mapa para la izquierda* (pp. 73-92). Traficantes de Sueños.
- Freitas, F., & Serrano, F. (2015). Growth Rate and Level Effects, the Stability of the Adjustment of Capacity to Demand and the Sraffian Supermultiplier. *Review of Political Economy*, 27(3), 258-281. <https://doi.org/10.1080/09538259.2015.1067360>
- Frenkel, R., & Rapetti, M. (2010). A Concise History of Exchange Rate Regimes in Latin America. *Documentos de Trabajo Del CEDES*, 67.
- Gadea, M. D., & Sanz-Villarroya, I. (2020). Growth and business cycle in Argentina. A long-run approach, 1870–2015. *Applied Economic Analysis*, 28(84), 197-220. <https://doi.org/10.1108/AEA-03-2020-0024>
- Galiani, S., & Somaini, P. (2018). Path-dependent import-substitution policies: The case of Argentina in the twentieth century. *Latin American Economic Review*, 27(5). <https://doi.org/10.1007/s40503-017-0047-4>
- Garegnani, P. (1970). Heterogeneous capital, the production function and the theory of distribution. *The Review of Economic Studies*, 37(3).
- Garegnani, P. (1978). Notes on consumption, investment and effective demand: I. *Cambridge Journal of Economics*, 2(4).
- Garegnani, P. (1984). Value and distribution in the classical economists and Marx. *Oxford Economic Papers*, 36(2).
- Gerchunoff, P., & Fajgelbaum, P. (2006). *¿Por qué Argentina no fue Australia? Una hipótesis sobre un cambio de rumbo*. Siglo XXI Editores Argentina.
- Gerchunoff, P., & Hora, R. (2021). *La moneda en el aire*. Siglo XXI Editores Argentina.
- Gerchunoff, P., & León, de, G. (2018). Entre la flexibilidad y la institucionalización: Una historia circular del mercado de trabajo argentino (1914-2016). In R. Cortes Conde & G. Della Paolera, *Nueva Historia Económica de la Argentina*. Edhasa.
- Gerchunoff, P., & Llach, L. (2003). Ved en Trono a la Noble Igualdad. Crecimiento, equidad y política económica en la Argentina, 1880-2003. *Fundación PENT*.
- Gerchunoff, P., & Llach, L. (2009). Equality or growth: A 20th century Argentine dilemma. *Revista de Historia Económica-Journal of Iberian and Latin American Economic History*, 27(3), 397-426.
- Gerchunoff, P., & Llach, L. (2019). *El ciclo de la ilusión y el desencanto. Políticas económicas argentinas de 1880 a nuestros días* (4.^a ed.). Crítica.
- Gerchunoff, P., & Rapetti, M. (2016). La Economía Argentina y su Conflicto Distributivo Estructural (1930-2015). *El Trimestre Económico*, 83(330), 225-272.
- Gerchunoff, P., Rapetti, M., & León, G. (2020). La paradoja populista. *Desarrollo Económico*, 59(229).
- Gerchunoff, S. (2019). Hegemonía y acumulación: Transformismo y valorización financiera del capital en el análisis de Eduardo Basualdo. *Revista de la Red de Intercátedras de Historia de América Latina Contemporánea*, 11.
- Glaeser, E., Di Tella, R., & Llach, L. (2018). Introduction to Argentine exceptionalism. *Latin American Economic Review*, 27(1). <https://doi.org/10.1007/s40503-017-0055-4>

- González, M. L. (2012). Las modificaciones en el patrón de crecimiento y su consecuencia sobre la evolución de los salarios reales en Argentina desde 1950. *Diálogos-Revista do Departamento de História e do Programa de Pós-Graduação em História*, 16(1), 39-68. <https://doi.org/10.4025/dialogos.v16i1.587>
- Graña, J., & Kennedy, D. (2008). Salario real, costo laboral y productividad. Argentina 1947-2006. Análisis de información y metodología de estimación. *Documentos de Trabajo del CEPED*, 12.
- Graña, J., & Kennedy, D. (2011). Producción de la riqueza social y su distribución: Productividad y salarios en Argentina. Medio siglo en perspectiva latinoamericana e internacional. *Cuestiones de sociología*, 7.
- Graña, J., & Kennedy, D. (2017). Rezago productivo y sus fuentes de compensación. La vigencia de los limitantes estructurales del ciclo económico argentino al comienzo del siglo XXI. *Cuadernos del CENDES*, 34(95).
- Graña, J. M. (2015). Evolución comparada del sector industrial argentino y estadounidense, entre el rezago productivo y el deterioro salarial. *H-industria. Revista de historia de la industria y el desarrollo en América Latina*, 17, Article 17.
- Graña, J. M., & Terranova, L. (2022). Neither mechanical nor premature: Deindustrialization and the New International Division of Labour (1970-2019). *Revista de Historia Industrial — Industrial History Review*, 31(86), Article 86. <https://doi.org/10.1344/rhihr.36591>
- Grimson, A. (2020). *¿Qué es el peronismo?* (1.^a ed.). Siglo XXI Editores Argentina.
- Grinberg, N. (2022). From Populism to Neoliberalism: The Political Economy of Latin American Import-Substitution Industrialization: Argentina, Brazil, Mexico and Colombia in Comparative Perspective. *Latin American Perspectives*, 49(2), 183-206. <https://doi.org/10.1177/0094582X211063503>
- Guissarri, A. (1989). *La Argentina informal. Realidad de la vida económica*. Emecé Editores.
- Harriague, M., & Rayes, A. (2018). Fuentes para el estudio de la historia económica argentina. In R. Cortes Conde & G. Della Paolera, *Nueva Historia Económica de la Argentina*. Edhasa.
- Heredia, M. (2004). El proceso como bisagra. Emergencia y consolidación del liberalismo tecnocrático. FIEL, FM y CEMA. In A. Pucciarelli, *Empresarios, tecnócratas y militares. La trama corporativa de la última dictadura militar* (p. 385). Siglo XXI Editores.
- Hobsbawm, E. (1994). *Historial del siglo XX. 1914-1991*. Crítica.
- Hopenhagen, H., & Neumeyer, P. A. (2004). Explaining Argentina's Great Depression of 1975-1990. *International Development Bank*.
- Hora, R. (2010). *Historia Economía de la Argentina en el siglo XIX*. Siglo XXI Editores. <https://sigloxxieditores.com.ar/libro/historia-economica-de-la-argentina-en-el-siglo-xix/>
- Hora, R. (2018). ¿Cómo pensó Tulio Halperin Donghi la política de entreguerras? *Estudios Sociales*, 54. <https://ri.conicet.gov.ar/handle/11336/177299>
- Iñigo Carrera, J. (1998). La acumulación de capital en la Argentina. *Centro para la Investigación como Crítica Práctica*, Buenos Aires. https://cicpint.org/wp-content/uploads/2017/03/1998_JIC_La-acumulacion-de-capital-en-la-Argentina.pdf

- Kacef, O., Robbio, J., & Vitto, C. (2022). Conflictividad política e inestabilidad macroeconómica: La economía argentina entre 1973 y 1976. In P. Gerchunoff, D. Heymann, & A. Jáuregui, *Medio siglo entre tormentas. Fluctuaciones, crisis y políticas macroeconómicas en la Argentina (1948-2002)*. Eudeba.
- Kalecki, M. (1943). Political aspects of full employment. *The Political Quarterly*. <https://doi.org/10.1111/j.1467-923X.1943.tb01016.x>
- Kennedy, D. (2011). *Economía política de la contabilidad social: Vínculos entre la teoría de la riqueza social y sus formas de cuantificación*. Universidad de Buenos Aires. Facultad de Ciencias Económicas.
- Kennedy, D. (Ed.). (2018a). *Debates en torno a las condiciones actuales de reproducción de la fuerza de trabajo en Argentina en perspectiva histórica*. Universidad de Buenos Aires. Facultad de Ciencias Económicas. Secretaría de Investigación.
- Kennedy, D. (2018b). El rol del deterioro estructural del salario real argentino desde mediados de los setenta. In D. Kennedy, *Debates en torno a las condiciones actuales de reproducción de la fuerza de trabajo argentina en perspectiva histórica* (1.^a ed.). Universidad de Buenos Aires. Facultad de Ciencias Económicas. Secretaría de Investigación.
- Kennedy, D., & Aguila, N. (2021). From Individual Wage to Household's Labor Income: Consequences of Women's Labor Force Incorporation in Argentina, 1974–2018. *Review of Radical Political Economics*, 53(3), 423-441. <https://doi.org/10.1177/0486613420971072>
- Kidyba, S., & Suárez, L. (2020). Aplicación de los Índices Encadenados al empalme de series Argentina 1935-2020. *Documentos de Trabajo del PICNA*, 5.
- Kulfas, M., & Schorr, M. (2003). Deuda externa y valorización financiera en la Argentina actual- Factores explicativos del crecimiento del endeudamiento externo y perspectivas ante el proceso de renegociación. *Realidad Económica*, 198.
- Ladeux, J., & Schiaffino, P. (2022). El gigante con pies de barro. ¿Fue la Argentina realmente rica? El capital humano durante la primera globalización en una perspectiva comparada. *Revista de Historia Económica / Journal of Iberian and Latin American Economic History*, 40(3), 439-473. <https://doi.org/10.1017/S0212610920000208>
- Lavoie, M. (2015). *Post-Keynesian economics: New foundations* (Paperback ed. reprinted with amendments). Elgar. <https://doi.org/10.4337/9781783475827>
- Libman, E., Palazzo, G., & Rodríguez, M. (2022). El dilema de la estabilización: Deuda, inflación y conflictividad política en tiempos de Alfonsín (1983-1989). In P. Gerchunoff, D. Heymann, & A. Jáuregui (Eds.), *Medio siglo entre tormentas. Fluctuaciones, crisis y políticas macroeconómicas en la Argentina (1948-2022)*. Eudeba.
- Llach, J. J. (1985). *La Argentina que no fue [Tomo 1: Las fragilidades de la Argentina agroexportadora (1918-1930)]*. Ediciones del IDES.
- Llach, L. (2013). Newly Rich, Not Modern Yet: Argentina Before the Depression. In G. Di Tella, E. Glaeser, & L. Llach, *Exceptional Argentina*.
- Maddison Project Database. (2020). Maddison style estimates of the evolution of the world economy. A new 2020 update. *Bolt, Jutta and Jan Luiten van Zanden (2020)*.
- Manzanelli, P., Barrera, M. A., Wainer, A., & Bona, L. (2015). Deuda externa, fuga de capitales y restricción externa. Desde la última dictadura militar hasta la actualidad. *Documentos de Trabajo del CEFIDAR*, 68.

- Manzano, V. (2017). *La era de la juventud en Argentina. Cultura, política y sexualidad desde Perón hasta Videla*. Fondo de Cultura Económica.
- Marzetti, M., & Spruk, R. (2023). Long-Term Economic Effects of Populist Legal Reforms: Evidence from Argentina. *Comparative Economic Studies*, 65(1), 60-95. <https://doi.org/10.1057/s41294-022-00193-4>
- Médici, F. (2010). *La demanda efectiva como determinante del producto: Un análisis de cointegración para Argentina (1980-2007)* [Magister en Economía, Universidad Nacional de La Plata]. <https://doi.org/10.35537/10915/3311>
- Mesyngier, L. (1998). *La política implícita. Instrumentos económicos y transformación social en la última dictadura*. Eudeba.
- Míguez, E. (2005). «El fracaso argentino». Interpretando la evolución económica en el «corto siglo XX». *Desarrollo Económico*, 44(176), 483-514.
- Mirowski, P., & Plehwe, D. (Eds.). (2015). *The Road from Mont Pèlerin. The Making of the Neoliberal Thought Collective*. Harvard University Press.
- Morresi, S. (2010). El liberalismo conservador y la ideología del Proceso de Reorganización Nacional. *Sociohistórica*, 27, 103-135.
- Müller, A. (2001). Un quiebre olvidado: La política económica de Martínez de Hoz. *Ciclos*, XI(21).
- Müller, A. (2011). ¿La culpa es de Martínez de Hoz? *Documentos de Trabajo del CESP*, 26.
- Nochteff, H. (1994). Los senderos perdidos del subdesarrollo. Elite económica y restricciones al desarrollo en la Argentina. In D. Azpiazu & H. Nochteff, *El desarrollo ausente. Restricciones al desarrollo, neoconservadurismo y élite económica en la Argentina. Ensayos de Economía Política*. FLACSO.
- Ortiz, R., & Schorr, M. (2006). La economía política del gobierno de Alfonsín: Creciente subordinación al poder económico durante la década perdida. In A. Pucciarelli, *Los años de Alfonsín. ¿El poder de la democracia o la democracia del poder?* Siglo XXI.
- Ortiz, R., & Schorr, M. (2021). ¿Década perdida? Los grupos económicos en el gobierno de alfonsín. In M. Schorr, *El viejo y el nuevo poder económico en la Argentina*. Siglo XXI Editores Argentina.
- Petri, F. (1989). *Teorías del valor y la distribución. Una comparación entre clásicos y neoclásicos*. Universidad Nacional de Moreno.
- Petri, F. (2002). A ‘Sraffian’ critique of general equilibrium theory, and the classical-Keynesian alternative. In F. Petri & F. Hahn, *General Equilibrium. Problems and Prospects*. Routledge.
- Petri, F. (2004). *General Equilibrium, Capital and Macroeconomics: A key to recent controversies in equilibrium theory*. Edward Elgar Publishing.
- Petri, F. (2021). *Microeconomics for the Critical Mind: Mainstream and Heterodox Analyses*. Springer International Publishing. <https://doi.org/10.1007/978-3-030-62070-7>
- Polanyi-Levitt, K. (2013). On Transformations: Past, Present and Future? In *From the Great Transformation to the Great Financialization*. Fernwood Publishing.
- Prados De La Escosura, L. (2016). Mismeasuring long-run growth: The bias from splicing national accounts—the case of Spain. *Cliometrica*, 10(3), 251-275. <https://doi.org/10.1007/s11698-015-0131-4>
- Pucciarelli, A. (2004). *Empresarios, tecnócratas y militares. La trama corporativa de la última dictadura militar*. Siglo XXI Editores.

- Rappoport, M. (2008). *Historia de la economía argentina del siglo XX*. Página 12.
- Roth, J. (2022). Pre-test with Caution: Event-study Estimates After Testing for Parallel Trends. *American Economic Review: Insights*, 4(3). <https://doi.org/10.1257/aeri.20210236>
- Sábato, J. E., & Schvarzer, J. (1985). Funcionamiento de la economía y poder político en la Argentina: Trabas para la democracia. In A. Rouquié, *¿Cómo renacen las democracias?* Emecé.
- Salvadori, N. (2003). *The Theory of Economic Growth: A 'Classical' Perspective*. Edward Elgar.
- Sánchez, G. (2018). Crecimiento y producción en Argentina en el siglo XX. In R. Cortes Conde & G. Della Paolera, *Nueva Historia Económica de la Argentina*. Edhasa.
- Sáenz-Villarroya, I. (2005). The convergence process of Argentina with Australia and Canada. *Explorations in Economic History*, 42, 439-458.
- Semán, E. (2021). *Breve Historia del Antipopulismo* (1.^a ed.). Siglo XXI Editores Argentina.
- Serrano, F. (1995). Long period effective demand and the Sraffian supermultiplier. *Contributions to Political Economy*, 14(1), 67-90.
- Sidicaro, R. (2004). Coaliciones golpistas y dictaduras militares: El “proceso” en perspectiva comparada. In A. Pucciarelli, *Empresarios, tecnócratas y militares. La trama corporativa de la última dictadura militar* (p. 385). Siglo XXI Editores.
- Spruk, R. (2019). The rise and fall of Argentina. *Latin American Economic Review*, 28(1), 16. <https://doi.org/10.1186/s40503-019-0076-2>
- Tafunell, X., & Ducoing, C. (2015). Non-Residential Capital Stock in Latin America. 1875 – 2008. *Australian Economic History Review*. <https://doi.org/10.1111/aehr.12076>
- Taylor, A. (2018). The Argentina Paradox: Microexplanations and Macropuzzles. *Latin American Economic Review*, 27(3). <https://doi.org/10.1007/s40503-017-0051-8>
- Torre, J. C. (2021). *Diario de una temporada en el quinto piso* (1.^a ed.). Edhasa.
- Trímboli, J. (2023). *Tulio Halperín Donghi. La herencia está ahí: Diez entrevistas comentadas*. Omnívora.
- Vernengo, M. (2020). The State of Post-Keynesian Economics and its connections with other heterodox perspectives. *American Review of Political Economy*, 14(1). <https://doi.org/10.38024/arpe.209>
- Zack, G., & Pryluka, P. (2022). Dictadura y reforma económica. Argentina en el nuevo orden internacional. In P. Gerchunoff, D. Heymann, & A. Jáuregui (Eds.), *Medio siglo entre tormentas. Fluctuaciones, crisis y políticas macroeconómicas en la Argentina (1948-2022)*. Eudeba.

8. Appendix

This appendix develops the model of Dvoskin and Torchinsky Landau (2023) used for the qualitative analysis of the effects of the dictatorship. First, in the model growth is limited by demand, which is in turn composed, as usual, of consumption (C_t), investment (I_t), public spending (G_t) and exports (X_t).

$$Y_t + M_t = C_t + I_t + G_t + X_t \quad (1)$$

These last two components make up autonomous demand (Z_t), that is, the expenditures that do not necessarily depend on the level of activity of the period.

$$Z_t = G_t + X_t \quad (2)$$

Consumption, imports and investment, by contrast, are determined by the level of income of the period. In the case of imports (M_t), they are assumed to be determined by the level of activity in a fixed proportion (m). In the case of consumption, it is assumed to be equal to the wage bill, made up of an exogenous wage (w) and the level of employment, determined by the employment requirements per unit of output (v_l). The price level is normalised to 1, so that w represents the real wage.

$$C_t = w v_l Y_t \quad (3)$$

Unlike other growth models of the *demand-led* type, in this one investment is a component induced by demand, in line with the Sraffian supermultiplier (Serrano, 1995). That is, investment decisions adjust so as to satisfy a given level of utilisation of installed capacity, regarded as “normal” and determined by the technique in use given the distribution of income (Fagundes & Freitas, 2019). Thus, for example, a positive shock to the level of demand can be met in the short run by an increase in the utilisation of installed capacity, which will eventually return to its normal level (u_n). The reason is that deviations of the capital-output ratio from the one that minimises costs can only be temporary. As a consequence, the response of investment (I_t) to an increase in demand (Y_t) adjusts flexibly to accommodate utilisation, according to the following expression:

$$I_t = h_t Y_t \quad \text{where } \dot{h}_t = f(u_t - u_n) \quad (3)$$

where h_t represents the elasticity of investment with respect to demand and u_t the utilisation of installed capacity (Y_t/Y_t^K). The investment equation links demand with supply and, in particular, with the maximum level of output of the economy given its capital stock (Y_t^K). Given the technology, this level is given by the amount of capital per unit of output (v), so that $Y_t^K = \frac{1}{v} K_t$. The dynamics of the capital stock (g_t^K) are given by investment:

$$g_t^K = \frac{I_t}{K_t} \quad (4)$$

Substituting (3) into (4) and using the definitions of u_t , v and h_t , it follows that the growth of the capital stock is a function of the investment accelerator and the utilisation of installed capacity:

$$g_t^K = \frac{h_t}{v} u_t \quad (5)$$

With these equations, the long-run level of output is determined by:

$$Y^i = \frac{1}{1 - w v_l - \frac{g_z v}{u_n} + m} \cdot Z_t \quad (6)$$

Finally, the structuralist component of the model addresses the specificities of a small open developing economy by formalising the components of the balance of payments. Exports, since they respond to external demand, are exogenous, while imports (M_t) are determined by the level of activity in a fixed proportion (m). There are also outflows for interest payments (at a rate i^i) on the stock of gross external debt (D_t). At this point we introduce a single adaptation to the original model, which will be convenient for the analysis of the period. It is proposed that, in a context of free capital mobility, a part (f) of income²⁷ is allocated to the formation of external assets (F_t), increasing financing needs²⁸.

$$F_t = f Y_t \quad (7)$$

The parameter f will be regarded as exogenous, but it is not fixed; it could vary significantly according to the differential between domestic and external interest rates and the expectations of depreciation of the local currency, among others²⁹. Countries can thus enter a trajectory of unsustainable external indebtedness. The dynamics of external indebtedness ($\dot{D}_t$) are determined by the balance-of-payments outcome.

$$\dot{D}_t = M_t - X_t + F_t + i^i D_t \quad (8)$$

In the model, the limit to external indebtedness is formalised by means of a threshold for the level of external debt as a percentage of long-run exports ($\bar{d}$), given that the capacity to pay the external debt is given by the foreign-exchange earnings from exports (Bhering et al., 2019). The dynamics of this indicator are given by:

$$\left(\frac{\dot{D}_t}{X_t} \right) = \frac{M_t + F_t}{X_t} - 1 + (i^i - g_x) \frac{D_t}{X_t} \quad (9)$$

²⁷ Strictly, the private sector allocates to saving in domestic currency a fraction of its surplus, not of total output.

²⁸ Net flows remain constant, but gross indebtedness must increase to offset the financing provided by residents of the domestic economy to the rest of the world.

²⁹ This behaviour could be motivated by different factors, among which it is worth highlighting the perception, on the part of some creditors, that the interest rate on domestic assets might not compensate for the expected rate of depreciation of the domestic currency. It could also be explained by the principle of portfolio risk diversification, even if returns in domestic currency were high enough. Burdisso and Corso (2011) estimate that the optimal proportion of assets in foreign currency stood, between 1976 and 1991, at between 40% and 80%.

If explosive dynamics of the external debt are ruled out in the long run (that is, if it holds that $\dot{d} \leq 0$), the following stability condition must be met:

$$\dot{d} < g_x \quad (10)$$

On the other hand, the economy faces an external constraint given by the threshold of the level of debt.

$$\dot{d} < \bar{d} \quad (11)$$

Given these conditions, a level of output is determined beyond which capital flows are interrupted and the balance of payments becomes unsustainable. The growth rate ($\dot{g}$) is determined by the average growth of the autonomous components (g_z), but in the long run the growth rate of public spending (g_G) cannot exceed that of exports (g_x) without the level of output exceeding the one compatible with external equilibrium.

$$\dot{g} = g_z \text{ subject to } g_G \leq g_x \quad (12)$$